Thermal actuators

An actuator uses thermal expansion of its active element to generate the actuating force. Use Figure 4.12 to identify the material with the largest expansion coefficient.

Answer

Neoprene, at the upper left of the chart, has a larger value of expansion coefficient than any other on the chart.

large strains), giving them low expansion coefficients. Composites, even though they have polymer matrices, can have low values of α because the reinforcing fibers, particularly carbon, expand very little.

The chart shows contours of λ/α, a quantity important in designing against thermal distortion. An extra material, Invar (a nickel alloy), has been added to the chart because of its uniquely low expansion coefficient at and near room temperature, a consequence of a trade-off between normal expansion and a contraction associated with a magnetic transformation. An application that uses the chart is developed in Chapter 6, Section 6.16.

The thermal expansion–modulus chart

Thermal stress is the stress that appears in a body when it is heated or cooled but prevented from expanding or contracting. It depends on the expansion coefficient, α, of the material and on its modulus, E. A standard development of the theory of thermal expansion leads to the relationship

$$\alpha = \frac{\gamma_G \rho C_p}{3E} \tag{4.18}$$

where γ_G is Gruneisen's constant. It has values between 0.4 and 4, but for most solids it is near 1. Since ρC_p is almost constant (see Equation (4.15)), the equation tells us that α is proportional to $1/E$. Figure 4.13 shows that this is broadly so. Ceramics, with the highest moduli, have the lowest coefficients of expansion; elastomers with the lowest moduli expand the most. Some materials with a low coordination number (silica and some diamond-cubic or zinc-blend structured materials) can absorb energy preferentially in transverse modes, leading to very small or negative values of γ_G and a low expansion coefficient (silica, SiO_2, is an example). Others, like Invar, contract as they lose their ferromagnetism when heated through the Curie temperature; over a narrow range of temperature, they too show near-zero expansion, useful in the manufacture of precision equipment and in glass-metal seals.

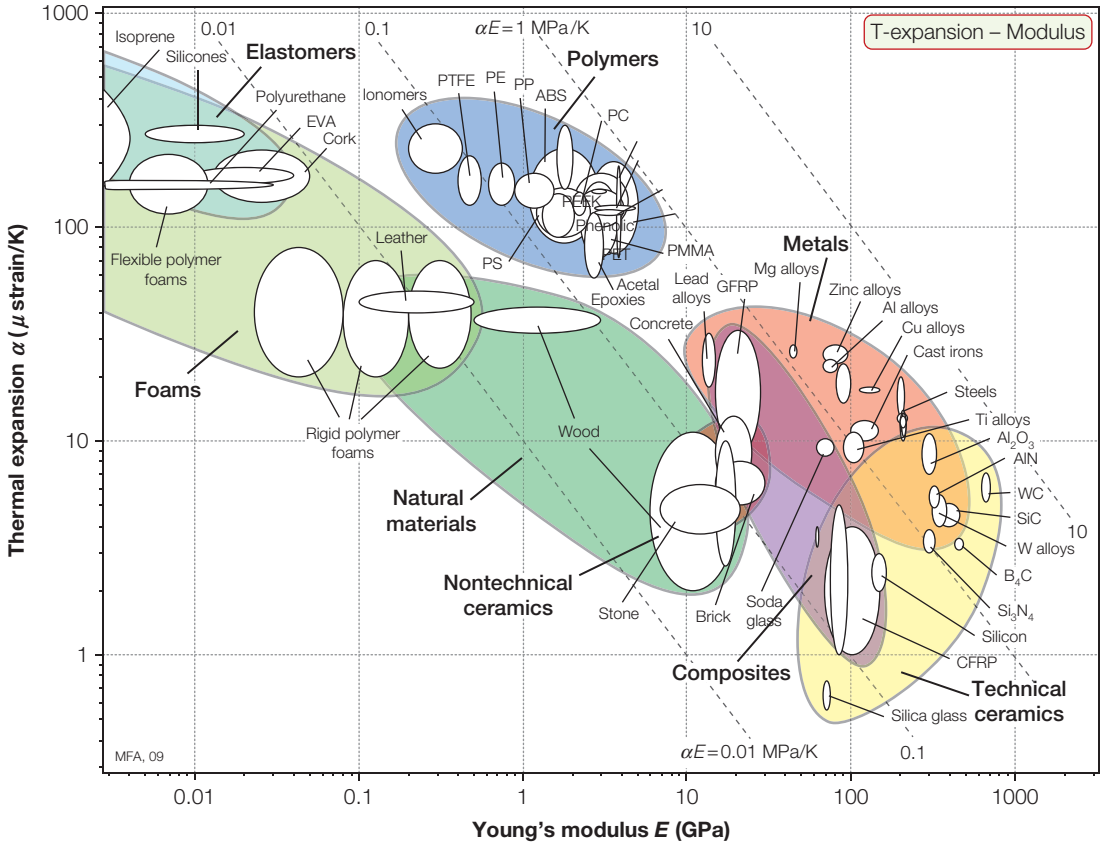


FIGURE 4.13

The linear expansion coefficient α plotted against Young's modulus E. The contours show the thermal stress created by a temperature change of 1°C if the sample is axially constrained. A correction factor C is applied for biaxial or triaxial constraint (see text).

One more useful fact. The moduli of materials scale approximately with their melting point, T_m:

$$E \approx \frac{100\,k\,T_m}{\Omega} \tag{4.19}$$

where k is Boltzmann's constant and Ω is the volume per atom in the structure. Substituting this and equation (4.15) for ρC_p into equation (4.18) for α gives

$$\alpha = \frac{\gamma_G}{100\,T_m} \tag{4.20}$$

—the expansion coefficient varies inversely with the melting point. Equivalently, the thermal strain for all solids, just before they melt, depends only

Thermal stress

Approximately what stress will appear in a rod of steel, rigidly clamped at its ends, if its temperature is changed by 100°C? Use Figure 4.13 to find out.

Answer

The figure shows that, for steel, $\alpha E \approx 3\,\text{MPa/K}$. Thus, a temperature change of 100°C will create a stress of approximately 300 MPa.

on γ_G, and this is roughly a constant at about 1%. Equations (4.18), (4.19), and (4.20) are examples of property correlations, useful for estimating and checking material properties (Appendix A, Section A.12).

Whenever the thermal expansion or contraction of a body is prevented, thermal stresses appear; if large enough, these stresses cause yielding, fracture, or elastic collapse (buckling). It is common to distinguish between thermal stress caused by external constraint (a rod rigidly clamped at both ends, for example) and that which appears without external constraint because of temperature gradients in the body. All scale as the quantity αE, shown as a set of diagonal contours in Figure 4.13. More precisely, the stress $\Delta\sigma$ produced by a temperature change of 1°C in a constrained system, or the stress per °C caused by a sudden change of surface temperature in one that is not constrained, is given by

$$C\,\Delta\sigma = \alpha E \tag{4.21}$$

where $C = 1$ for axial constraint, $(1 - \nu)$ for biaxial constraint or normal quenching, and $(1 - 2\nu)$ for triaxial constraint, where ν is Poisson's ratio. These stresses are large: typically 1 MPa/K. They can cause a material to yield, crack, spall, or buckle when it is suddenly heated or cooled.

The maximum service temperature chart

Temperature affects material performance in many ways. As the temperature is raised, the material may creep, limiting its ability to carry loads. It may degrade or decompose, changing its chemical structure in ways that make it unusable. And it may oxidize or interact in other ways with the environment in which it is used, leaving it unable to perform its function. The approximate temperature at which, for any of these reasons, it is unsafe to use a material is called its *maximum service temperature* T_{max}. Figure 4.14 show this plotted as a bar chart.

The chart gives a birds-eye view of the regimes of temperature in which each material class is usable. Note that few polymers can be used above 200ºC, few metals above 800ºC, and only ceramics offer strength above 1,500ºC.

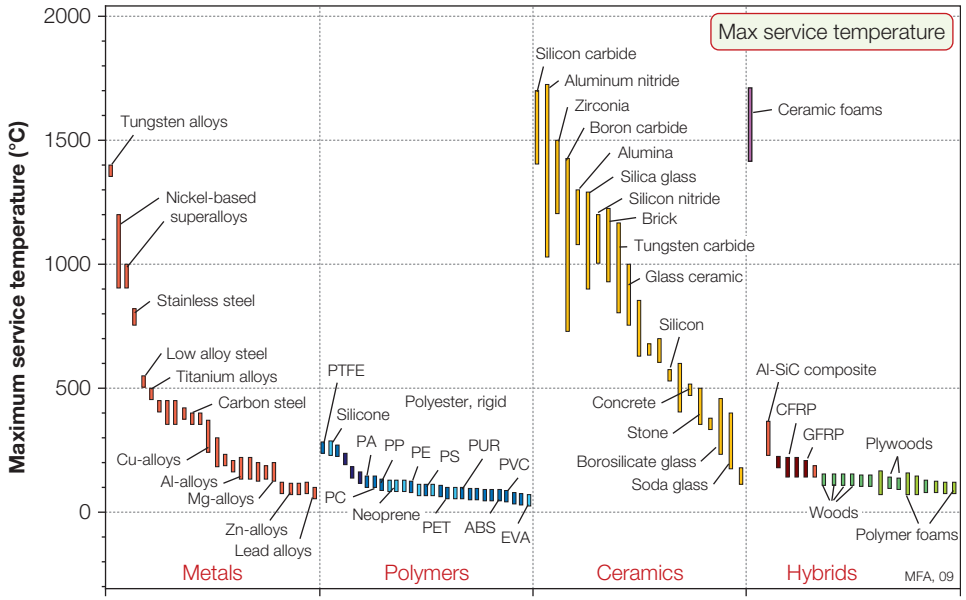


FIGURE 4.14
The maximum service temperature—the temperature above which a material becomes unusable.

Friction and wear

God, it is said, created materials but the devil made surfaces. Surfaces are the source of many problems. When surfaces touch and slide, there is friction; where there is friction, there is wear. Tribologists—those who study friction and wear—are fond of citing the enormous cost, through lost energy and worn equipment, for which these two phenomena are responsible. It is certainly true that, if friction could be eliminated, the efficiency of engines, gear boxes, drive trains, and the like, would increase. If wear could be eradicated, they would also last longer. But before accepting this negative image, one should remember that without wear pencils would not write on paper or chalk on blackboards; without friction, one would slither off the slightest incline.

Use temperature for stainless steel

Stainless steel is proposed for use as part of a structure operating at 500°C. Is it safe to do so?

Answer

Figure 4.14 shows that the maximum use temperature for stainless steel is in the range 700 to 1,100°C. Use at 500°C appears to be practical.

Tribological properties are not attributes of one material alone but of one material sliding on another with, almost always, a third in between. The number of combinations is far too great to allow choice in a simple, systematic way. The selection of materials for bearings, drives, and sliding seals relies heavily on experience. This experience is captured in reference sources (for which see Appendix D). In the end it is these that must be consulted. But it does help to have a feel for the magnitude of friction coefficients and wear rates and to have an idea of how these relate to material class.

When two surfaces are placed in contact under a normal load F_n and one is made to slide over the other, a force F_s opposes the motion. This force is proportional to F_n but does not depend on the area of the surface. This is the single most significant result of studies of friction, since it implies that surfaces do not contact completely but only touch over small patches, the area that is independent of the apparent, nominal area of contact A_n. The *coefficient friction* μ is defined by

$$\mu = \frac{F_s}{F_n} \tag{4.22}$$

Approximate values for μ for dry, unlubricated, sliding of materials on a steel counterface are shown in Figure 4.15. Typically, $\mu \approx 0.5$. Certain materials show much higher values, either because they seize when rubbed together (a soft metal rubbed on itself with no lubrication, for instance) or

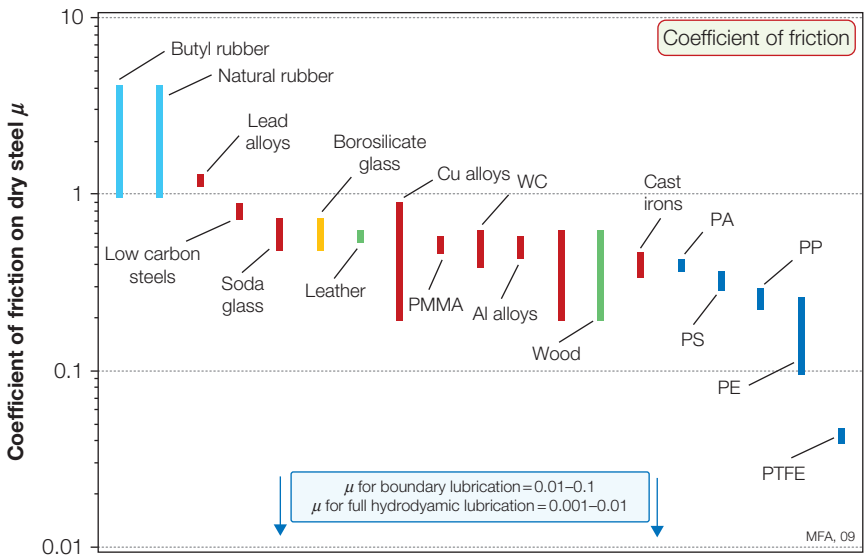


FIGURE 4.15
The friction coefficient μ of materials sliding on an unlubricated steel counterface.

because one surface has a sufficiently low modulus that it conforms to the other (rubber on rough concrete). At the other extreme are sliding combinations with exceptionally low coefficients of friction, such as PTFE or bronze bearings loaded with graphite sliding on polished steel. Here the coefficient of friction falls as low as 0.04, although this is still high compared with friction for lubricated surfaces, as noted at the bottom of the diagram.

When surfaces slide, they wear. Material is lost from both surfaces, even when one is much harder than the other. The *wear rate* W (units: m^2) is conventionally defined as

$$W = \frac{\textit{Volume of material removed from contact surface}}{\textit{Distance slid}} \tag{4.23}$$

A more useful quantity, for our purposes, is the specific wear rate

$$\Omega = \frac{W}{A_n} \tag{4.24}$$

which is dimensionless. It increases with bearing pressure P (which is the normal force F_n divided by the nominal area A_n), such that the ratio

$$k_a = \frac{W}{F_n} = \frac{\Omega}{P} \tag{4.25}$$

is roughly constant. The *wear-rate constant* k_a (units: $(\text{MPa})^{-1}$) is a measure of the propensity of a sliding couple for wear: High k_a means rapid wear at a given bearing pressure.

The bearing pressure P is the quantity specified by the design. The ability of a surface to resist a static contact pressure is measured by its hardness H, so we anticipate that the maximum bearing pressure $P_{\max}$ should scale with the hardness of the softer surface:

$$P_{\max} = CH$$

where C is a constant. Thus the wear rate of a bearing surface can be written as

$$\Omega = k_a P = C\left(\frac{P}{P_{\max}}\right) k_a H \tag{4.26}$$

Two material properties appear in this equation: the wear-rate constant k_a and the hardness, H. They are plotted in Figure 4.16. The dimensionless quantity

$$K = k_a H \tag{4.27}$$

is shown as a set of diagonal contours. Note first that materials of a given class (metals, for instance) tend to lie along a downward-sloping diagonal across the figure, reflecting the fact that low wear rate is associated with high hardness. The best materials for bearings for a given bearing pressure

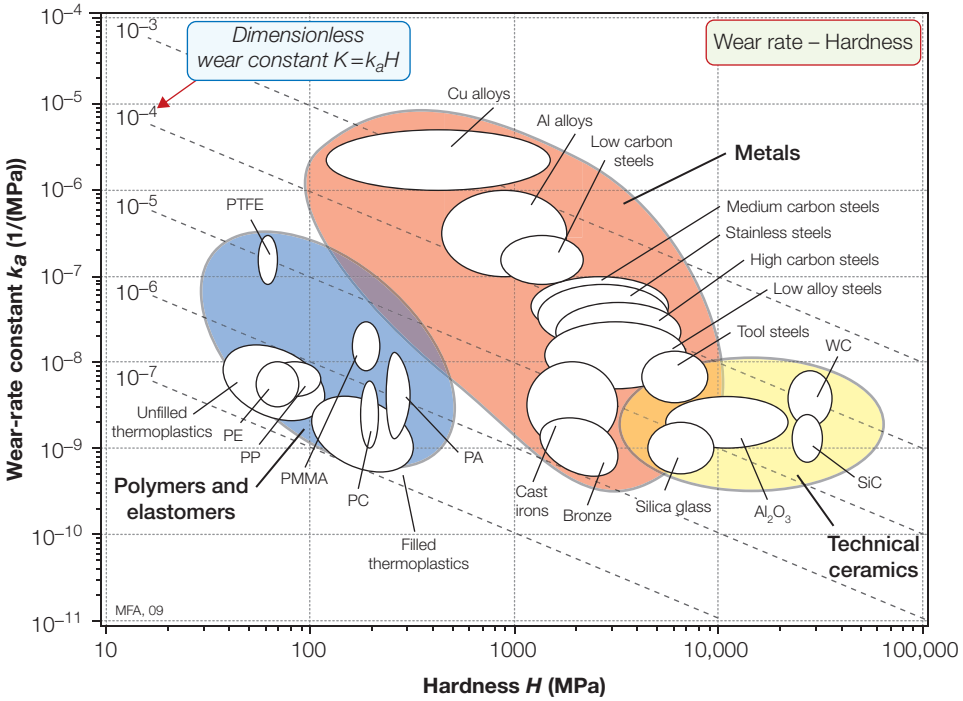


FIGURE 4.16

The normalized wear rate k_A plotted against hardness H, here expressed in MPa rather than Vickers (H in MPa = 10 H_V). The chart gives an overview of the way in which common engineering materials behave.

P are those with the lowest value of k_a, that is, those nearest the bottom of the diagram. On the other hand, an efficient bearing, in terms of size or weight, will be loaded to a safe fraction of its maximum bearing pressure, that is, to a constant value of P/P_{max}; for these, materials with the lowest values of the product $k_a H$ are best.

Cost bar charts

Properties like modulus, strength, and conductivity do not change with time. Cost is bothersome because it does. Supply, scarcity, speculation, and

Materials for bearings

Use Figure 4.16 to find two metals and two polymers that offer good wear resistance at constant bearing pressure.

Answer

The figure suggests bronze, cast iron, polycarbonate (PC), and nylon (PA) as good choices.

inflation contribute to considerable fluctuations in the cost per kilogram of a commodity like copper or silver. Data for cost per kg are tabulated for some materials in daily papers and trade journals; those for others are harder to come by. Approximate values for the cost of materials per kg, and their cost per m^3, are plotted in Figures 4.17(a) and (b). Most commodity

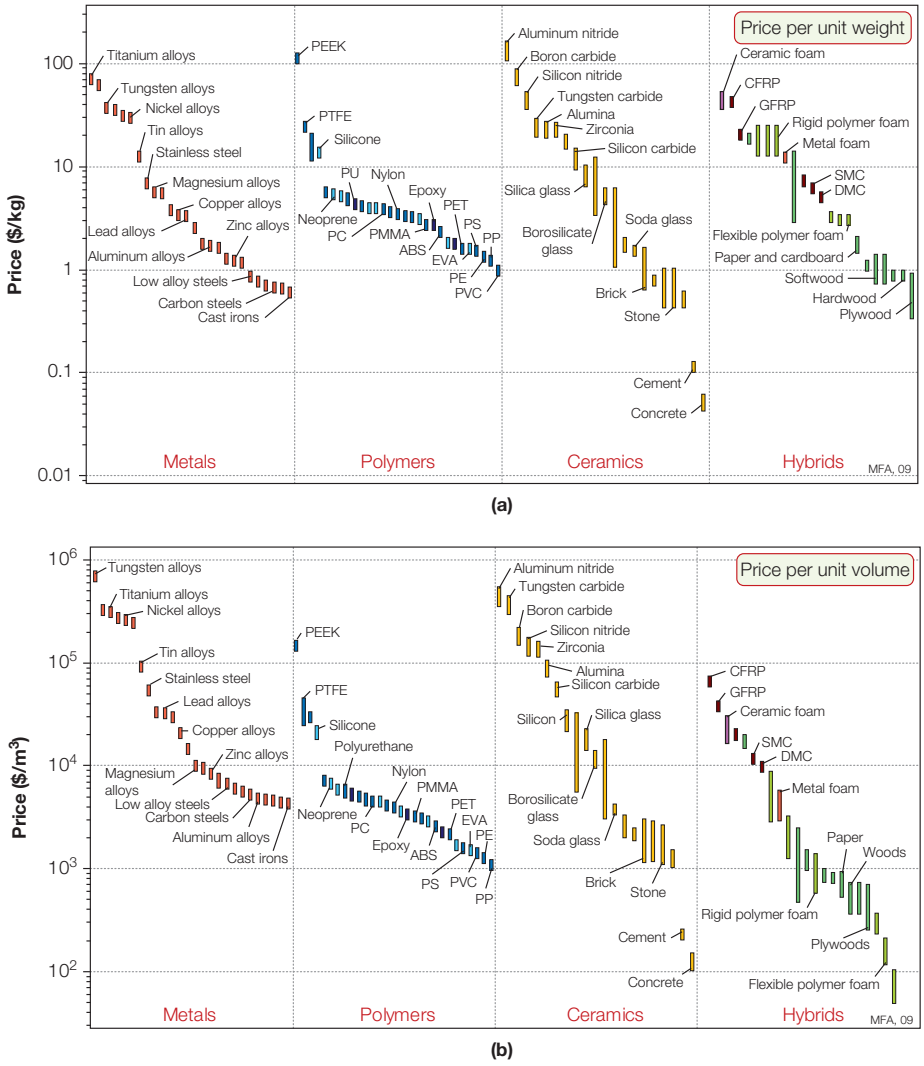


FIGURE 4.17

(a) The approximate price/kg of materials. Commodity materials cost about $1/kg; special materials cost much more. (b) The approximate price/m^3 of materials. Polymers, because they have low densities, cost less per unit volume than most other materials.

Polymer prices

Which four polymers are the least expensive?

Answer

Figure 4.17 shows that the four least expensive polymers, either by weight or by volume, are polypropylene (PP), polyethylene (PE), polyvinylchloride (PVC), and polystyrene (PS). These are the commodity polymers—the ones used in the greatest quantities.

Relative price

Are polypropylene and polyethylene more or less expensive than steel per unit weight? What about per unit volume?

Answer

Figure 4.17 reveals that the two polymers are more expensive than steel per unit weight, but because they have much lower densities, they are cheaper per unit volume. If we want a proper comparison, what should be the basis of comparison? The answer, developed in later chapters, is *per unit function*.

materials (glass, steel, aluminum, and the common polymers) cost between 0.5 and 2.0 \$/kg. Because they have low densities, the cost/m^3 of commodity polymers is less than that of metals.

The modulus–relative cost chart

In design for minimum cost, material selection is guided by indices that involve modulus, strength, and cost per unit volume. To make some correction for the influence of inflation and the units of currency in which cost is measured, we define a *relative cost per unit volume* $C_{v,R}$.

$$C_{v,R} = \frac{Cost/kg \times Density\ of\ material}{Cost/kg \times Density\ of\ mild\ steel\ rod} \tag{4.28}$$

At the time of writing, steel reinforcing rod costs about US\$0.3/kg.

Figure 4.18 shows the modulus E plotted against relative cost per unit volume $C_{v,R}\rho$ where ρ is the density. Cheap, stiff materials lie toward the top left. Guide lines for selecting materials that are stiff and cheap are plotted in the figure.

The strength–relative cost chart

Cheap strong materials are selected using Figure 4.19. It shows failure strength, defined as before, plotted against relative cost per unit volume. The qualifications for the definition of strength, given earlier, apply here also.

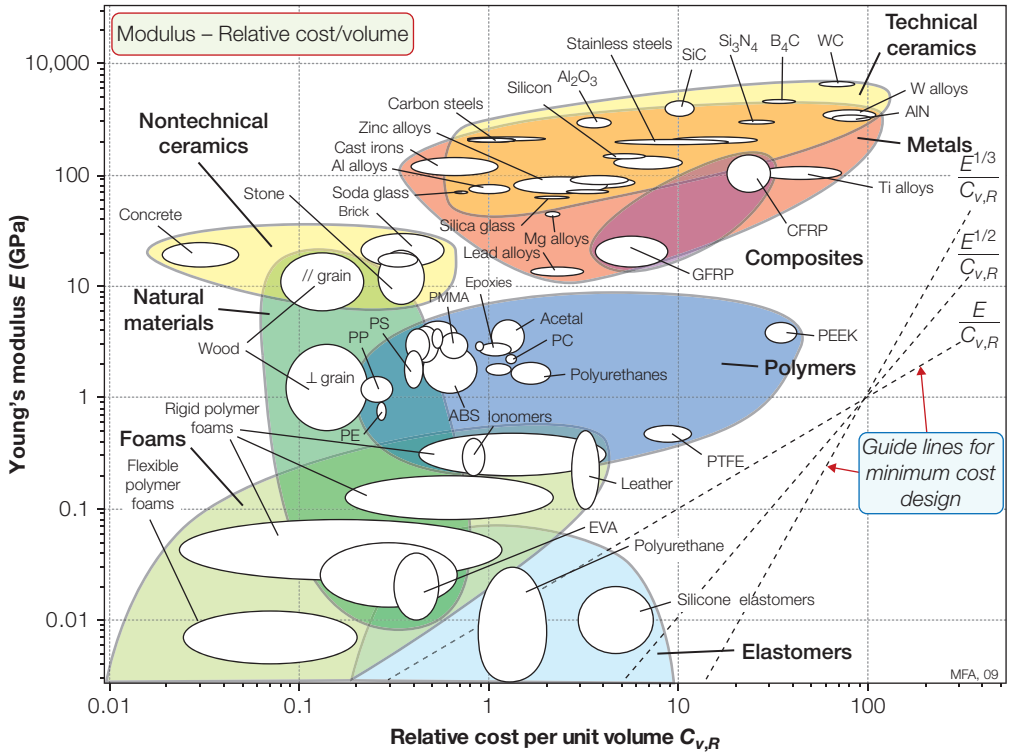


FIGURE 4.18
Young's modulus E plotted against relative cost per unit volume $C_{v,R}$. The design guide lines help selection to maximize stiffness per unit cost.

It must be emphasized that the data plotted here and in Figure 4.18 are less reliable than those in other charts and are subject to unpredictable change. Despite this dire warning, the two charts are genuinely useful. They allow selection of materials using the criterion of "function per unit cost." An example is given in Section 6.5.

4.4 SUMMARY AND CONCLUSIONS

The engineering properties of materials are usefully displayed as material selection charts. Eighteen of them are introduced in this chapter; more appear in later ones. The charts summarize material properties in a compact, easily accessible way, showing the range spanned by each material family and class. By choosing the axes in a sensible way, more information can be displayed. A chart of modulus E against density ρ reveals the longitudinal wave velocity $(E/\rho)^{1/2}$. A chart of fracture toughness K_{1c} against modulus E shows the

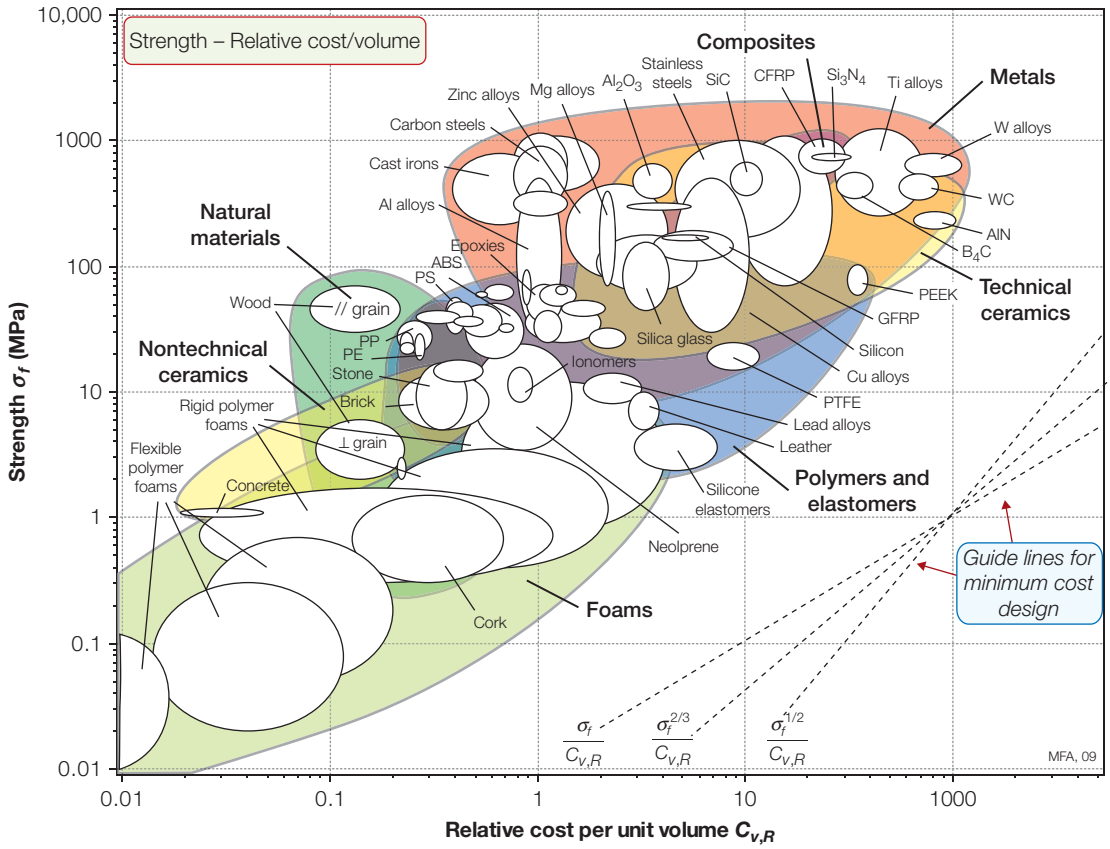


FIGURE 4.19

Strength σ_f plotted against relative cost per unit volume $C_{v,R}$. The design guide lines help selection to maximize strength per unit cost.

toughness G_{1c}. A chart of thermal conductivity λ against diffusivity, a, also gives the volume-specific heat ρC_v. A chart of strength, σ_f, against modulus, E, shows the energy-storing capacity, σ_f^2/E, and there are many more.

The most striking feature of the charts is the way in which members of a material class cluster together. Despite the wide range of modulus and density of metals (as an example), the clusters occupy a field that is distinct from that of polymers or that of ceramic, or that of composites. The same is true of strength, toughness, thermal conductivity, and the rest: The fields frequently overlap, but they always have a characteristic place within the whole picture.

The position of the fields and their relationship can be understood in simple physical terms. The nature of the bonding, the packing density, the lattice resistance, and the vibrational modes of the structure are themselves a function of

bonding and packing. It may seem odd that so little mention has been made of microstructure in determining properties. But the charts clearly show that the first-order difference between the properties of materials has its origins in the mass of the atoms, the nature of the inter-atomic forces, and the geometry of packing. Alloying, heat treatment, and mechanical working, to which we return in Chapter 13, all influence microstructure and, through this, properties, giving the elongated bubbles shown on many of the charts; but the magnitude of their effect is less, by factors of 10, than that of bonding and structure.

All the charts have one thing in common: Parts of them are populated with materials and parts are not. Some parts are inaccessible for fundamental reasons that relate to the size of their atoms and the nature of the forces that bind their atoms together. But other parts are empty even though, in principle, they are accessible. If they were accessed, the new materials that lie there could allow novel design possibilities. Ways of doing this are explored further in Chapters 11 and 12.

The charts have numerous applications. One is the checking and validation of data (Appendix A); here use is made of both the range covered by the envelope of material properties and the numerous relations between them (like $E\Omega = 100\,kT_m$), described in Section 4.3. Another concerns the development of, and identification of uses for, new materials; materials that fill gaps in one or more of the charts generally offer some improved design potential. But most important of all, the charts form the basis for a procedure for materials selection. That is developed in the following chapters.

4.5 FURTHER READING

The best general book on the physical origins of the mechanical properties of materials remains that by Cottrell (1964).

Ashby, M.F., Shercliff, H.R., & Cebon, D. (2009). *Materials: Engineering, science, processing and design* (2nd ed.). Butterworth-Heinemann, ISBN 978-1-85617-895-2.
An elementary text that introduces materials through material property charts and develops the selection methods through case studies.

Budinski, K.G., & Budinski, M.K. (2010). *Engineering materials, properties and selection* (9th ed.). Prentice Hall, ISBN 978-0-13-712842-6.
A well-established materials text that deals well with both material properties and processes.

Cottrell, A.H. (1964). *Mechanical properties of matter*. Wiley, Library of Congress Number 65-14262.
An inspirational book, clear, full of insights and simple derivations of the basic equations describing the mechanical behavior of solids, liquids, and gases.

Dieter, G.E. (1999). *Engineering design, a materials and processing approach* (3rd ed.). McGraw-Hill, ISBN 9-780-073-66136-0.
A well-balanced and respected text focusing on the place of materials and processing in technical design.

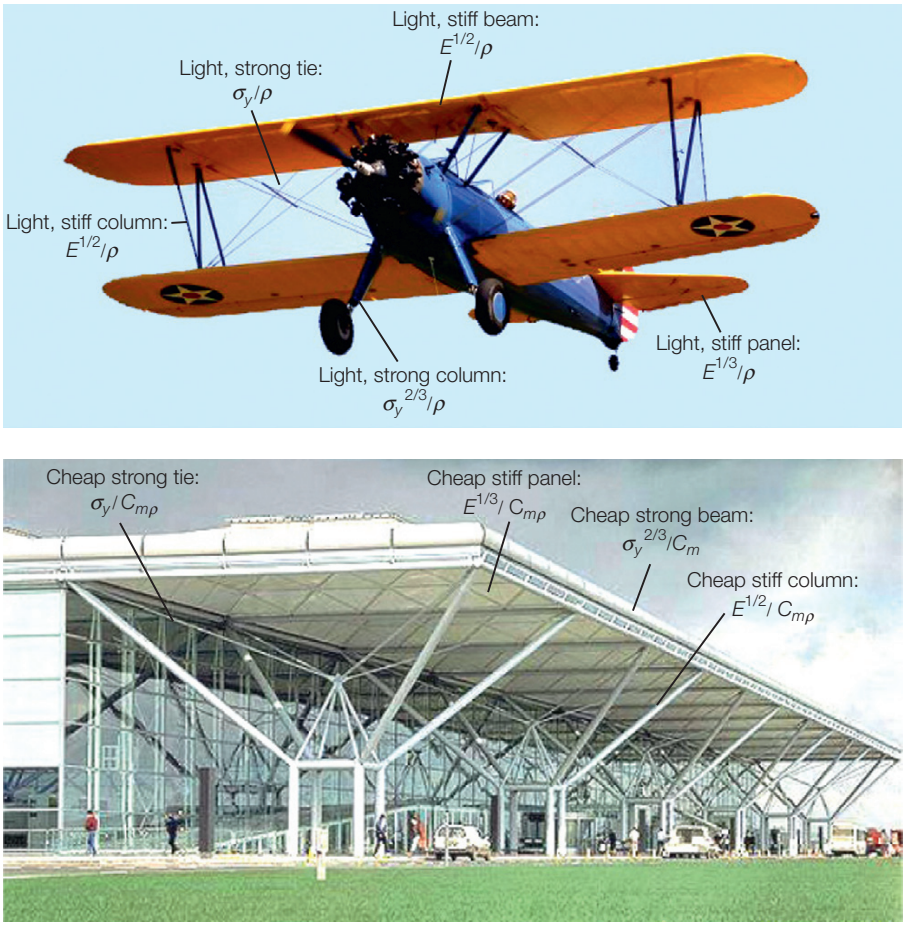
Farag, M.M. (2008). *Materials and process selection for engineering design* (2nd ed.). CRC Press, Taylor and Francis, ISBN 9-781-420-06308-0.
A materials science approach to the selection of materials.

Shackelford, J.F. (2009). *Introduction to materials science for engineers* (7th ed.). Prentice Hall, ISBN 978-0-13-601260-3.
A well-established materials text with a design slant.

Tabor, D. (1978). *Properties of matter*. Penguin Books.
This text, like that of Cottrell, is notable for its clarity and physical insight.

CHAPTER 5

Materials Selection—The Basics



Structures with components loaded in tension, bending, torsion, and compression with material indices. (Image of Stansted Airport courtesy of Norman Foster and Partners, London.)

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5.1 INTRODUCTION AND SYNOPSIS

This chapter sets out the basic procedure for selection, establishing the link between material and function (Figure 5.1). A material has *attributes*: its density, strength, cost, resistance to corrosion, and so forth. A design demands a certain profile of these: a low density, a high strength, a modest cost, and resistance to sea water, perhaps. It is important to start with the full menu of materials as options; failure to do so may mean a missed opportunity. If an innovative choice is to be made, it must be identified early in the design process. Later, too many decisions will have been taken and commitments made to allow radical change: It is now or never.

The task of selection, stated in two lines, is that of

1. identifying the desired attribute profile, and then
2. comparing this with those of real engineering materials to find the best match.

The first step in tackling selection is that of *translation*—examining the design requirements to identify the constraints that they impose on material

choice. The immensely wide choice is narrowed, first, by *screening out* the materials that cannot meet the constraints. Further narrowing is achieved by *ranking* the candidates by their ability to maximize performance. This chapter explains how to do both.

The material property charts introduced in Chapter 4 are designed for use with these criteria. Constraints and objectives can be plotted on them, isolating the subset of materials that are the best choice for the design. The whole procedure can be implemented in software as a design tool, allowing computer-aided selection. The procedure is fast and makes for lateral ("what if...?") thinking. Examples of the method are given in Chapter 6.

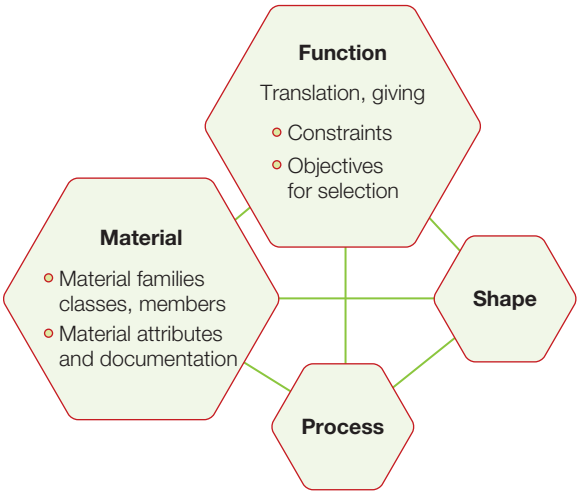


FIGURE 5.1
Material selection is determined by function. Shape sometimes influences selection. This chapter and the next deal with materials selection when this is independent of shape.

5.2 THE SELECTION STRATEGY

Material attributes

Figure 5.2 illustrates how the universe of materials is divided into families, classes, subclasses, and members. Each member is characterized by a set of *attributes*: its properties. As an example, the materials universe contains the family "metals," which in turn contains the class "aluminum alloys," the subclass "6000 series," and finally the particular member "Alloy 6061." It, and every other member of the universe, is characterized by a set of attributes that include its mechanical, thermal, electrical, optical, and chemical properties; its processing characteristics; its cost and availability; and the environmental consequences of its use. We call this its *property profile*. Selection involves seeking the best match between the property profiles of the materials in the universe and the property profile required by the design.

Selection strategies

This chapter is about strategies for selection. It is simpler to start with the selection of a product than with a material; the ideas are the same, but the material has added complications.

You need a new car. To meet your needs it must be a mid-sized four-door family sedan with a gas engine delivering at least 150 horsepower—enough to tow your power boat. Given all of these, you wish it to cost as little to own and run as possible (Figure 5.3, *left*). The three *constraints* are listed on page 101, but they are not all of the same type.

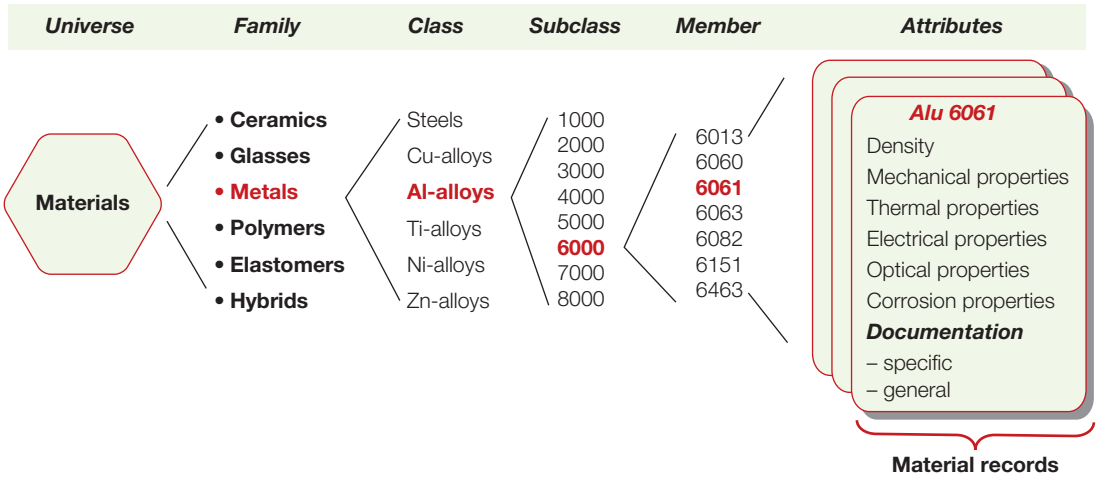


FIGURE 5.2
The taxonomy of the universe of materials and their attributes. Computer-based selection software stores data in a hierarchical structure like this.

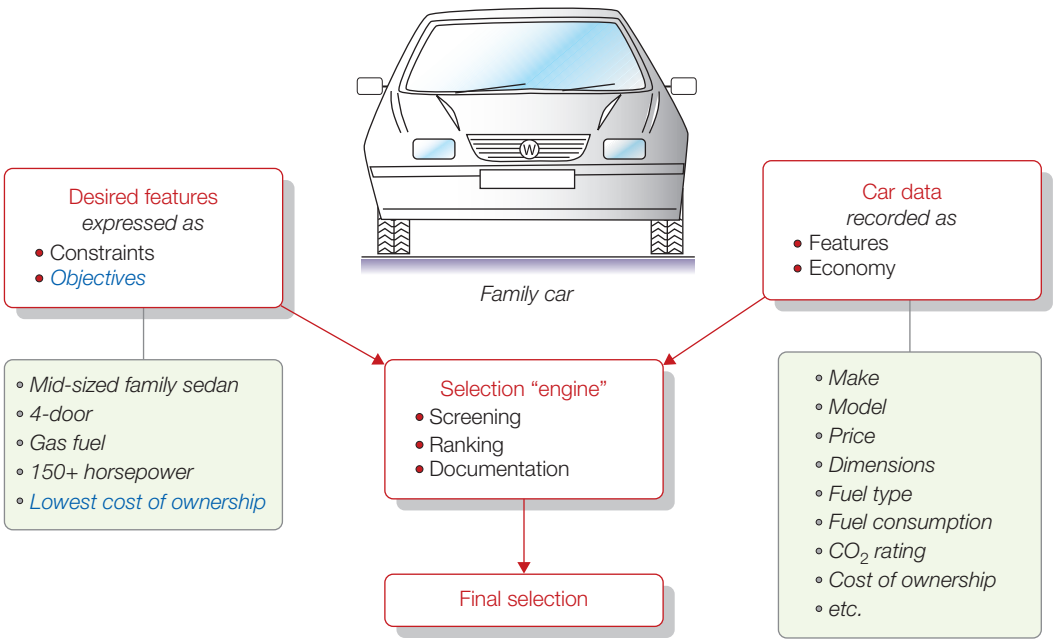


FIGURE 5.3
Choosing a car—an example of the selection strategy. Required features are constraints; they are used to screen out unsuitable cars. The survivors are ranked by cost of ownership.

- The requirements of *four-door family sedan* and *gas power* are *simple constraints*; a car *must* have these to be a candidate.
- The requirement of *at least 150 hp* places a lower limit but no upper limit on power; it is a *limit constraint*; any car with 150 hp or more is acceptable.

The wish for minimum cost of ownership is an *objective*, a criterion of excellence. The most desirable cars, from among those that meet the constraints, are those that minimize this objective.

To proceed you need information about available cars (Figure 5.3, *right*). Car magazines, car makers' websites, and dealers list such information. It includes car type and size, number of doors, fuel type, engine power, and price; car magazines go further and estimate the cost of ownership (meaning the sum of running costs, tax, insurance, servicing, and depreciation), listing it as $/mile or €/km.

Now: decision time (Figure 5.3, *center*). The selection engine (you in this example) uses the constraints to *screen out*, from all the available cars, those that are not four-door gas-powered family sedans with at least 150 hp. Many cars meet these constraints, so the list is still long. You need a way to order it so that the best choices are at the top. That is what the objective is for: It allows you to *rank* the surviving candidates by cost of ownership—those with the lowest values are ranked most highly. Rather than just choosing the one that is cheapest, it is better to keep the top three or four and seek further *documentation*, exploring their other features in depth (delivery time, size of trunk, comfort of seats, guarantee period, and so on) and weighing the small differences in cost against the desirability of these features.

Methods like this are used as a tool for decision making in many fields: in deciding between design options for new products, in optimizing the operating methods for a new plant, in guiding the selection of a site for a new town … and in selecting materials. We turn to that next.

Selecting materials involves seeking the best match between design requirements and the properties of the materials that might be used to make the design. Figure 5.4 shows the strategy of the last section applied to selecting materials for the protective visor of a safety helmet. On the left are requirements that the material must meet, expressed as constraints and objectives. The constraints: ability to be molded and, of course, transparency. The objective: If the visor is to protect the face, it must be as shatterproof as possible, meaning it must have as high a fracture toughness as possible.

On the right is the database of material attributes, drawn from suppliers' data sheets, handbooks, web-based sources, or software specifically designed

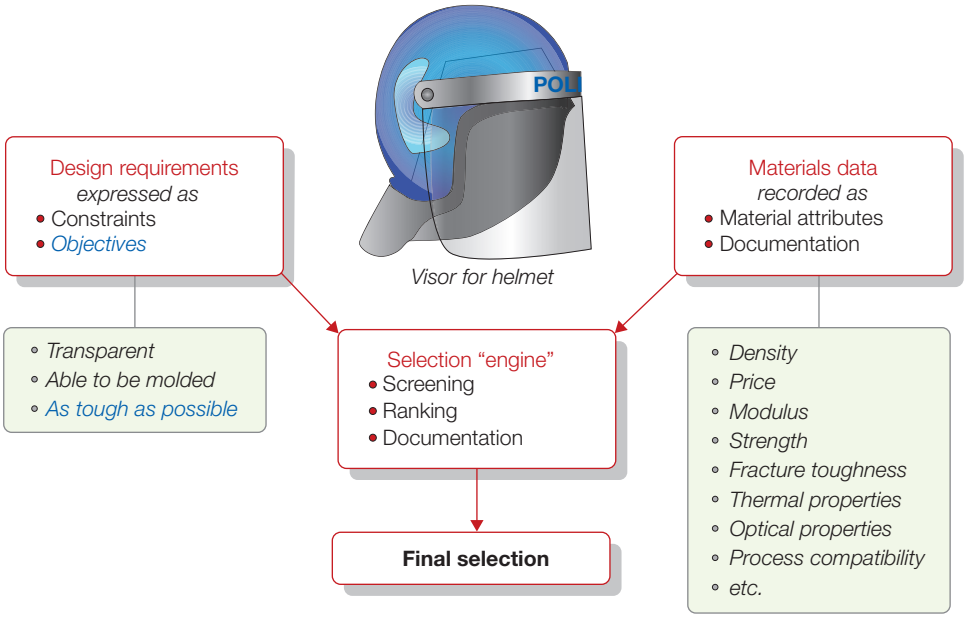


FIGURE 5.4

Choosing a material. Design requirements are first expressed as constraints and objectives. The constraints are used for screening. The survivors are ranked by the objective, expressed as a material index.

for materials selection. The comparison "engine"—you again—applies the constraints on the left to the materials on the right, screening out materials that fail to meet them and delivering a list of viable candidates, just as with cars. The list is then ranked by the fracture toughness. The three or so materials that meet the constraints and have the highest fracture toughness are then explored in depth by seeking documentation for them.

There is, however, a complication. The requirements for the car were straightforward—doors, fuel type, power; all of these are explicitly listed by the manufacturer. The design requirements for a component of a product specify what it should *do* but not what *properties* its materials should have. So the first step in selecting materials is one of *translation*: converting the design requirements (often vague) into constraints and objectives that can be applied to the materials database (Figure 5.5, *top*). The next task is *screening*, as with cars, eliminating the materials that cannot meet the constraints. This is followed by *ranking*, ordering the survivors by their ability to meet a criterion of excellence, such as minimizing cost or maximizing impact resistance. The final task is to explore the most promising candidates in depth, examining how they are used at present, case histories of failures, and how best to design with them; this step is called *documentation*. Now a closer look at each step.

Translation

How are the design requirements for a component (defining what it must do) translated into a prescription for a material? Any engineering component has one or more *functions*: to support a load, to contain a pressure, to transmit heat, and so on. This must be achieved subject to *constraints*: that certain dimensions are fixed, that the component must carry the design loads or pressures without failure, that it insulates or conducts, that it can function in a certain range of temperature and in a given environment, and many more. In designing the component, the designer has an *objective*: to make it as cheap as possible, perhaps, or as light, or as safe, or perhaps some combination of these.

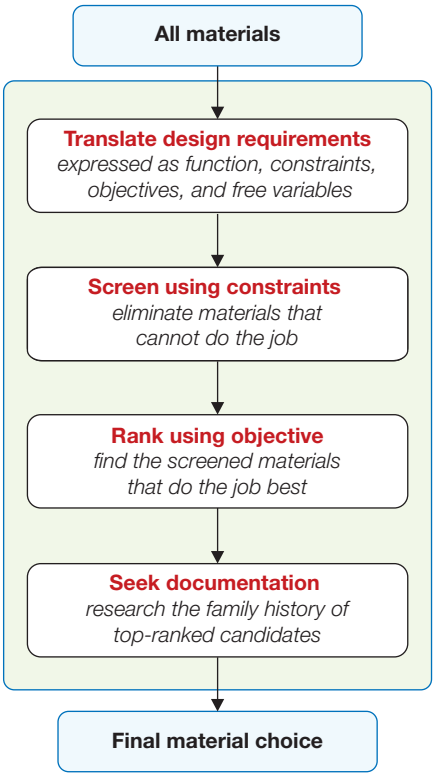


FIGURE 5.5
The strategy for materials selection. The four main steps—translation, screening, ranking, and documentation—are shown here.

Certain parameters can be adjusted to optimize the objective; the designer is free to vary dimensions that have not been constrained by design requirements and, most importanty, free to choose the material for the component. We refer to these as *free variables*. Function, constraints, objectives, and free variables (Table 5.1) define the boundary conditions for selecting a material and—in the case of load-bearing components—a shape for its cross-section. The first step in relating design requirements to material properties is a clear statement of function, constraints, objectives, and free variables.

Table 5.1 Function, Constraints, Objectives, and Free Variables

Function	What does the component do?
Constraints*	What nonnegotiable conditions must be met? What negotiable but desirable conditions must be met?
Objective	What is to be maximized or minimized?
Free variable	Which parameters of the problem is the designer free to change?

**It is sometimes useful to distinguish between "hard" and "soft" constraints. Stiffness and strength might be absolute requirements (hard constraints); cost might be negotiable (soft constraint).*

Design requirements for the helmet visor

A material is required for the visor of a safety helmet to provide maximum facial protection.

Translation

To allow clear vision the visor must be optically transparent. To protect the face from the front, from the sides, and from below it must be doubly curved, requiring that the material can be molded. We thus have two constraints: transparency and ability to be molded.

Fracture of the visor would expose the face to damage; maximizing safety therefore translates into maximizing resistance to fracture. The material property that measures resistance to fracture is the fracture toughness, K_{1c}. The objective is therefore to maximize K_{1c}.

Screening: Attribute limits

Unbiased selection requires that all materials be considered candidates until shown to be otherwise, using the steps in the boxes below "Translate" in Figure 5.5. The first of these, *screening*, eliminates candidates that cannot do the job at all because one or more of their attributes lies outside the limits set by the constraints. As examples, the requirement that "the component must function in boiling water" or that "the component must be transparent" imposes obvious limits on the attributes of *maximum service temperature* and *optical transparency* that successful candidates must meet. We refer to these as *attribute limits*.

Ranking: Material indices

Attribute limits do not, however, help with ordering the candidates that remain. To do this we need optimization criteria. These are found in the material indices, developed next, which measure how well a candidate that has passed the screening step can perform. Performance is sometimes limited by a single property, sometimes by a combination of them. Thus the best materials for buoyancy are those with the lowest density, ρ; those best for thermal insulation are the ones with the smallest values of the thermal conductivity, λ. It is not always the smallest that we want; the best material for a heat exchanger, for instance, is the one with the largest value of λ. Here maximizing or minimizing a single property maximizes performance. But—as we shall see—it is more usual that performance is limited not by one property but by a combination of them. Thus the best materials for a light strong tie-rod or cable are those with the greatest value of the specific strength, σ_f/ρ, where σ_f is failure strength. The best materials for a spring are those with the greatest value of σ_f^2/E, where E is Young's modulus. The property or property group that maximizes performance for a given design is called its *material index*. There are many such indices, each associated with maximizing some aspect of

performance.[1] They provide *criteria of excellence* that allow ranking of materials by their ability to perform well in the given application.

To summarize: Screening isolates candidates that are capable of doing the job; ranking identifies those among them that can do the job best.

Screening and ranking for the helmet visor

A search for transparent materials that can be molded delivers the following list. The first four are thermoplastics; the last two, glasses. Fracture toughness values can be found in Appendix A.

Material	Average Fracture Toughness K_{1c} MPa.m$^{1/2}$
Polycarbonate (PC)	3.4
Cellulose acetate (CA)	1.7
Polymethyl methacrylate (acrylic, PMMA)	1.2
Polystyrene (PS)	0.9
Soda-lime glass	0.6
Borosilicate glass	0.6

The constraints have reduced the number of viable materials to six candidates. When ranked by fracture toughness, the top-ranked candidates are PC, CA, and PMMA.

Documentation

The outcome of the steps so far is a ranked short-list of candidates that meet the constraints and that maximize or minimize the criterion of excellence, whichever is required. You could just choose the top-ranked candidate, but what secret vices might it have? What are its strengths and weaknesses? Does it have a good reputation? What, in a word, is its credit rating? To proceed further we seek a detailed profile of each candidate: its *documentation* (Figure 5.5, *bottom*).

Documentation differs greatly from the structured property data used for screening. Typically, it is descriptive, graphical, or pictorial: case studies of previous uses of the material, failure analyses and details of its corrosion, information about availability and pricing, and the like. Such information is found in handbooks, suppliers' data sheets, case studies of use, and failure analyses. Documentation helps narrow the short-list to a final choice, allowing a definitive match to be made between design requirements and material attributes.

Why are all these steps necessary? Without screening and ranking, the candidate pool is enormous and the volume of documentation overwhelming. Dipping

[1] Maximizing performance often means minimizing something: Cost is the obvious example; mass, in transport systems, is another. A low-cost or light component, here, improves performance.

into it, hoping to stumble on a good material, gets you nowhere. But once a small number of potential candidates have been identified by the screening–ranking steps, detailed documentation can be sought for these few alone, and the task becomes viable.

Documentation for materials for the helmet visor

At this point it helps to know how the three top-ranked candidates listed in the last examples box are used. A quick web search reveals the following.

Polycarbonate
Safety shields and goggles; lenses; light fittings; safety helmets; laminated sheet for bullet-proof glazing.

Cellulose Acetate
Spectacle frames; lenses; goggles; tool handles; covers for television screens; decorative trim, steering wheels for cars.

PMMA, Plexiglas
Lenses of all types; cockpit canopies and aircraft windows; containers; tool handles; safety spectacles; lighting, automotive taillights.

This is encouraging: All three materials have a history of use for goggles and protective screening. The one that ranked highest in our list—polycarbonate—has a history of use for protective helmets. We select this material, confident that with its high fracture toughness it is the best choice.

Local conditions

The final choice between competing candidates will often depend on local conditions: in-house expertise or equipment, the availability of local suppliers, and so forth. A systematic procedure cannot help here; the decision must instead be based on local knowledge. This does not mean that the result of the systematic procedure is irrelevant. It is always important to know which material is best, even if for local reasons you decide not to use it.

We explore documentation more fully later. Here we focus on the derivation of property limits and indices.

5.3 MATERIAL INDICES

Constraints set property limits. Objectives define material indices, for which we seek extreme values. When the objective is not coupled to a constraint, the material index is a simple material property. When, instead, the two [better] are coupled, the index becomes a group of properties like those cited above. Where do they come from? This section explains.

Think for a moment of the simplest of mechanical components. The loading on a component can generally be decomposed into some combination of axial tension, bending, torsion, and compression. Almost always, one mode dominates. So common is this that the functional name given to the component describes the way it is loaded: *Ties* carry tensile loads; *beams* and *panels* carry bending moments; *shafts* carry torques; *columns* carry compressive axial loads. The words "tie," "beam," "shaft," and "column" each imply a function. Here we explore constraints, objectives, and resulting material indices for some of these.

The life energy and emissions for transport systems are dominated by the fuel consumed during use. The lighter the system is made, the less fuel it consumes and the less carbon it emits. So a good starting point is *minimum weight design*, subject, of course, to the other necessary constraints, of which the most important here have to do with stiffness and strength. We consider the generic components shown in Figure 5.6: a tie, a panel, and beams, loaded as shown.

Minimizing mass: A light, strong tie A design calls for a tie like those of the biplane in the cover picture. It must carry a tensile force F^* without failure and be as light as possible (Figure 5.6(a)). The length L is specified but the cross-section area A is not. Here, "maximizing performance" means "minimizing the mass while still carrying the load F^* safely." The design requirements, translated, are listed in Table 5.2.

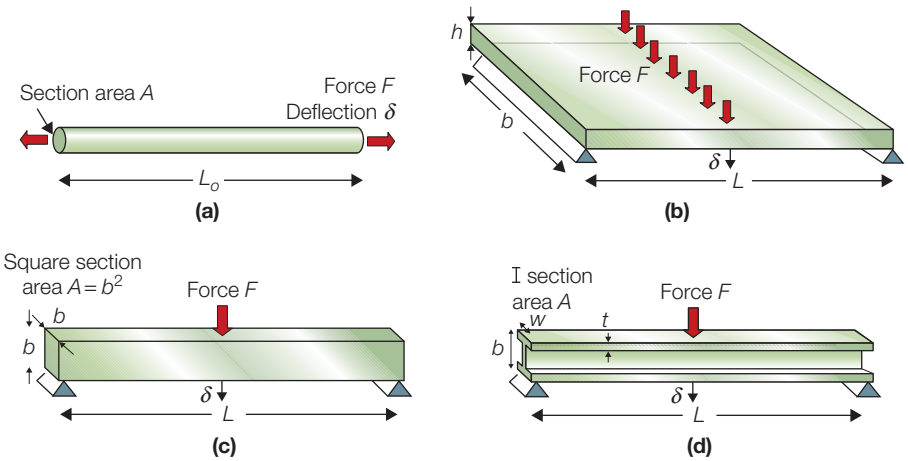


FIGURE 5.6

Generic components: (a) a tie, a tensile component; (b) a panel, loaded in bending; (c) and (d) beams, loaded in bending.

Table 5.2 Design Requirements for the Light, Strong Tie

Function	Tie rod
Constraints	Length L is specified (geometric constraint) Tie must support axial tensile load F^* without failing (functional constraint)
Objective	Minimize the mass m of the tie
Free variables	Cross-section area A Choice of material

We first seek an equation describing the quantity to be maximized or minimized. Here it is the mass m of the tie and it is a minimum that we seek. This equation, called *the objective function*, is

$$m = A\,L\,\rho \tag{5.1}$$

where A is the area of the cross-section and ρ is the density of the material of which it is made. The length L and force F are specified and are therefore fixed; the cross-section A, is free. We can reduce the mass by reducing the cross-section, but there is a constraint: A must be sufficient to carry F^*, requiring that

$$\frac{F^*}{A} \leq \sigma_f \tag{5.2}$$

where σ_f is the failure strength. Eliminating A between these two equations gives

$$m \geq (F^*)(L)\left(\frac{\rho}{\sigma_f}\right) \tag{5.3}$$

(ρ: *Material properties*; F^*: *Functional constraint*; L: *Geometric constraint*)

Note the form of this result. The first bracket contains the specified load F. The second bracket contains the specified geometry (length L of the tie). The last bracket contains the material properties. The lightest tie that will carry F^* safely[2] is that made of the material with the smallest value of ρ/σ_f. We could define this as the material index of the problem, seeking a minimum, but it is more usual when dealing with specific properties to express them in a form for which a maximum is sought. We therefore invert the material

[2] In reality a safety factor, S_f, is always included in such a calculation, so that Equation (5.2) becomes $F/A = \sigma_f/S_f$. If the same safety factor is applied to each material, its value does not influence the choice. We omit it here for simplicity.

properties in Equation (5.3) and define the material index M_t (subscript t for tie) as:

$$M_{t1} = \frac{\sigma_f}{\rho} \tag{5.4}$$

The lightest tie-rod that will carry F^* without failing is that with the largest value of this index, the "specific strength," plotted in Figure 4.6. A similar calculation for a light *stiff* tie (one for which the stiffness S rather then the strength σ_f is specified) leads to the index

$$M_{t2} = \frac{E}{\rho} \tag{5.5}$$

where E is Young's modulus. This time the index is the "specific stiffness," also shown in Figure 4.6. The material group (rather than just a single property) appears as the index in both cases because minimizing the mass m—the objective—was coupled to the constraints of carrying the load F without failing or deflecting too much.

Note the procedure. The length of the rod is specified but we are free to choose the cross-section area A. The objective is to minimize its mass m. We write an equation for m: It is the objective function. But there is a constraint: The rod must carry the load F without yielding (in the first example) or bending too much (in the second). Use this to eliminate the free variable A and read off the combination of properties, M, to be maximized. It sounds easy and it is, so long as you are clear from the start what the constraints are, what you are trying to maximize or minimize, which parameters are specified, and which are free.

That was easy. Now for some slightly more difficult (and important) examples.

Minimizing Mass: A light, stiff panel A *panel* is a flat slab, like a table top. Its length L and width b are specified but its thickness is free. It is loaded in bending by a central load F (see Figure 5.6(b)). The stiffness constraint requires that it must not deflect more than δ . The objective is to achieve this with minimum mass, m. Table 5.3 summarizes the design requirements.

Table 5.3 Design Requirements for a Light, Stiff Panel

Function	Panel
Constraints	Bending stiffness S^* specified (functional constraint) Length L and width b specified (geometric constraints)
Objective	Minimize mass m of the panel
Free variables	Panel thickness h Choice of material

The objective function for the mass of the panel is the same as that for the tie:

$$m = A L \rho = b h L \rho$$

Its bending stiffness S must be at least S^*:

$$S = \frac{C_1 EI}{L^3} \geq S^* \tag{5.6}$$

Here C_1 is a constant that depends only on the distribution of the loads—we don't need its value (you can find it in Appendix B). The second moment of area, I, for a rectangular section is

$$I = \frac{b h^3}{12} \tag{5.7}$$

We can reduce the mass by reducing h, but only so far that the stiffness constraint is still met. Using the last two equations to eliminate h in the objective function gives

$$m = \left(\frac{12 S^*}{C_1 b}\right)^{1/3} (bL^2) \left(\frac{\rho}{E^{1/3}}\right) \tag{5.8}$$

Material properties

Functional constraint

Geometric constraints

The quantities S^*, L, b, and C_1 are all specified; the only freedom of choice left is that of the material. The index is the group of material properties, which we invert such that a maximum is sought: The best materials for a light, stiff panel are those with the greatest values of

$$M_{p1} = \frac{E^{1/3}}{\rho} \tag{5.9}$$

Repeating the calculation with a constraint of strength rather than stiffness leads to the index

$$M_{p1} = \frac{\sigma_y^{1/2}}{\rho} \tag{5.10}$$

These don't look much different from the previous indices, E/ρ and σ_y/ρ, but they are: They lead to different choices of material, as we shall see in a moment.

Now for another bending problem in which shape plays a role.

Minimizing mass: A light, stiff beam Beams come in many shapes: solid rectangles, cylindrical tubes, I-beams, and more. Some of these have too many free geometric variables to apply the previous method directly. However, if we constrain the shape to be *self-similar* (such that all dimensions of the cross-section change in proportion as we vary the overall size), the problem becomes tractable again. We therefore consider beams in two stages: first, we identify the optimum materials for a light, stiff beam of a prescribed simple shape (a square section); second, we explore how much lighter it could be made for the same stiffness by using a more efficient shape.

Consider a beam of square section $A = b \times b$ that may vary in size but the square shape is retained. It is loaded in bending over a span of fixed length L with a central load F (see Figure 5.6(c)). The stiffness constraint is again that it must not deflect more than δ under F, with the objective that the beam should again be as light as possible. Table 5.4 summarizes the design requirements.

Proceeding as before, the objective function for the mass is

$$m = AL\rho = b^2 L\rho \tag{5.11}$$

The bending stiffness S of the beam must be at least S^*:

$$S = \frac{C_2 EI}{L^3} \geq S^* \tag{5.12}$$

where C_2 is a constant (Appendix B). The second moment of area, I, for a square section beam is

$$I = \frac{b^4}{12} = \frac{A^2}{12} \tag{5.13}$$

For a given L, S^* is adjusted by altering the size of the square section. Now eliminating b (or A) in the objective function for the mass gives

$$m = \left(\frac{12S^*L^3}{C_2}\right)^{1/2} (L)\left(\frac{\rho}{E^{1/2}}\right) \tag{5.14}$$

Table 5.4 Design Requirements for a Light, Stiff Beam

Function	Beam
Constraints	Length L is specified (geometric constraint) Section shape square (geometric constraint) Beam must support bending load F without deflecting too much, meaning that bending stiffness S is specified as S^* (functional constraint)
Objective	Minimize mass m of the beam
Free variables	Cross-section area A Choice of material

The quantities S^*, L, and C_2 are all specified or constant; the best materials for a light, stiff beam are those with the largest values of index M_b, where

$$M_{b_1} = \frac{E^{1/2}}{\rho} \tag{5.15}$$

Repeating the calculation with a constraint of strength rather than stiffness leads to the index

$$M_{b_2} = \frac{\sigma_y^{2/3}}{\rho} \tag{5.16}$$

This analysis was for a square beam, but the result in fact holds for any shape so long as the shape is held constant. This is a consequence of Equation (5.13); for a given shape, the second moment of area I can always be expressed as a constant times A^2, so changing the shape just changes the constant C_2 in Equation (5.14), not the resulting index.

As noted above, real beams have section shapes that improve their efficiency in bending, requiring less material to get the same stiffness. By shaping the cross-section it is possible to increase I without changing A. This is achieved by locating the material of the beam as far from the neutral axis as possible, as in thin-walled tubes or I-beams (see Figure 5.6 (d)). Some materials are more amenable than others to being made into efficient shapes. Comparing materials on the basis of the index in M_b therefore requires some caution; materials with lower index values may "catch up" by being made into more efficient shapes. We examine this in more detail in Chapter 9.

Minimizing material cost: Cheap ties, panels, and beams When the objective is to minimize cost rather than mass, the indices change again. If the material price is C_m \$/kg, the cost of the material to make a component of mass m is just mC_m. The objective function for the material cost C of the tie, panel or beam then becomes

$$C = m\,C_m = A\,L\,C_m\,\rho \tag{5.17}$$

Proceeding as before leads to indices that have the form of Equations (5.4), (5.5), (5.9), (5.10), (5.15), and (5.16), with ρ replaced by $C_m\,\rho$. Thus the index guiding material choice for a tie of specified strength and minimum material cost is

$$M = \frac{\sigma_f}{C_m\,\rho} \tag{5.18}$$

where C_m is the material price per kg. The index for a cheap stiff panel is

$$M_{p1} = \frac{E^{1/3}}{C_m \rho} \tag{5.19}$$

and so forth (It must be remembered that the material cost is only part of the cost of a shaped component; there is also the manufacturing cost—the cost to shape, join, and finish it.)

Associating material indices with components The components in the cover images of this chapter are labeled with the type of loading support and with the index that guides the choice of material to make them. The biplane typifies lightweight design, meaning that its materials are chosen to carry the design loads at minimum mass. The airport structure uses very large quantities of materials: Here the objective is to carry the design loads safely while minimizing the cost of the material. The guiding indices for each structure are derived from a single objective: minimizing mass in one case, minimizing material cost in the other. Often a design involves more than one objective: in choosing materials for the frame of a bicycle you might wish to minimize both the weight *and* the cost. That requires trade-off methods, the subject of Chapter 7.

How general are material indices?

This is a good moment to describe the method in more general terms. *Structural elements* are components that perform a physical function: They carry loads, transmit heat, store energy and so on. In short, they satisfy *functional requirements*. We have already identified examples: A tie must carry a specified tensile load; a spring must provide a given restoring force or store a given energy; a heat exchanger must transmit heat with a given heat flux, and so on.

The performance of a structural element is determined by three things: the functional requirements, the geometry, and the properties of the material of which it is made.[3] The performance P of the element is described by an equation of the form

$$P = \left[\begin{pmatrix} \textit{Functional} \\ \textit{requirements}, F \end{pmatrix}, \begin{pmatrix} \textit{Geometric} \\ \textit{parameters}, G \end{pmatrix}, \begin{pmatrix} \textit{Material} \\ \textit{properties}, M \end{pmatrix} \right]$$

or

$$P = f(F, G, M) \tag{5.20}$$

[3] In Chapter 9 we introduce a fourth: section shape.

where P, the *performance metric*, describes some aspect of the performance of the component: its mass, volume, cost, or life, for example; and f means "*a function of*." *Optimum design* is the selection of the material and geometry that maximize or minimize P, according to its desirability or otherwise.

The three groups of parameters in Equation (5.20) are said to be *separable* when the equation can be written

$$P = f_1(F) \cdot f_2(G) \cdot f_3(M) \tag{5.21}$$

where f_1, f_2, and f_3 are separate functions that are simply multiplied together. It turns out that, commonly, they are. When this is so the optimum choice of material becomes independent of the details of the design; it is the same for *all* geometries, G, and for all values of the function requirement, F. Then the optimum subset of materials can be identified without solving the complete design problem, or even knowing all the details of F and G. This enables enormous simplification: The performance for *all* F and G is maximized by maximizing f_3 (M), which is called the material efficiency coefficient, or *material index* for short. The remaining bit, f_1 $(F) \cdot f_2$ (G), is related to *the structural efficiency coefficient*, or *structural index*. We don't need it now, but we will examine it briefly in Section 5.6.

Each combination of function, objective, and constraint leads to a material index (Figure 5.7); the index is characteristic of the combination and thus of the function the component performs. The method is general and, in later

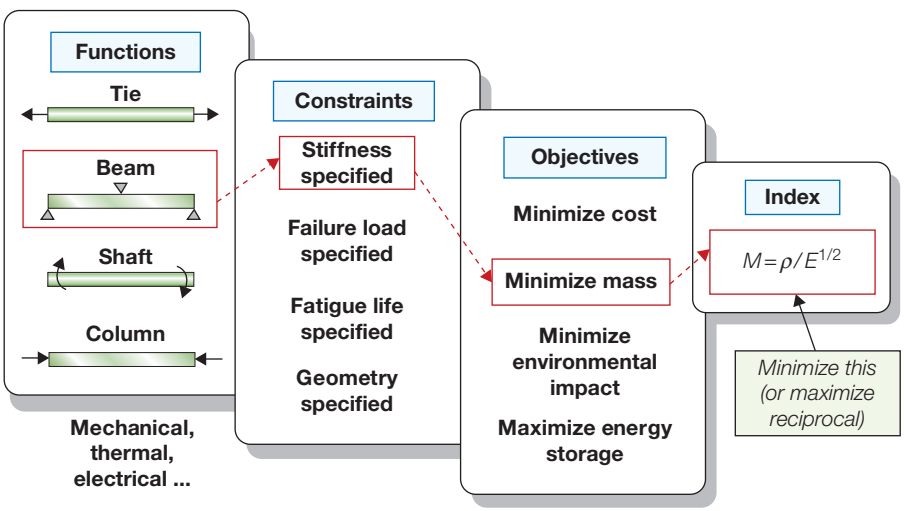


FIGURE 5.7

The specification of function, objective, and constraint leads to a materials index. The combination in the highlighted boxes leads to the index $E^{1/2}/\rho$.

Table 5.5 Examples of Material Indices

Function, Objective, and Constraints	Index
Tie, minimum weight, stiffness prescribed	$\frac{E}{\rho}$
Beam, minimum weight, stiffness prescribed	$\frac{E^{1/2}}{\rho}$
Beam, minimum weight, strength prescribed	$\frac{\sigma_y^{2/3}}{\rho}$
Beam, minimum cost, stiffness prescribed	$\frac{E^{1/2}}{C_m\rho}$
Beam, minimum cost, strength prescribed	$\frac{\sigma_y^{2/3}}{C_m\rho}$
Column, minimum cost, buckling load prescribed	$\frac{E^{1/2}}{C_m\rho}$
Spring, minimum weight for given energy storage	$\frac{\sigma_y^2}{E\rho}$
Thermal insulation, minimum cost, heat flux prescribed	$\frac{1}{\lambda C_p \rho}$
Electromagnet, maximum field, temperature rise prescribed	$\frac{C_p\,\rho}{\rho_e}$

ρ = density; E = Young's modulus; σ_y = elastic limit; C_m = cost/kg; λ = thermal conductivity; ρ_e = electrical resistivity; C_p = specific heat

chapters, is applied to a wide range of problems. Table 5.5 gives examples of indices and the design problems that they characterize. A fuller catalog of indices is given in Appendix C. New problems throw up new indices, as the case studies of the next chapter will show.

5.4 THE SELECTION PROCEDURE

We can now assemble the four steps into a systematic procedure.

Translation and deriving the index

Table 5.6 lists the steps. Simplified: Identify the material attributes that are constrained by the design, decide what you will use as a criterion of excellence (to be minimized or maximized), substitute for any free variables using one of the constraints, and read off the combination of material properties that optimize the criterion of excellence.

Screening: Applying attribute limits

Any design imposes certain non-negotiable demands ("constraints") on the material of which it is made. We have explained how these are translated

Table 5.6 Translation

Step No.	Action
1	Define the design requirements: *Function:* What does the component do? *Constraints:* Essential requirements that must be met: e.g., stiffness, strength, corrosion resistance, forming characteristics, etc. *Objective:* What is to be maximized or minimized? *Free variables:* Which are the unconstrained variables of the problem?
2	List the constraints (no yield, no fracture, no buckling, etc) and develop an equation for them if necessary.
3	Develop an equation for the objective in terms of the functional requirements, the geometry, and the material properties (*objective function*).
4	Identify the free (unspecified) variables.
5	Substitute the free variables from the constraint equations into the objective function.
6	Group the variables into three groups: functional requirements F, geometry G, and material properties M; thus Performance metric $P \leq f_1(F) \cdot f_2(G) \cdot f_3(M)$ or performance metric $P \leq f_1(F) \cdot f_2(G) \cdot f_3(M)$
7	Read off the material index, expressed as a quantity M that optimizes the performance metric P. M is the criterion of excellence.

into attribute limits. Attribute limits plot as horizontal or vertical lines on material selection charts, illustrated in Figure 5.8. It shows a schematic $E - \rho$ chart, in the manner of Chapter 4. We suppose that the design imposes limits on these of $E > 10$ GPa and $\rho < 3{,}000$ kg/m^3, shown in the figure. The optimizing search is restricted to the window boxed by the limits, labeled "Search region."

Less quantifiable properties such as corrosion resistance, wear resistance, or formability can all appear as limits, which take the form

$$A > A^*$$

or

$$A < A^* \tag{5.22}$$

where A is an attribute (service temperature, for instance) and A^* is a critical value of that attribute, set by the design, that must be exceeded or (in the case of corrosion rate) must *not* be exceeded.

One should not be too hasty in applying attribute limits; it may be possible to engineer a route around them. A component that gets too hot can be cooled; one that corrodes can be coated with a protective film. Many designers apply attribute limits for fracture toughness, K_{1c}, and ductility, ϵ_f, insisting on materials with, as rules of thumb, $K_{1c} > 15$ MPa.m$^{1/2}$ and $\epsilon_f > 2\%$ in order to guarantee adequate tolerance to stress concentrations.

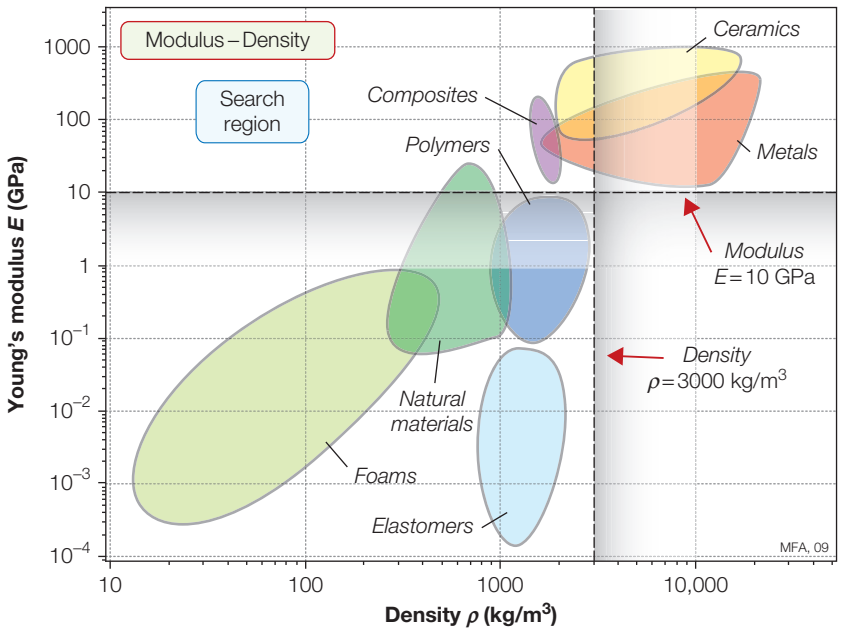


FIGURE 5.8
A schematic $E - \rho$ chart showing a lower limit for E and an upper limit for ρ.

By doing this they eliminate materials that the more innovative designer is able to use to good purpose (the limits just cited for K_{1c} and ϵ_f eliminate all polymers and all ceramics, a rash step too early in the design). At this stage, keep as many options open as possible.

Ranking: Indices on charts

The next step is to seek, from the subset of materials that meet the property limits, those materials that maximize performance. We will use the design of light, stiff components as an example; the other material indices are used in a similar way.

Figure 5.9 shows, as before, the modulus E plotted against density ρ on log scales. The material indices E/ρ, $E^{1/2}/\rho$, and $E^{1/3}/\rho$ can be plotted onto the figure. The condition

$$\frac{E}{\rho} = C$$

or, taking logs,

$$\text{Log}(E) = \text{Log}(\rho) + \text{Log}(C) \qquad (5.23)$$

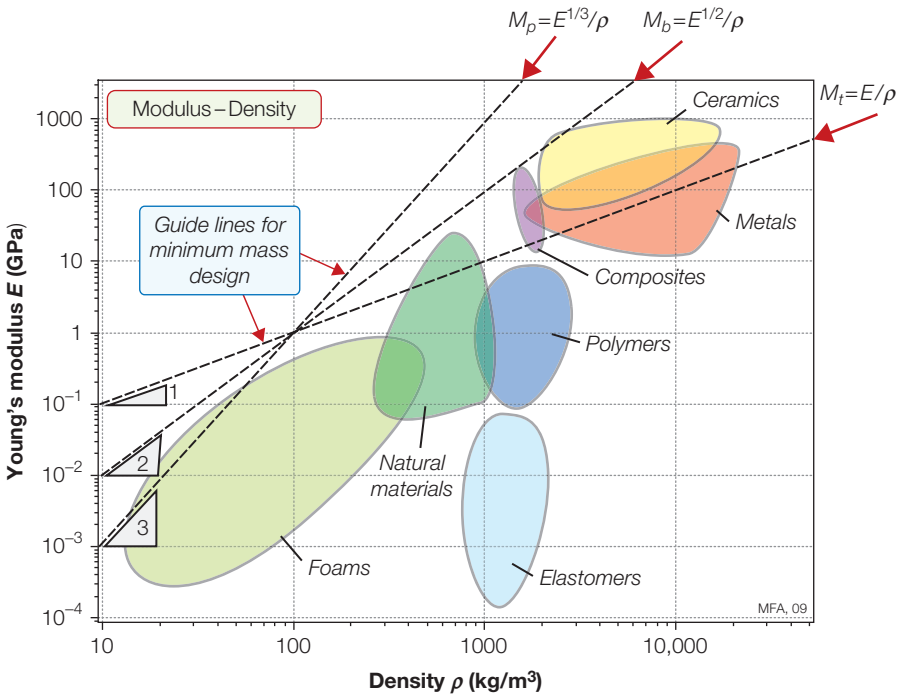


FIGURE 5.9
A schematic $E - \rho$ chart showing guide lines for the three material indices for stiff, lightweight design.

describes a family of straight parallel lines of slope 1 on a plot of Log(E) against Log(ρ), each line corresponding to a value of the constant C. The condition

$$\frac{E^{1/2}}{\rho} = C \tag{5.24}$$

or, taking logs again,

$$\text{Log}(E) = 2\ \text{Log}(\rho) + 2\ \text{Log}(C) \tag{5.25}$$

gives another set, this time with a slope of 2, and

$$\frac{E^{1/3}}{\rho} = C \tag{5.26}$$

gives yet another set, with slope 3. We shall refer to these lines as *selection guide lines*. They give the slope of the family of parallel lines belonging to that index. Where appropriate, the charts of Chapter 4 show guide lines like these.

It is now easy to read off the subset materials that optimally maximize performance for each loading geometry. All the materials that lie on a line of

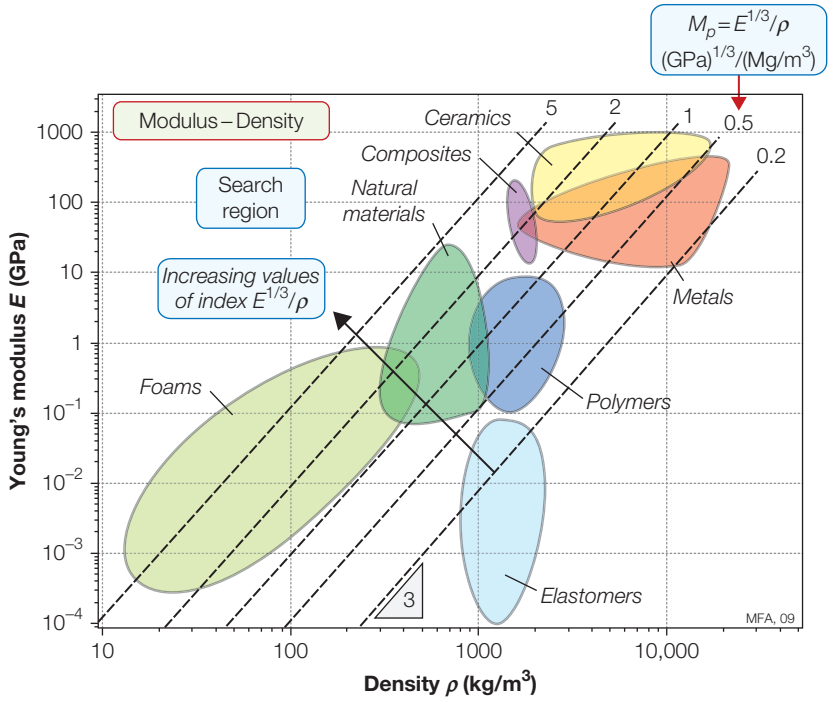


FIGURE 5.10
A schematic $E - \rho$ chart showing a grid of lines for the material index $M = E^{1/3}/\rho$.

constant $E^{1/3}/\rho$ perform equally well as a light, stiff panel; those above the line are better; those below, worse. Figure 5.10 shows a grid of lines corresponding to values of $E^{1/3}/\rho$ from 0.2 to 5 in units of GPa$^{1/3}$/(Mg.m^{-3}). A material with $M = 2$ in these units gives a panel that has one-tenth the weight of one with $M = 0.2$ that has the same stiffness. The subset of materials with particularly good index values is identified by picking a line that isolates a search area containing a reasonably small number of candidates, as shown schematically in Figure 5.11 as a diagonal selection line. Attribute limits can be added, narrowing the search window: That corresponding to $E > 50$ GPa is shown as a horizontal line. The materials lying in the search region meet both criteria. Their number is expanded or contracted by moving the index line down or up.

Documentation

We now have a ranked shortlist of potential candidate materials. The last step is to explore their character in depth. The list of constraints usually contains some that cannot be expressed as simple attribute limits. Many of

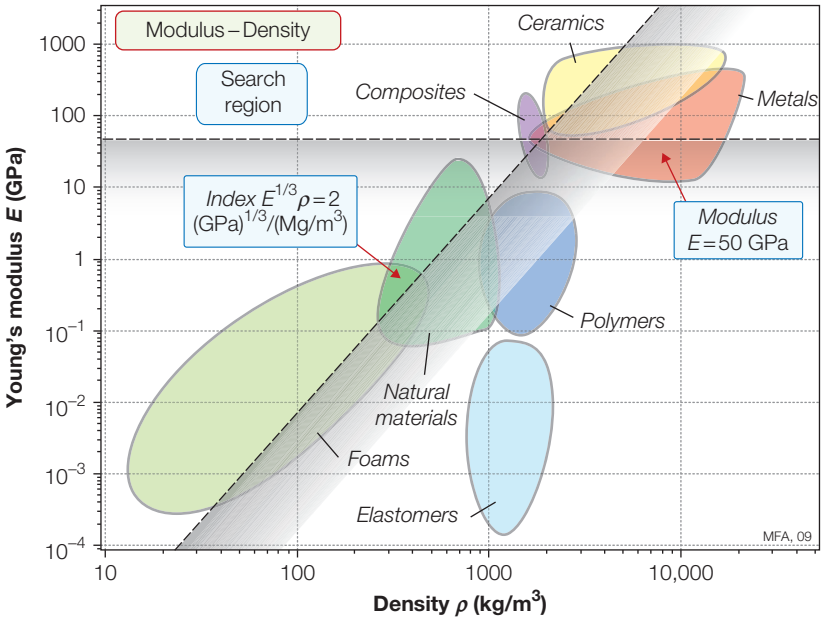


FIGURE 5.11
A selection based on the index $M = E^{1/3}/\rho > 2$ $(GPa)^{1/3}$ (Mg/m^3) together with the property limit $E > 50$ GPa. The materials contained in the search region become the candidates for the next stage of the selection process.

these relate to the behavior of the material in a given environment or to aspects of the ways in which the material can be shaped, joined, or finished. Such information can be found in handbooks, manufacturers' data sheets and computer-based sources.

And there will be constraints that, at this point, have been overlooked simply because they were not seen as such. Confidence is built by seeking design guidelines, case studies, or failure analyses that document each candidate, building a dossier of its strengths and weaknesses and ways in which these can be overcome. All of these come under the heading of documentation, sources of which are listed in Appendix D.

5.5 COMPUTER-AIDED SELECTION

The charts in Chapter 4 give an overview of material properties, but the number of materials that can be shown on any one of them is obviously limited. Selection using them is practical when there are very few constraints. When there are many—as there usually are—it becomes cumbersome. Both problems are overcome by computer implementation of the methods.

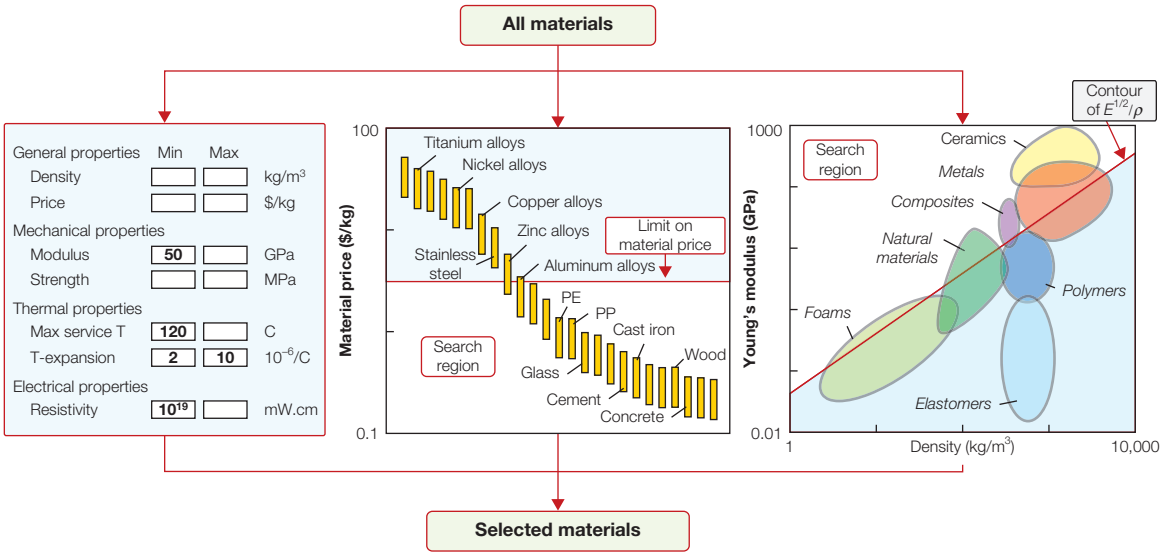


FIGURE 5.12

Computer-aided selection using the CES software. The schematic shows the three types of selection window. They can be used in any order and any combination. The selection engine isolates the subset of materials that passes all the selection stages.

The *CES* material and process selection software[4] is an example of such an implementation. A database contains records for materials, organized in the hierarchical manner shown earlier in Figure 5.2. Each record contains structured attribute data for a material, each attribute stored as a range spanning its typical (or often its permitted) range. It also contains limited documentation in the form of text, images, and references to sources of information about the material. The data are interrogated by a search engine that offers search interfaces shown schematically in Figure 5.12. On the left is a simple query interface for screening of single properties (the cover image of Chapter 3 is an example). The desired upper or lower limits for constrained attributes are entered; the search engine rejects all materials with attributes that lie outside the limits. In the center is shown a second way of interrogating the data: a bar chart like that shown earlier in Figure 4.1.

The bar chart, and the bubble chart in Figure 5.12 (*right*), are ways of both applying constraints and ranking. Used for ranking, a selection line or box is superimposed on the charts with edges that lie at the constrained values of the attributes, eliminating the material in the shaded areas and leaving the materials that meet all the constraints. If ranking is sought instead (having already applied all of the necessary constraints), the line or box is

[4] Granta Design Ltd., Cambridge, U.K. (*www.grantadesign.com*).

positioned so that a few materials—say three—are left in the selected area; these are the top-ranked candidates.

5.6 THE STRUCTURAL INDEX

Books on optimal design of structures (e.g., Shanley, 1960) make the point that the efficiency of material usage in mechanically loaded components depends on the product of three factors: the material index, as defined here; a factor describing section shape, the subject of Chapter 9; and a *structural index*,[5] which contains elements of the G and F of Equation (5.21). The subjects of this book—material and process selection—focus on the material index and on shape; however, we should examine the structural index briefly, partly to make the connection with the classical theory of optimal design and partly because it becomes useful (even to us) when structures are scaled in size.

In design for minimum mass (Equations (5.3), (5.8), and (5.14)), a measure of the efficiency of the design is given by the quantity m/L^3. Equation (5.3) for the light strong tie, when divided by L^3, becomes

$$\frac{m}{L^3} \geq \left(\frac{F^*}{L^2}\right)\left(\frac{\rho}{\sigma_f}\right) \tag{5.27}$$

Equation (5.8) for the light stiff panel becomes

$$\frac{m}{L^3} \geq \left(\frac{12}{C_1}\right)^{1/3}\left(\frac{b^2 S^*}{L^3}\right)^{1/3}\left(\frac{\rho}{E^{1/3}}\right) \tag{5.28}$$

and Equation (5.14) for the light stiff beam becomes

$$\frac{m}{L^3} \geq \left(\frac{12}{C_2}\right)^{1/2}\left(\frac{S^*}{L}\right)^{1/2}\left(\frac{\rho}{E^{1/2}}\right) \tag{5.29}$$

This m/L^3 has the dimensions of density; the lower this pseudodensity the lighter the structure for a given scale and thus the greater the structural efficiency. The first bracketed terms on the right-hand side of Equations (5.28) and (5.29) are constants. The last bracketed term in all three equations is the material index. The remaining term, F^*/L^2 in Equation (5.27), S^*b^2/L^3, in (5.28) and S^*/L in (5.29), is called the *structural index*. It has the dimensions of stress; it is a measure of the intensity of loading. Design proportions that are optimal, minimizing material usage, are optimal for structures of any size provided they all have the same structural index. The performance equations

[5] Also called the "structural loading coefficient," the "strain number," or the "strain index."

are written here in a way that isolates the structural index, a convention we shall follow in the case studies of Chapter 6.

The structural index for a component of minimum cost is the same as that for one of minimum mass.

5.7 SUMMARY AND CONCLUSIONS

Material selection is tackled in four steps:

- *Translation*—reinterpreting the design requirements in terms of function, constraints, objectives, and free variables.
- *Screening*—deriving attribute limits from the constraints and applying these to isolate a subset of viable materials.
- *Ranking*—ordering the viable candidates by the value of a material index, the criterion of excellence that maximizes or minimizes some measure of performance.
- *Documentation*—seeking documentation for the top-ranked candidates, exploring aspects of their history, their established uses, their behavior in relevant environments, their availability, and more, until a sufficiently detailed picture is built up that a final choice can be made.

Hard-copy material charts allow a first go at the task, and have the merit of maintaining breadth of vision: All material classes are in the frame, so to speak. But the number of materials is large, they have many properties, and the number of combinations of these appearing in indices is very much larger. It is impractical to print charts to include them all. These problems are overcome by computer implementation, allowing freedom to explore the whole universe of materials and providing detail when required.

The selection procedure described here is extended in Chapter 7 to deal with multiple constraints and objectives, and in Chapter 9 to include section shape. Before moving on to these, it is a good idea to consolidate the ideas so far by applying them to a number of case studies. These follow in Chapter 6.

5.8 FURTHER READING

The following books discuss optimization methods and their application in materials engineering.

Arora, J. S. (1989). *Introduction to optimum design*. McGraw-Hill, ISBN 0-07-002460-X.
An introduction to the terminology and methods of optimization theory.

Ashby, M. F., Shercliff, H. R., & Cebon, D. (2007). *Materials: Engineering, science, processing and design*. Butterworth-Heinemann, ISBN 978-0-7506-8391-3.
An elementary text that introduces materials through material property charts and develops the selection methods through case studies.

Dieter, G. E. (1999). *Engineering design, a materials and processing approach* (3rd ed.). McGraw-Hill, ISBN 9-780-073-66136-0.
A well-balanced and respected text focusing on the place of materials and processing in technical design.

Gordon, J. E. (1976). *The new science of strong materials, or why you don't fall through the floor* (2nd ed.). Penguin Books, ISBN 0-1402-0920-7.
This very readable book presents ideas about plasticity and fracture and describes ways of designing materials to prevent them.

Gordon, J. E. (1978). *Structures, or why things don't fall down*. Penguin Books, ISBN 0-1402-1961-7.
A companion to the other book by Gordon (above), this time introducing structural design.

Shanley, F. R. (1960). *Weight-strength analysis of aircraft structures* (2nd ed.). Dover Publications, Library of Congress Number 60-50107.
A remarkable text, no longer in print, on the design of lightweight structures.

CHAPTER 6

Case Studies: Materials Selection



Image of rower on the Cam in Cambridge, UK, courtesy of Andrew Dunn.

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6.1 INTRODUCTION AND SYNOPSIS

Here we have a collection of case studies illustrating the selection methods of Chapter 5. They are deliberately simplified to avoid obscuring the method under layers of detail. In most cases little is lost by this: The best choice of material for the simple example is the same as that for the more complex, for the reasons given in Section 5.3.

Each case study is laid out in the following way:

- *The problem statement*, setting the scene
- *The translation*, identifying function, constraints, objectives, and free variables, from which emerge the attribute limits and material indices

- *The selection,* in which the full menu of materials is reduced by screening and ranking to a shortlist of viable candidates
- *The postscript,* allowing a commentary on results and philosophy

The first few examples are straightforward, chosen to illustrate the method. Later examples are less obvious and require clear thinking to identify and distinguish objectives and constraints. Confusion here can lead to bizarre and misleading conclusions. Always apply common sense: Does the selection include the traditional materials used for that application? Are some members of the subset obviously unsuitable? If they are, it is usually because a constraint has been overlooked or an objective misapplied. The answer is to rethink them.

Most of the case studies use the hard-copy charts of Chapter 4; those at the end illustrate computer-based methods.

6.2 MATERIALS FOR OARS

Credit for inventing the rowed boat seems to belong to the Egyptians. Boats with oars appear in carved relief on monuments built in Egypt between 3300 and 3000 B.C. Boats, before steam power, could be propelled by poling, by sail, or by oar. Oars gave more control than the other two, the military potential of which was well understood by the Romans, the Vikings, and the Venetians.

Records of rowing races on the Thames in London extend back to 1716. Originally the competitors were watermen, rowing the ferries used to carry people and goods across the river. Gradually gentlemen became involved (notably the young gentlemen of Oxford and Cambridge), sophisticating both the rules and the equipment. The real stimulus for development of boats and oars came in 1900 with the establishment of rowing as an Olympic sport. Since then both have drawn to the fullest on the craftsmanship and materials of their day. Consider, as an example, the oar.

The translation Mechanically speaking, an oar is a beam, loaded in bending. It must be strong enough to carry, without breaking, the bending moment exerted by the oarsman; it must have a stiffness to match the rower's own characteristics; and it must give the right "feel." Meeting the strength constraint is easy. Oars are designed on *stiffness*, that is, to give a specified elastic deflection under a given load.

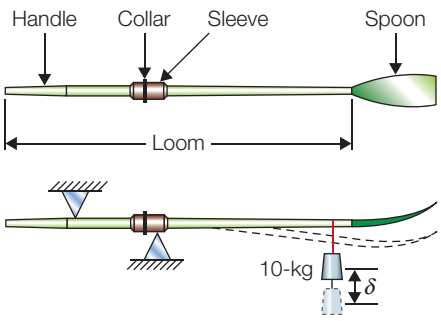


FIGURE 6.1
An oar. Oars are designed on stiffness, measured in the way shown in the *lower* figure, and they must be light.

Figure 6.1 (*top*) shows an oar: A blade or "spoon" is bonded to a shaft or

"loom" that carries a sleeve and collar to give positive location in the rowlock. The lower part of the figure shows how the oar stiffness is measured: A 10-kg weight is hung on the oar 2.05 m from the collar and the deflection δ at this point is measured. A soft oar will deflect nearly 50 mm; a hard one only 30. A rower, ordering an oar, will specify how hard it should be.

In addition, the oar must be light; extra weight increases the wetted area of the hull and the drag that goes with it. So there we have it: an oar is a beam of specified stiffness and minimum weight. The material index we want was derived in Chapter 5 as Equation (5.15). It is that for a light, stiff beam:

$$M = \frac{E^{1/2}}{\rho} \tag{6.1}$$

where E is Young's modulus and ρ is the density. There are other obvious constraints. Oars are dropped, and blades sometimes clash. The material must be tough enough to survive this, so brittle materials (those with a toughness G_1c less than 1 kJ/m^2) are unacceptable. Given these requirements, summarized in Table 6.1, what materials would you choose to make oars?

The selection Figure 6.2 shows the appropriate chart: that in which Young's modulus is plotted against density ρ. The selection line for the index M has a slope of 2, as explained in Section 5.4; it is positioned so that a small group of materials is left above it. They are the materials with the largest values of M and represent the best choice, provided they satisfy the other constraint (a simple attribute limit on toughness). They contain three classes of material: woods, carbon-reinforced polymers, and certain ceramics (Table 6.2). Ceramics are brittle; the toughness-modulus chart in Figure 4.7 shows that all fail to meet the requirements of the design. The recommendation is clear. Make your oars out of wood or—better—out of CFRP.

Postscript Now we know what oars should be made of. What, in reality, is used? Racing oars and sculls are made either of wood or of a high performance composite: carbon-fiber–reinforced epoxy.

Wooden oars are made today, as they were 100 years ago, by craftsmen working mainly by hand. The shaft and blade are of Sitka spruce from the

Table 6.1 Design Requirements for the Oar

Function	Oar—meaning light, stiff beam
Constraints	Length L specified Bending stiffness S^* specified Toughness $G_{1c} > 1$ kJ/m^2
Objective	Minimize the mass m
Free variables	Shaft diameter Choice of material

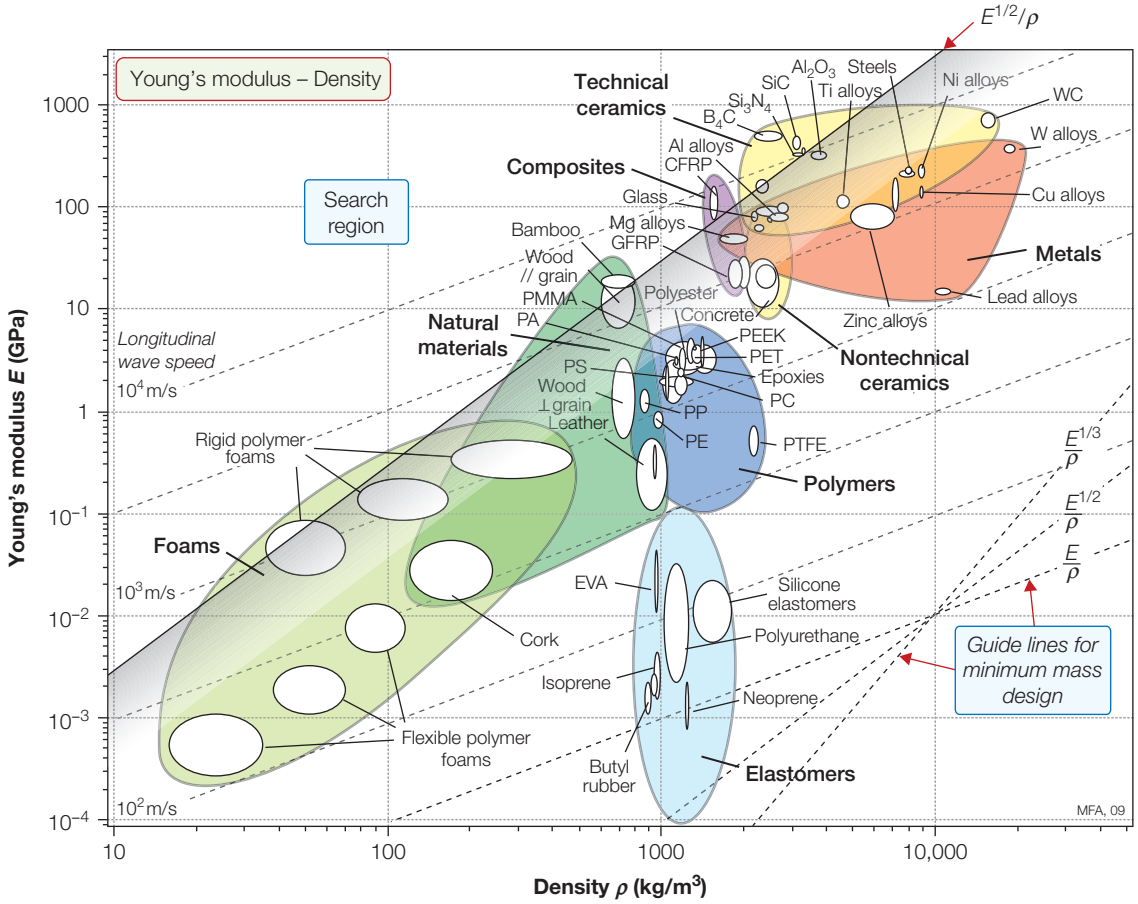


FIGURE 6.2
Materials for oars. CFRP is better than wood because the structure can be controlled.

Table 6.2 Materials for Oars

Material	Index M $(GPa)^{1/2}/(Mg/m^3)$	Comment
Bamboo	4.0–4.5	The traditional material for oars for canoes
Woods	3.4–6.3	Inexpensive, traditional, but with natural variability
CFRP	5.3–7.9	As good as wood, more control of properties
Ceramics	4–8.9	Good M but toughness low and cost high

northern United States or Canada, the further north the better because the short growing season gives a finer grain. The wood is cut into strips, four of which are laminated together to average the stiffness, and the blade is glued to the shaft. The rough oar is then shelved for some weeks to settle down, and finished by hand-cutting and polishing. When finished, a spruce oar weighs between 4 and 4.3 kg.

Composite blades are a little lighter than wood for the same stiffness. The component parts are fabricated from a mixture of carbon and glass fibers in an epoxy matrix, assembled and glued. The advantage of composites lies partly in the saving of weight (typical weight: 3.9 kg) and partly in the greater control of performance: The shaft is molded to give the stiffness specified by the purchaser. Until recently a CFRP oar cost more than a wooden one, but the price of carbon fibers has fallen sufficiently that the two cost about the same.

Could we do better? The chart shows that wood and CFRP offer the lightest oars, at least when normal construction methods are used. Novel composites, not shown on the chart, might permit further weight saving; and functional-grading (a thin, very stiff outer shell with a low-density core) might do it. But both appear, at present, unlikely.

Related reading

Redgrave, S. (1992). *Complete book of rowing*. Partridge Press.

Related case studies

6.3 "Mirrors for large telescopes"
6.4 "Materials for table legs"
10.2 "Spars for human-powered planes"
10.3 "Forks for a racing bicycle"

6.3 MIRRORS FOR LARGE TELESCOPES

There are some very large optical telescopes in the world. The newer ones employ complex and cunning tricks to maintain their precision as they track across the sky—more on that in the Postscript. But if you want a simple telescope, you make the reflector as a single rigid mirror. The largest such telescope is sited on Mount Semivodrike, near Zelenchukskaya in the Caucasus Mountains of Russia. The mirror is 6 m (236 inches) in diameter. To be sufficiently rigid, the mirror is made of glass about 1 m thick and weighs 70 tonnes.

The total cost of a large (236-inch) telescope is, like the telescope itself, astronomical—about US$300 m. The mirror itself accounts for only about 5% of this cost; the rest of the cost is the mechanism that holds, positions, and moves it as it tracks across the sky. This mechanism must be stiff enough to position the mirror relative to the collecting system with a precision about

equal to that of the wavelength of light. It might seem, at first sight, that doubling the mass m of the mirror would require that the sections of the support structure be doubled too, so as to keep the stresses (and hence the strains and displacements) the same; but the heavier structure then deflects under its own weight. In practice, the sections have to increase as m^2, and so does the cost.

A century ago, mirrors were made of speculum metal (density: about 8000 kg/m^3). Since then, they have been made of glass (density: 2,300 kg/m^3), silvered on the front surface, so none of the optical properties of the glass are used. Glass is chosen for its mechanical properties only; the 70 tonnes of glass are just a very elaborate support for 100 nm (about 30 grams) of silver. Could one, by taking a radically new look at materials for mirrors, suggest possible routes to the construction of lighter, cheaper telescopes?

The translation At its simplest, the mirror is a circular disk with diameter $2R$ and mean thickness t, simply supported at its periphery (Figure 6.3). When horizontal, it will deflect under its own weight m; when vertical it will not deflect significantly. This distortion (which changes the focal length and introduces aberrations) must be small enough that it does not interfere with performance; in practice, this means that the deflection δ of the midpoint of the mirror must be less than the wavelength of light. Additional requirements are high dimensional stability (no creep) and low thermal expansion (Table 6.3).

The mass of the mirror (the property we wish to minimize) is

$$m = \pi R^2 t \rho \qquad (6.2)$$

where ρ is the density of the material of the disk. The elastic deflection, δ, of the center of a

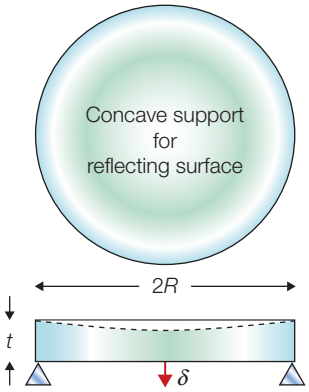


FIGURE 6.3
The mirror of a large optical telescope is modeled as a disk, simply supported at its periphery. It must not sag by more than a wavelength of light at its center.

Table 6.3 Design Requirements for the Telescope Mirror

Function	Precision mirror
Constraints	Radius R specified Must not distort more than δ under self-weight High dimensional stability: no creep, low thermal expansion
Objective	Minimize the mass, m
Free variables	Thickness of mirror, t Choice of material

horizontal disk due to its own weight is given, for a material with Poisson's ratio of 0.3 (Appendix B), by

$$\delta = \frac{3}{4\pi}\frac{m g R^2}{E t^3} \tag{6.3}$$

The quantity g in this equation is the acceleration due to gravity: 9.81 m/s^2; E, as before, is Young's modulus. We require that this deflection be less than (say) 10 μm. The diameter $2R$ of the disk is specified by the telescope design, but the thickness t is a free variable. Solving for t and substituting this into the first equation gives

$$m = \left(\frac{3g}{4\delta}\right)^{1/2} \pi R^4 \left[\frac{\rho}{E^{1/3}}\right]^{3/2} \tag{6.4}$$

The lightest mirror is the one with the greatest value of the material index

$$M = \frac{E^{1/3}}{\rho} \tag{6.5}$$

We treat the remaining constraints as attribute limits, requiring a melting point greater than 500°C to avoid creep, zero moisture take up, and a low thermal expansion coefficient ($\alpha < 20 \times 10^{-6}$/K).

The selection Here we have another example of elastic design for minimum weight. The appropriate chart is again that relating Young's modulus E and density ρ—but the line we now construct on it has a slope of 3, corresponding to the condition $M = E^{1/3}/\rho$ = constant (Figure 6.4). Glass lies at the value M = 1.7 (GPa)$^{1/3}$.m^3/Mg. Materials that have larger values of M are better; those with lower, worse. Glass is much better than steel or speculum metal (that is why most mirrors are made of glass), but it is less good than magnesium, several ceramics, carbon-fiber– and glass-fiber–reinforced polymers, or—an unexpected finding—stiff foamed polymers. The shortlist before applying the attribute limits is given in Table 6.4.

One must, of course, examine other aspects of this choice. The mass of the mirror, calculated from Equation (6.4), is listed in the table. The CFRP mirror is less than half the weight of the glass one, and its support structure could thus be as much as four times less expensive. The possible saving by using foam is even greater. But could they be made?

Some of the choices—polystyrene foam or CFRP—may at first seem impractical. But the potential cost saving (the factor of 16) is so vast that they are worth examining. There are ways of casting a thin film of silicone rubber or of epoxy onto the surface of the mirror backing (the polystyrene or the CFRP) to give an optically smooth surface that could be silvered. The most obvious obstacle is the lack of stability of polymers—they change dimensions with age, humidity, temperature and so on. But glass itself can be reinforced with carbon

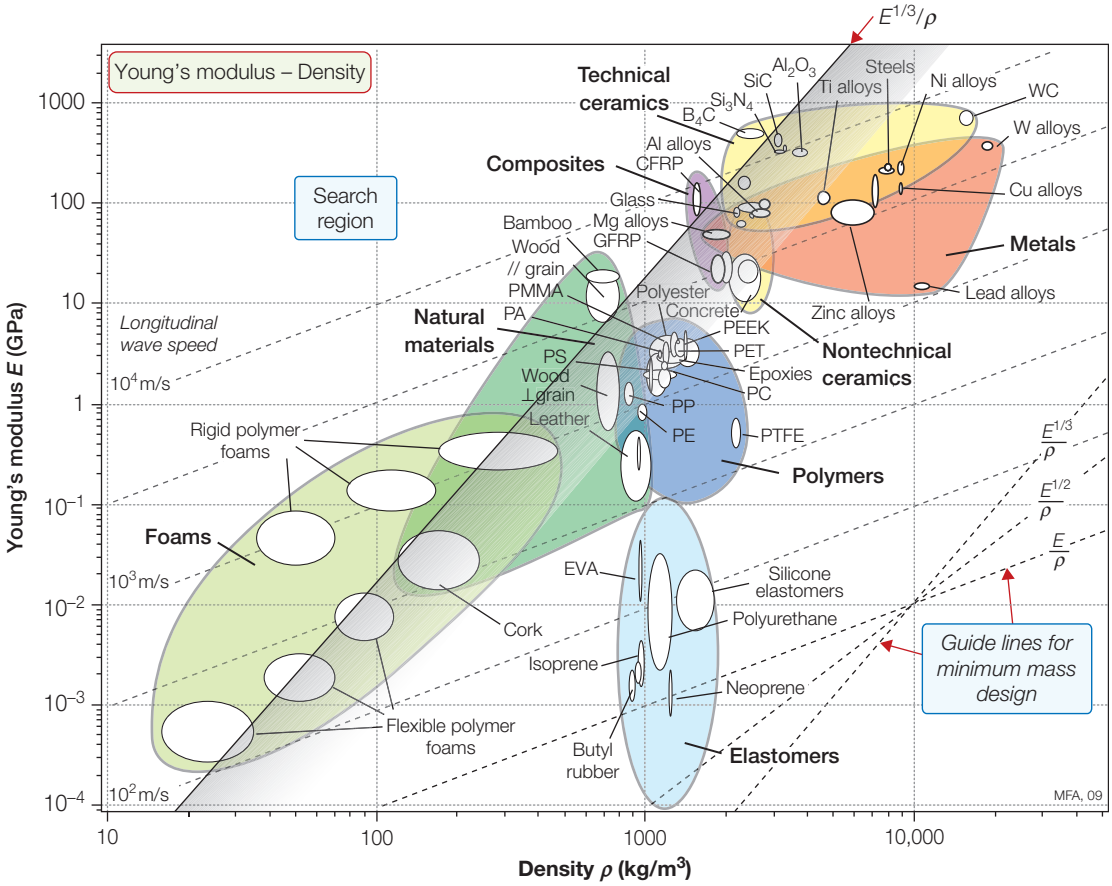


FIGURE 6.4

Materials for telescope mirrors. Glass is better than most metals, among which magnesium is a good choice. Carbon-fiber–reinforced polymers give, potentially, the lowest weight of all, but may lack adequate dimensional stability. Foamed glass is a possible candidate.

fibers; and it can also be foamed to give a material that is denser than polystyrene foam but much lighter than solid glass. Both foamed and carbon-reinforced glass have the same chemical and environmental stability as solid glass. They could provide a route to large cheap mirrors.

Postscript There are, of course, other things you can do. The stringent design criterion ($\delta < 10\,\mu$m) can be partially overcome by engineering design without reference to the material choice. The 8.2-m Japanese telescope on Mauna Kea, Hawaii, and the Very Large Telescope (VLT) at Cerro Paranal Silla in Chile each have a thin glass reflector supported by an array of hydraulic or piezo-electric jacks that exert distributed forces over the back surface,

Table 6.4 Mirror Backing for 200-inch (5.1-m) Telescope

Material	$M = E^{1/3}/\rho$ $(GPa)^{1/3}.m^3/Mg$	m (tonne) $2R = 5.1$ m (from Eq. 6.4)	Comment
Steel (or speculum)	0.74	73.6	Very heavy—the original choice
GFRP	1.5	25.5	Not dimensionally stable enough—use for radio telescope
Al-Alloys	1.6	23.1	Heavier than glass, and with high thermal expansion
Glass	1.7	21.6	The present choice
Mg-Alloys	1.9	17.9	Lighter than glass but high thermal expansion
CFRP	3.0	9	Very light, but not dimensionally stable; use for radio telescopes
Foamed polystyrene	4.5	5	Very light, but dimensionally unstable. Foamed glass?

controlled to vary with the attitude of the mirror. The Keck telescope, also on Mauna Kea, is segmented, with each segment independently positioned to give optical focus. But the limitations of this sort of mechanical system still require that the mirror meet a stiffness target. While stiffness at minimum weight is the design requirement, the material-selection criteria remain unchanged.

Radio telescopes do not have to be quite as precisely dimensioned as optical ones because they detect radiation with a longer wavelength, about 0.25 mm rather than the 0.02 mm of light waves. But they are much bigger (60 m rather than 6) and they suffer from similar distortional problems. A recent 45 m radio telescope built for the University of Tokyo has a parabolic reflector made up of 6000 CFRP panels, each servo-controlled to compensate for macrodistortion. Radio telescopes are now routinely made from CFRP, for exactly the reasons we deduced.

Related case study

6.16 "Minimizing thermal distortion in precision devices"

6.4 MATERIALS FOR TABLE LEGS

Luigi Tavolino, furniture designer, conceives of a lightweight table of daring simplicity: a flat sheet of toughened glass supported on slender, unbraced cylindrical legs (Figure 6.5). The legs must be solid (to make them thin) and as light as possible (to make the table easier to move). They must support the table top and whatever is placed upon it without buckling (Table 6.5). What materials could one recommend?

The translation This is a problem with two objectives[1]: Weight is to be minimized and slenderness maximized. There is one constraint: resistance to buckling. Consider minimizing weight first.

The leg is a slender column of material of density ρ and modulus E. Its length, L, and the maximum load, F, it must carry are determined by the design: They are fixed. The radius r of a leg is a free variable. We wish to minimize the mass m of the leg, given by the objective function

$$m = \pi r^2 L \rho \tag{6.6}$$

subject to the constraint that it supports a load P without buckling. The elastic buckling load F_{crit} of a column of length L and radius r (see Appendix B) is

$$F_{crit} = \frac{\pi^2 E I}{L^2} = \frac{\pi^3 E r^4}{4 L^2} \tag{6.7}$$

using $I = \pi r^4/4$ where I is the second moment of the area of the column. The load F must not exceed F_{crit}. Solving for the free variable, r, and substituting it into the equation for m gives

$$m \geq \left(\frac{4F}{\pi}\right)^{1/2} (L)^2 \left[\frac{\rho}{E^{1/2}}\right] \tag{6.8}$$

The material properties are grouped together in the last pair of brackets. The weight is minimized by selecting the subset of materials with the greatest value of the material index

$$M_1 = \frac{E^{1/2}}{\rho}$$

(a result we could have taken directly from Appendix C).

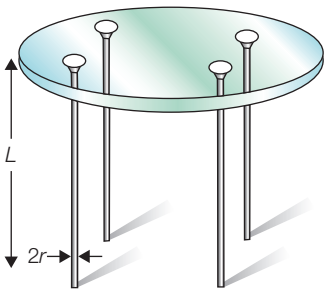


FIGURE 6.5
A lightweight table with slender cylindrical legs. Lightness and slenderness are independent design goals, both constrained by the requirement that the legs must not buckle when the table is loaded. The best choice is a material with high values of both $E^{1/2}/\rho$ and E.

Table 6.5 Design Requirements for Table Legs

Function	Column (supporting compressive loads)
Constraints	Length L specified Must not buckle under design loads Must not fracture if accidentally struck
Objectives	Minimize mass, m Maximize slenderness
Free variables	Diameter of legs, $2r$ Choice of material

[1] Formal methods for dealing with multiple objectives are developed in Chapter 7.

Now slenderness. Inverting Equation (6.7) with F_{crit} set equal to F gives an equation for the thinnest leg that will not buckle:

$$r \geq \left(\frac{4F}{\pi^3}\right)^{1/4} (L)^{1/2} \left[\frac{1}{E}\right]^{1/4} \tag{6.9}$$

The thinnest leg is that made of the material with the largest value of the material index

$$M_2 = E$$

The selection We seek the subset of materials that have high values of $E^{1/2}/\rho$ and E. We need the $E - \rho$ chart again (Figure 6.6). A guideline of slope 2 is drawn on the diagram; it defines the slope of the grid of lines for values of $E^{1/2}/\rho$. The guideline is displaced upward (retaining the slope) until a reasonably small subset of materials is isolated above it; it is shown at the position M_1 = 5 GPa$^{1/2}$/(Mg/m^3). Materials above this line have higher values of M_1. They are identified in the figure as *woods* (the traditional material for table legs), *composites* (particularly CFRP), and certain *engineering ceramics*. Polymers are out: They are not stiff enough; metals too: They are too heavy (even magnesium alloys, which are the lightest).

The choice is further narrowed by the requirement that, for slenderness, E must be large. A horizontal line on the diagram links materials with equal values of E; those above are stiffer. Figure 6.6 shows that placing this line at M_1 = 100 GPa eliminates woods and GFRP. If the legs must be really thin, then the shortlist is reduced to CFRP and ceramics: They give legs that weigh the same as the wooden ones but are barely half as thick. Ceramics, we know, are brittle: They have low values of fracture toughness. Table legs are exposed to abuse—they get knocked and kicked; common sense suggests that an additional constraint is needed, that of adequate toughness. This can be done using Figure 4.7; it eliminates ceramics, leaving CFRP. The cost of CFRP may cause Snr. Tavolino to reconsider his design, but that is another matter: He did not mention cost in his original specification.

It is a good idea to lay out the results as a table, showing not only the materials that are best but those that are second-best—they may, when other considerations are involved, become the best choice. Table 6.6 shows the way to do it.

Postscript Tubular legs, the reader will say, must be lighter than solid ones. True; but they will also be fatter. So it depends on the relative importance Snr. Tavolino attaches to his two objectives—lightness and slenderness—and only he can decide that. If he can be persuaded to live with fat legs, tubing

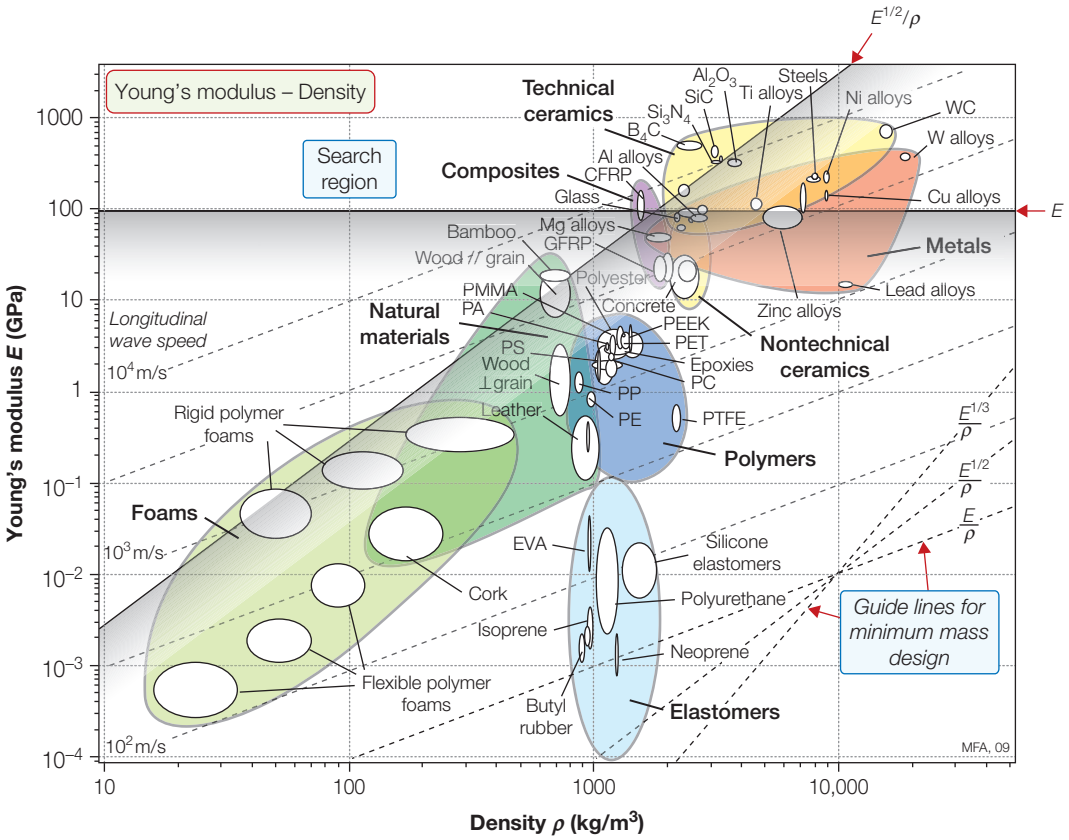


FIGURE 6.6
Materials for light, slender legs. Wood is a good choice; so is a composite such as CFRP, which, having a higher modulus than wood, gives a column that is both light and slender. Ceramics meet the stated design goals, but are brittle.

Table 6.6 Materials for Table Legs

Material	Typical M_1 ($GPa^{1/2}.m^3/Mg$)	Typical M_2 (GPa)	Comment
GFRP	2.5	20	Less expensive than CFRP, but lower M_1 and M_2
Woods	4.5	10	Outstanding M_1; poor M_2 Inexpensive, traditional, reliable
Ceramics	6.3	300	Outstanding M_1 and M_2 Eliminated by brittleness
CFRP	6.6	100	Outstanding M_1 and M_2, but expensive

can be considered—and the material choice may be different. Materials selection when section shape is a variable comes in Chapter 9.

Ceramic legs were eliminated because of low toughness. If (improbably) the goal is to design a light, slender-legged table for use at high temperatures, ceramics should be reconsidered. The brittleness problem can be bypassed by protecting the legs from abuse or by prestressing them in compression.

Related case studies

6.2 "Materials for oars"
6.3 "Mirrors for large telescopes"
8.5 "Conflicting objectives: Table legs again"
10.2 "Spars for human-powered planes"
10.3 "Forks for a racing bicycle"
10.5 "Table legs yet again: Thin or light?"

6.5 COST: STRUCTURAL MATERIALS FOR BUILDINGS

The most expensive thing that most people buy is the house they live in. Roughly half the expense of building a house is the cost of the materials of which it is made, and they are used in large quantities (family house: around 200 tonnes; large apartment block: around 20,000 tonnes). The materials are used in three ways: structurally to hold the building up; as cladding to keep the weather out; and as "internals" to insulate against heat and sound, and to decorate.

Consider the selection of materials for the structure (Figure 6.7). They must be stiff, strong, and cheap. Stiff, so that the building does not flex too much under wind and internal loads; strong, so that there is no risk of it collapsing. And cheap, because such a lot of material is used. The structural frame of a building is rarely exposed to the environment, and is not, in general, visible, so criteria of corrosion resistance or appearance are not important here.

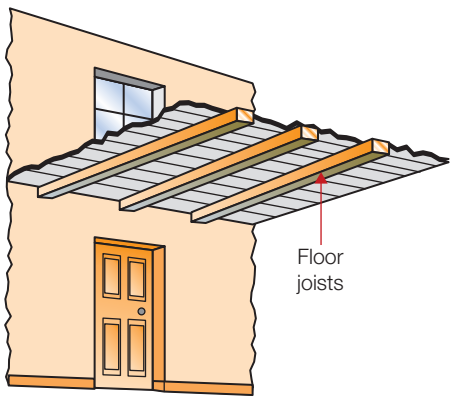


FIGURE 6.7
The materials of a building perform three broad roles. The frame gives mechanical support; the cladding excludes the environment; and the internal surfacing controls heat, light, and sound. The selection criteria depend on the function.

Table 6.7 Design Requirements for Floor Beams

Function	Floor beam
Constraints	Length L specified Stiffness: must not deflect too much under design loads Strength: must not fail under design loads
Objective	Minimize cost, C
Free variables	Cross-section area of beam, A Choice of material

The design goal is simple: strength and stiffness at minimum cost. To be more specific: Consider the selection of material for floor joists. Table 6.7 summarizes the requirements.

The translation Floor joists are beams; they are loaded in bending. The material index for a stiff beam of minimum mass, m, was developed in Chapter 5 (Equations (5.11) through (5.15)). The cost C of the beam is just its mass, m, times the cost per kg, C_m, of the material of which it is made:

$$C = m\,C_m = A\,L\rho\,C_m \tag{6.10}$$

which becomes the objective function of the problem. Proceeding as in Chapter 5, we find the index for a stiff beam of minimum cost to be

$$M_1 = \frac{E^{1/2}}{\rho\,C_m}$$

The index when strength rather than stiffness is the constraint was not derived earlier. Here it is. The objective function is still Equation (6.10), but the constraint is now that of strength: The beam must support F without failing. The failure load of a beam (Appendix B, Section B.4) is

$$F_f = C_2\frac{I\,\sigma_f}{\gamma_m\,L} \tag{6.11}$$

where C_2 is a constant, σ_f is the failure strength of the material of the beam, and γ_m is the distance between the neutral axis of the beam and its outer filament. We consider a rectangular beam of depth d and width b. We assume the proportions of the beam are fixed so that $d = \alpha b$ where α is the aspect ratio, typically 2 for wood beams. Using this and $I = bd^3/12$ to eliminate A in Equation (6.10) gives the cost of the beam that will just support the load F_f:

$$C = \left(\frac{6\sqrt{\alpha}}{C_2}\,\frac{F_f}{L^2}\right)^{2/3}(L^3)\left[\frac{\rho\,C_m}{\sigma_f^{2/3}}\right] \tag{6.12}$$

The mass is minimized by selecting materials with the largest values of the index

$$M_2 = \frac{\sigma_f^{2/3}}{\rho C_m}$$

The selection Stiffness first. Figure 6.8(a) shows the relevant chart: modulus E against relative cost per unit volume, $C_m\,\rho$ (the chart uses a relative cost C_R, defined in Chapter 4, in place of C_m but this makes no difference to the selection). The shaded band has the appropriate slope for M_1; it isolates concrete, stone, brick, woods, cast irons, and carbon steels. Figure 6.8(b) shows strength against relative cost. The shaded band—M_2 this time—gives almost the same selection. They are listed, with values, in

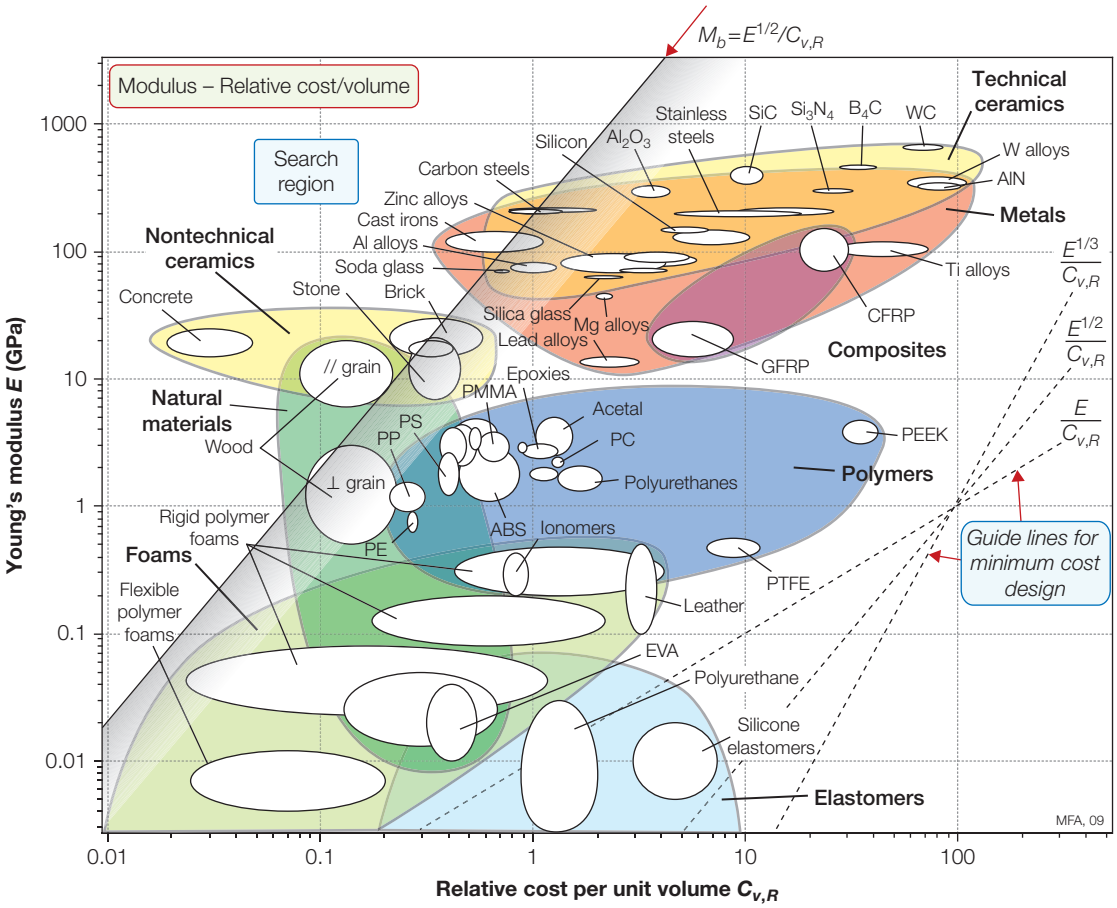


FIGURE 6.8(a)
The selection of cheap, stiff materials for the structural frames of buildings.

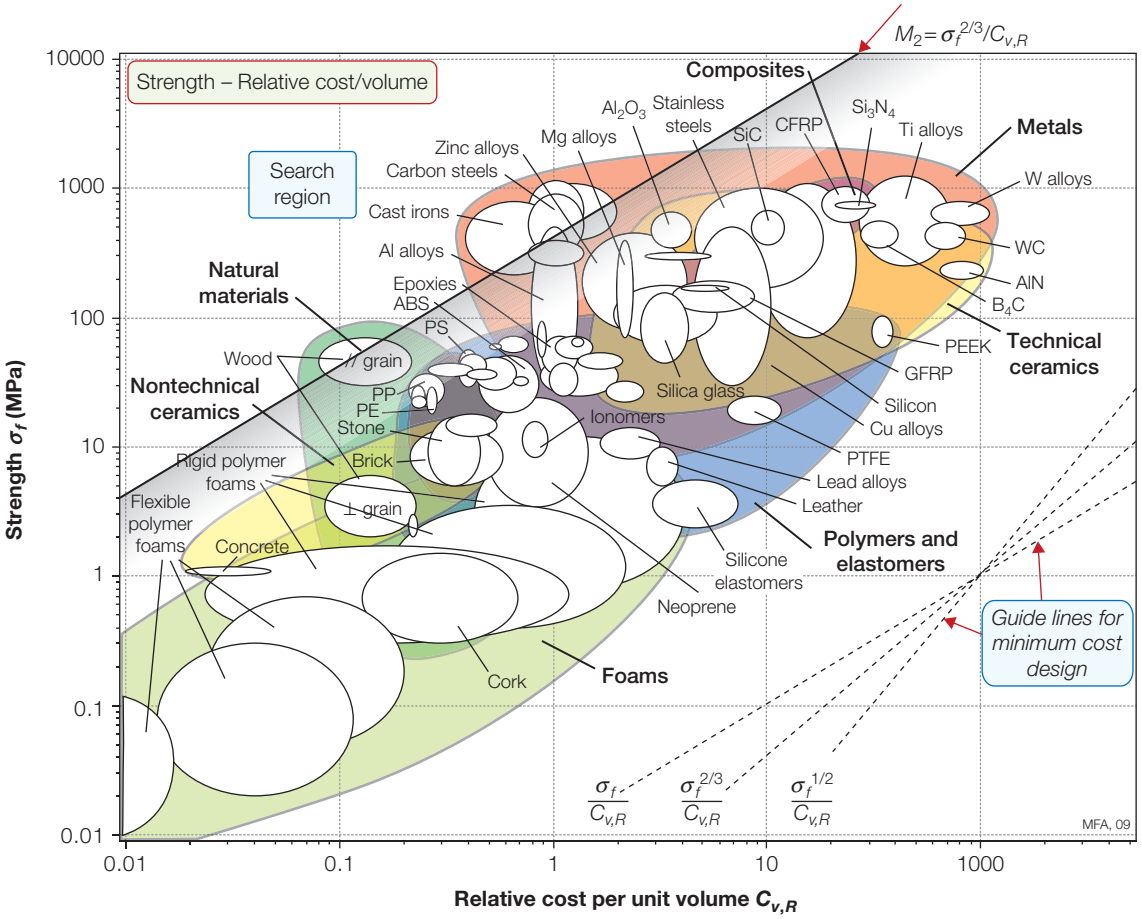


FIGURE 6.8(b)
The selection of cheap, strong materials for the structural frames of buildings.

Table 6.8. They are exactly the materials with which buildings have been, and are, made.

Postscript Concrete, stone, and brick have strength only in compression; the form of the building must use them in this way (columns, arches). Wood, steel, and reinforced concrete have strength in bending and tension as well as compression; steel, additionally, can be given efficient shapes (I-sections, box sections, tubes, discussed in Chapter 9). The form of the building made from these has much greater freedom.

It is sometimes suggested that architects live in the past; that in the twenty-first century they should be building with carbon (CFRP) and fiberglass (GFRP) composites, aluminum and titanium alloys, and stainless steel. Some do, but

Table 6.8 Structural Materials for Buildings

Material	M_1 (GPa$^{1/2}$)/ (kg/m^3)	M_2 (MPa$^{2/3}$)/ (kg/m^3)	Comment
Concrete	160	14	Use in compression only
Brick	12	12	
Stone	9.3	12	
Woods	21	90	Can support bending and tension as well as compression, allowing greater freedom of shape
Cast iron	17	90	
Steel	14	45	

the last two figures give an idea of the penalty involved: The cost of achieving the same stiffness and strength is between 5 and 50 times greater. Civil construction (buildings, bridges, roads, and the like) is materials-intensive: The cost of the material dominates the product cost, and the quantity used is enormous. Then only the cheapest of materials qualify, and the design must be adapted to use them.

Related reading

Cowan, H.J. & Smith, P.R. (1988). *The science and technology of building materials*. Van Nostrand-Reinhold.

Doran, D.K. (1992). The construction reference book. Butterworth-Heinemann.

Related case studies

6.2 "Materials for oars"
6.4 "Materials for table legs"
8.2 "Floor joists again"
10.4 "Floor joists: wood, bamboo or steel?"

6.6 MATERIALS FOR FLYWHEELS

Flywheels store energy. Small ones—the sort found in children's toys—are made of lead. Old steam engines and modern automobiles have flywheels too; they are made of cast iron. Flywheels have been proposed for power storage and regenerative braking systems for vehicles; a few have been built, some of high-strength steel, some of composites. Lead, cast iron, steel, composites—there is a strange diversity here. What *is* the best choice of material for a flywheel?

An efficient flywheel stores as much *energy per unit weight* as possible. As the flywheel is spun up, increasing its angular velocity ω, it stores more energy. But if the centrifugal stress exceeds the tensile strength of the flywheel, it flies apart. So strength sets an upper limit on the energy that can be stored.

Table 6.9(a) Design Requirements for a Maximum-energy Flywheel	
Function	Flywheel for energy storage
Constraints	Outer radius, R, fixed Must not burst Adequate toughness to give crack tolerance
Objective	Maximize kinetic energy per unit mass
Free variable	Choice of material

Table 6.9(b) Design Requirements for Fixed Velocity	
Function	Flywheel for child's toy
Constraint	Outer radius, R, fixed
Objective	Maximize kinetic energy per unit volume at fixed angular velocity
Free variable	Choice of material

The flywheel of a child's toy is not efficient in this sense. Its angular velocity is limited by the pulling power of the child and never remotely approaches the burst velocity. In this case, and for the flywheel of an automobile engine, we wish to maximize the *energy stored* at a given *angular velocity* in a flywheel with an outer radius, R, that is constrained by the size of the cavity in which it must sit.

The objective and constraints in flywheel design thus depend on its purpose. The two alternative sets of design requirements are listed in Tables 6.9(a) and 6.9(b).

The translation An efficient flywheel of the first type stores as much energy per unit weight as possible, without failing. Think of it as a solid disk of radius R and thickness t, rotating with angular velocity ω (Figure 6.9). The energy U stored in the flywheel (Appendix B) is

$$U = \frac{1}{2}J\omega^2 \qquad (6.13)$$

Here $J = \frac{\pi}{2}\rho R^4 t$ is the polar moment of inertia of the disk and ρ the density of the material of which it is made, giving

$$U = \frac{\pi}{4}\rho R^4 t\omega^2 \qquad (6.14)$$

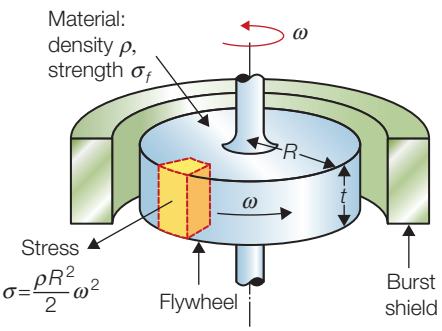


FIGURE 6.9
A flywheel. The maximum kinetic energy it can store is limited by its strength.

The mass of the disk is

$$m = \pi R^2 t \rho \tag{6.15}$$

The quantity to be maximized is the kinetic energy per unit mass, which is the ratio of the last two equations:

$$\frac{U}{m} = \frac{1}{4} R^2 \omega^2 \tag{6.16}$$

As the flywheel is spun up, the energy stored in it increases, but so does the centrifugal stress. The maximum principal stress in a spinning disk of uniform thickness (Appendix B again) is

$$\sigma_{max} = \left(\frac{3+\upsilon}{8}\right) \rho R^2 \omega^2 \approx \frac{1}{2} \rho R^2 \omega^2 \tag{6.17}$$

where υ is Poisson's ratio ($\upsilon \approx 1/3$). This stress must not exceed the failure stress σ_f (with an appropriate factor of safety, here omitted). This sets an upper limit to the product of the angular velocity, ω, and the disk radius, R (the free variables). Eliminating R_ω between the last two equations gives

$$\frac{U}{m} = \frac{1}{2} \left(\frac{\sigma_f}{\rho}\right) \tag{6.18}$$

The best materials for high-performance flywheels are those with high values of the material index

$$M = \frac{\sigma_f}{\rho} \tag{6.19}$$

It has units of kJ/kg.

And now the other sort of flywheel—that of the child's toy. Here we seek the material that stores the most energy per unit volume V at constant velocity, ω. The energy per unit volume at a given ω is (from Equation (6.2)):

$$\frac{U}{V} = \frac{1}{4} \rho R^2 \omega^2.$$

Both R and ω are fixed by the design, so the best material is now that with the greatest value of

$$M_2 = \rho \tag{6.20}$$

The selection Figure 6.10 shows the strength–density chart. Values of M_1 correspond to a grid of lines of slope 1. One such is plotted as a diagonal line at the value M_1 = 200 kJ/kg. Candidate materials with high values of M_1 lie in search region 1, which is toward the top left. The best choices are

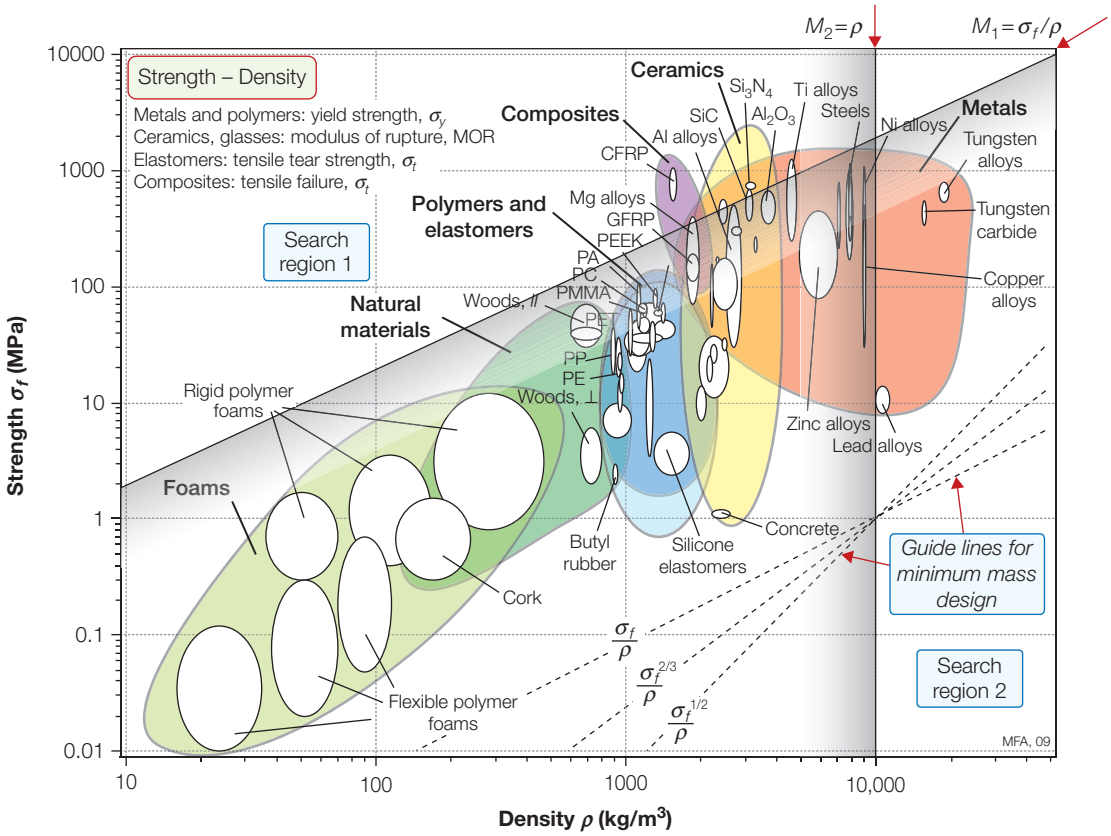


FIGURE 6.10

Materials for flywheels. Composites are the best choices. Lead and cast iron, traditional for flywheels, are good when performance is limited by rotational velocity, not strength.

unexpected ones: composites, particularly CFRP, high-strength titanium alloys, and some ceramics, but these are ruled out by their low toughness.

But what of the lead flywheels of children's toys? There could hardly be two more different materials than CFRP and lead: the one, strong and light; the other, soft and heavy. Why lead? It is because, in the child's toy, the constraint is different. Even a super-child cannot spin the flywheel of his or her toy up to its burst velocity. The angular velocity ω is limited instead by the drive mechanism (pull string, friction drive). Then, as we have seen, the best material is that with the largest density. The second selection line in Figure 6.10 shows the index M_2 at the value 10,000 kg/m^3. We seek materials in search region 2 to the right of this line. Lead is good. Tungsten is better, but more expensive. Cast iron is less good, but cheaper. Gold, platinum, and uranium (not shown on the chart) are the best of all, but may be thought unsuitable for other reasons.

Table 6.10 Energy Density of Power Sources

Source	Energy Density kJ/kg	Comment
Gasoline	20,000	Oxidation of hydrocarbon—mass of oxygen not included
Rocket fuel	5000	Less than hydrocarbons because oxidizing agent forms part of fuel
Flywheels	Up to 400	Attractive, but not yet proved
Lithium-ion battery	Up to 350	Expensive, limited life
Nickel-cadmium battery	170–200	Less expensive than lithium-ion
Lead-acid battery	50–80	Large weight for acceptable range
Springs, rubber bands	Up to 5	Much less efficient method of energy storage than flywheel

Postscript A CFRP rotor is able to store around 400 kJ/kg. A lead flywheel, by contrast, can store only 1 kJ/kg before disintegration; a cast-iron flywheel, about 30. All of these are small compared with the energy density in gasoline: roughly 20,000 kJ/kg. Even so, the energy density in the flywheel is considerable; its sudden release in a failure could be catastrophic. The disk must be surrounded by a burst shield and precise quality control in manufacture is essential to avoid out-of-balance forces. This has been achieved in a number of composite energy-storage flywheels intended for use in trucks and buses, and as an energy reservoir for smoothing wind-power generation.

And now a digression: the electric car. Hybrid gas-electric cars are already on the roads, using advanced battery technology to store energy. But batteries have their problems: The energy density they can contain is low (see Table 6.10); their weight limits both the range and the performance of the car. It is practical to build flywheels with an energy density about equal to that of the best batteries.

Consideration is now being given to a flywheel for electric cars. A pair of counter-rotating CFRP disks are housed in a steel burst shield. Magnets embedded in the disks pass near coils in the housing, inducing a current and allowing power to be drawn to the electric motor that drives the wheels. Such a flywheel could, it is estimated, give an electric car an adequate range at a cost that is competitive with the gasoline engine and with none of the local pollution.

Related reading

Christensen, R.M. (1979). *Mechanics of composite materials* (p. 213 et seq.). Wiley Interscience.

Lewis, G. (1990). *Selection of engineering materials* (Part 1, p. 1). Prentice-Hall.

Medlicott, P.A.C., & Potter, K.D. (1986). The development of a composite flywheel for vehicle applications. In K. Brunsch, H-D. Golden, & C-M. Horkert (Eds.), *High Tech—the way into the nineties* (p. 29). Elsevier.

Related case studies

6.7 "Materials for springs"
6.11 "Safe pressure vessels"
8.3 "Multiple constraints: con-rods for high-performance engines"

6.7 MATERIALS FOR SPRINGS

Springs come in many shapes (Figure 6.11 and Table 6.11) and have many purposes: think of tensile springs (a rubber band, for example), leaf springs, helical springs, spiral springs, torsion bars. Regardless of their shape or use, the best material for a spring of minimum volume is that with the greatest value of σ_f^2/E, and for minimum weight it is that with the greatest value of $\sigma_f^2/\rho E$ (derived in Equation (6.21)). We use springs as a way of introducing two of the most useful of the charts: Young's modulus E plotted against strength σ_f, and specific modulus E/ρ plotted against specific strength σ_f/ρ (Figures 4.5 and 4.6).

The translation The primary function of a spring is to store elastic energy and—when required—release it again. The elastic energy stored per unit volume in a block of material stressed uniformly to a stress σ is

$$W_v = \frac{1}{2}\frac{\sigma^2}{E} \qquad (6.21)$$

where E is Young's modulus. We wish to maximize W_v. The spring will be damaged if the stress σ exceeds the yield stress or failure stress σ_f; the constraint is $\sigma < \sigma_f$. Thus the maximum energy density is

$$W_v = \frac{1}{2}\frac{\sigma_f^2}{E} \qquad (6.22)$$

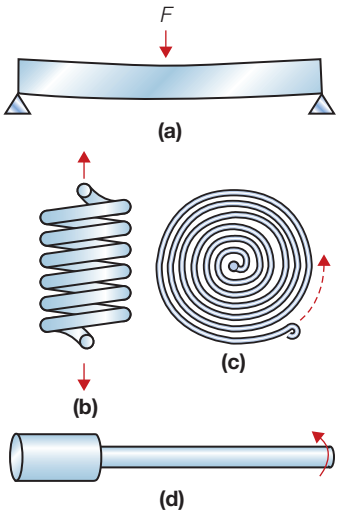


FIGURE 6.11
Springs store energy. The best material for any spring, regardless of its shape or the way in which it is loaded, is that with the highest value of σ_f^2/E or, if weight is important, $\sigma_f^2/\rho E$.

Table 6.11 Design Requirements for Springs

Function	Elastic spring
Constraint	No failure, meaning $\sigma < \sigma_f$ throughout the spring
Objective	Maximum stored elastic energy per unit volume, or maximum stored elastic energy per unit weight
Free variable	Choice of material

Torsion bars and leaf springs are less efficient than axial springs because much of the material is not fully loaded: The material at the neutral axis, for instance, is not loaded at all. For leaf springs

$$W_v = \frac{1}{4}\frac{\sigma_f^2}{E}$$

and for torsion bars

$$W_v = \frac{1}{3}\frac{\sigma_f^2}{E}$$

But—as these results show—this has no influence on the choice of material. The best stuff for a spring regardless of its shape is that with the biggest value of

$$M_1 = \frac{\sigma_f^2}{E} \tag{6.23}$$

If weight, rather than volume, matters, we must divide this by the density ρ (giving energy stored per unit weight), and seek materials with high values of

$$M_2 = \frac{\sigma_f^2}{\rho E} \tag{6.24}$$

The selection The choice of materials for springs of minimum volume is shown in Figure 6.12(a). A family of lines of slope 2 links materials with equal values of $M_1 = \sigma_f^2/E$; those with the highest values of M_1 lie toward the lower right. The heavy line is one of the family; it is positioned at 2.5 MJ/m^3 so that a subset of materials is left exposed. The best choice is a high-strength steel lying near the top end of the line. Other materials are suggested too: CFRP (now used for truck springs); titanium alloys (good but expensive); nylon, PA (children's toys often have nylon springs); and, of course, elastomers. Note how the procedure has identified a candidate from almost every class of materials: metals, polymers, elastomers, and composites. They are listed, with comments, in Table 6.12(a).

Materials selection for light springs is shown in Figure 6.12(b). A family of lines of slope 2 links materials with equal values of

$$M_2 = \left(\frac{\sigma_f}{\rho}\right)^2 \Big/ \left(\frac{E}{\rho}\right) = \frac{\sigma_f^2}{E\rho} \tag{6.25}$$

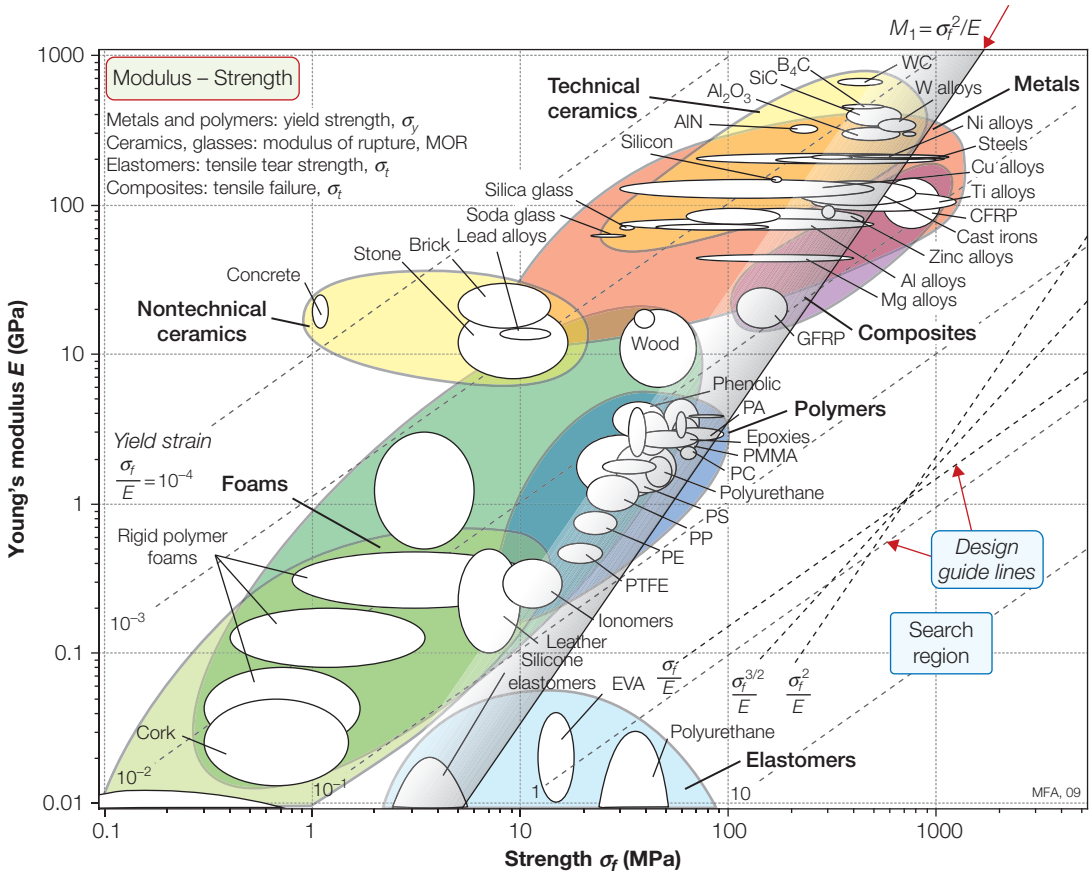


FIGURE 6.12(a)
Materials for small springs. High-strength ("spring") steel is good. Glass, CFRP, and GFRP all, under the right circumstances, make good springs. Elastomers are excellent. Ceramics are eliminated by their low tensile strength.

Table 6.12(a) Materials for Efficient Small Springs

Material	$M_1 = \sigma_f^2/E$ (MJ/m^3)	Comment
Ti alloys	4–12	Expensive, corrosion-resistant
CFRP	6–10	Comparable in performance with steel; expensive
Spring steel	3–7	The traditional choice: easily formed and heat treated
Nylon	1.5–2.5	Inexpensive and easily shaped, but high loss factor
Rubber	20–50	Better than spring steel, but high loss factor

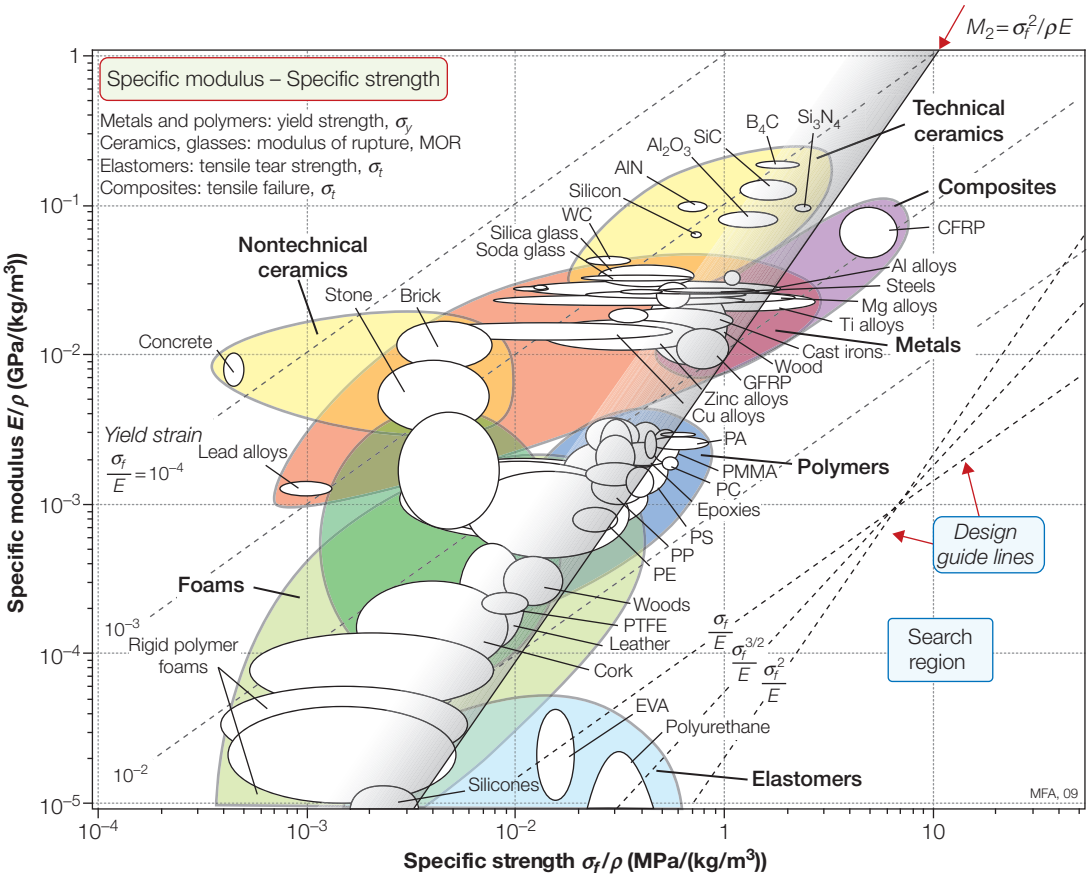


FIGURE 6.12(b)

Materials for light springs. Metals are disadvantaged by their high densities. Composites are good; so is wood. Elastomers are excellent.

One is shown at the value M_2 = 1 kJ/kg. Metals, because of their high density, are less good than composites, and much less good than elastomers. (You can store roughly eight times more elastic energy, per unit weight, in a rubber band than in the best spring steel.) Candidates are listed in Table 6.12(b). Wood—the traditional material for archery bows—now appears.

Postscript Many additional considerations enter the choice of a material for a spring. Springs for vehicle suspensions must resist fatigue and corrosion; engine valve springs must cope with elevated temperatures. A subtler property is the loss coefficient, shown in Figure 4.9. Polymers have a relatively high loss factor and dissipate energy when they vibrate; metals, if strongly hardened, do not. Polymers, because they creep, are unsuitable for springs that carry a steady load for long periods of time, though they are

Table 6.12(b) Materials for Efficient Light Springs

Material	$M_1 = \sigma_f^2/\rho E$ (kJ/kg)	Comment
Ti alloys	0.9–2.6	Better than steel; corrosion-resistant; expensive
CFRP	3.9–6.5	Better than steel; expensive
GFRP	1.0–1.8	Better than spring steel; less expensive than CFRP
Spring steel	0.4–0.9	Poor, because of high density
Wood	0.3–0.7	On a weight basis, wood makes good springs
Nylon	1.3–2.1	As good as steel, but with a high loss factor
Rubber	18–45	Outstanding; 20 times better than spring steel; but with high loss factor

still perfectly good for catches and locating springs that spend most of their time unstressed.

Related reading

Boiton, R.G. (1963). The mechanics of instrumentation. *Proc. I. Mech. E., 177*(10), 269–288.
Hayes, M. (1990). Materials update 2: springs. *Engineering*, May, p. 42.

Related case studies

6.8 "Elastic hinges and couplings"
10.7 "Shapes that flex: Leaf and strand structures"
10.8 "Ultra-efficient springs"
12.7 "Connectors that don't relax their grip"

6.8 ELASTIC HINGES AND COUPLINGS

Nature makes much use of elastic (or "natural") hinges: Skin, muscle, and cartilage all allow large, recoverable deflections. Humans, too, design with *flexure* and *torsion hinges*: ligaments that connect or transmit a load between components while allowing for limited relative movement between them by deflecting elastically (see Figure 6.13 and Table 6.13). Which materials make good hinges?

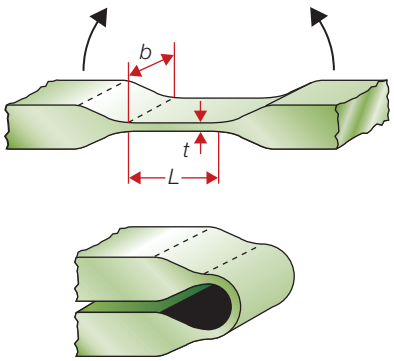


FIGURE 6.13
Elastic or "natural" hinges. The ligaments must bend repeatedly without failing. The cap of a shampoo bottle is an example; elastic hinges are used in high-performance applications too, and are found widely in nature.

The translation Consider the hinge for the lid of a box. The box, lid, and hinge are to be molded as a single unit. The hinge is a thin ligament that flexes elastically as the box is closed, as shown in the figure, but it carries no significant axial

Table 6.13 Design Requirements for Elastic Hinges	
Function	Elastic hinge
Constraint	No failure, meaning $\sigma < \sigma_f$ throughout the hinge
Objective	Maximize elastic flexure
Free variable	Choice of material

loads. Then the best material is the one that (for given ligament dimensions) bends to the smallest radius without yielding or failing. When a ligament of thickness t is bent elastically to a radius R, the surface strain is

$$\varepsilon = \frac{t}{2R} \tag{6.26}$$

and—since the hinge is elastic—the maximum stress is

$$\sigma = E\frac{t}{2R} \tag{6.27}$$

This must not exceed the yield or failure strength σ_f. Thus the minimum radius to which the ligament can be bent without damage is

$$R \geq \frac{t}{2}\left[\frac{E}{\sigma_f}\right] \tag{6.28}$$

The best material is the one that can be bent to the smallest radius, that is, the one with the greatest value of the index

$$M = \frac{\sigma_f}{E} \tag{6.29}$$

The selection We need the $\sigma_f - E$ chart again (Figure 6.14). Candidates are identified by using the guide line of slope 1; a line is shown at the position $M = \sigma_f/E = 2 \times 10^{-2}$. The best choices for the hinge lie to the right of this line: They are all polymeric materials. The shortlist (Table 6.14) includes polyethylene, polypropylene, nylon, and, best of all, elastomers, though these may be too flexible for the body of the box itself. Cheap products with this sort of elastic hinge are generally molded from polyethylene, polypropylene, or nylon. Spring steel and other metallic spring materials (like phosphor bronze) are possibilities: They combine usable σ_f/E with high E, giving flexibility with good positional stability (as in the suspensions of relays). The table gives further details.

Postscript Polymers give more design freedom than metals. The elastic hinge is one example of this, reducing the box, hinge, and lid (three components plus the fasteners needed to join them) to a single box-hinge-lid unit,

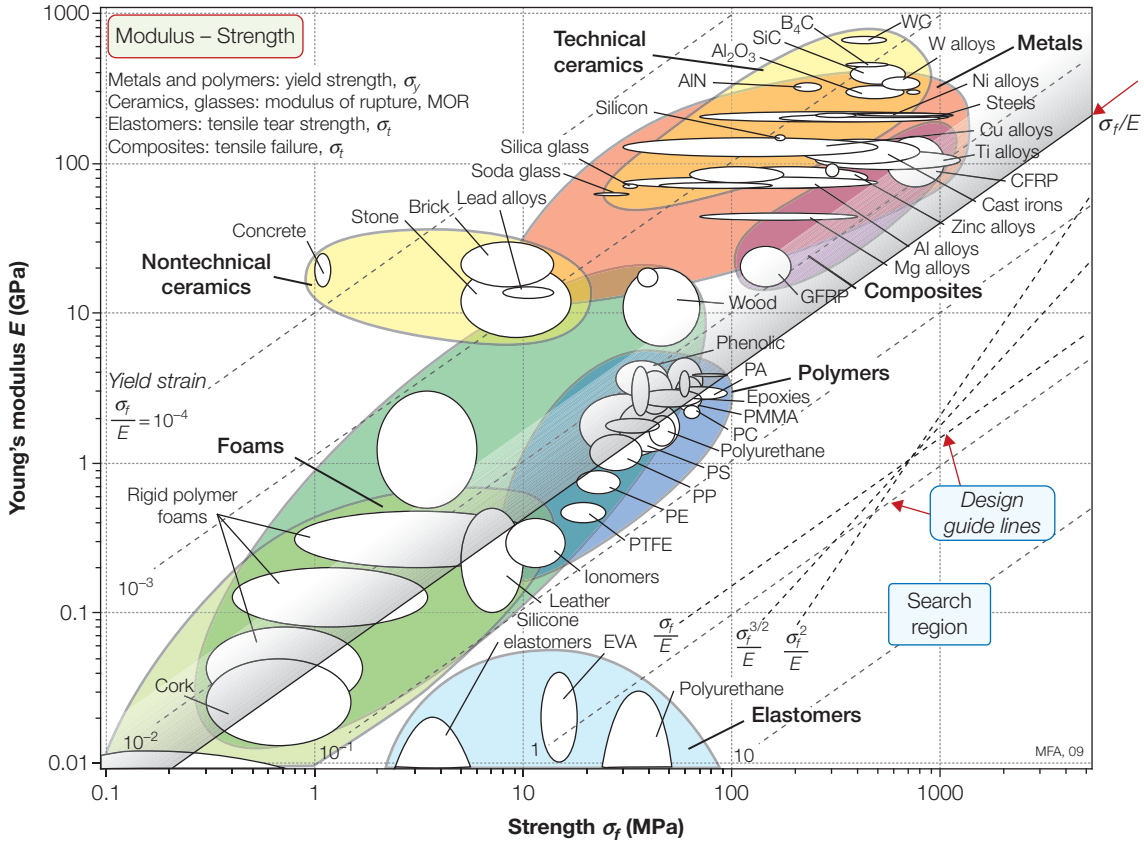


FIGURE 6.14
Materials for elastic hinges. Elastomers are best, but may not be rigid enough to meet other design needs. Polymers such as nylon, PTFE, and PE are better. Spring steel is less good, but much stronger.

Table 6.14 Materials for Elastic Hinges

Material	M ($\times 10^{-3}$)	Comment
Polyethylene	32	Widely used for cheap hinged bottle caps, etc.
Polypropylene	30	Stiffer than polyethylene; easily molded
Nylon	30	Stiffer than polyethylene; easily molded
PTFE	35	Very durable; more expensive than PE, PP, etc.
Elastomers	100–1000	Outstanding, but low modulus
High strength copper alloys	4	M less good than polymers; use when high tensile stiffness is required
Spring steel	6	

molded in one operation. Their spring-like properties allow snap-together, easily joined parts. Another example is the elastomeric coupling—a flexible universal joint, allowing high angular, parallel, and axial flexibility with good shock absorption characteristics. Elastomeric hinges offer many opportunities to be exploited in engineering design.

Related case studies

6.7 "Materials for springs"
6.9 "Materials for seals"
6.10 "Deflection-limited design with brittle polymers"
10.7 "Shapes that flex: Leaf and strand structures"

6.9 MATERIALS FOR SEALS

A reusable elastic seal consists of a cylinder of material compressed between two flat surfaces (Figure 6.15). The seal must form the largest possible contact width, b, while keeping the contact stress, σ, sufficiently low that it does not damage the flat surfaces; and the seal itself must remain elastic so that it can be reused many times. What materials make good seals? Elastomers—everyone knows that. But let us do the job properly; there may be more to be learned. We build the selection around the requirements of Table 6.15.

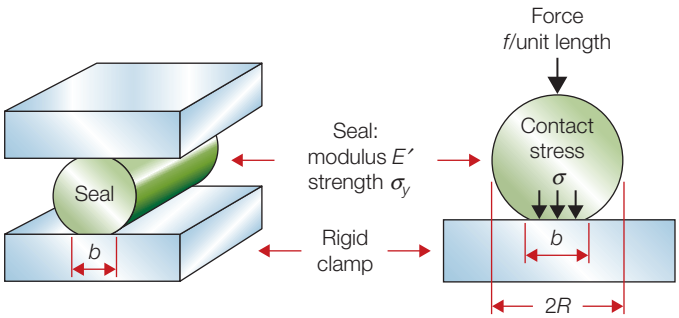


FIGURE 6.15
An elastic seal. A good seal gives a large conforming contact area without imposing damaging loads on itself or on the surfaces with which it mates.

Table 6.15 Design Requirements for Elastic Seals

Function	Elastic seal
Constraints	Limit on contact pressure Low cost
Objective	Maximum conformability to surface
Free variable	Choice of material

The translation A cylinder of diameter $2R$ and modulus E, pressed onto a rigid flat surface by a force f per unit length, forms an elastic contact of width b (Appendix B) where

$$b \approx 2.3\left(\frac{fR}{E}\right)^{1/2} \tag{6.30}$$

This is the quantity to be maximized: the objective function. The contact stress, both in the seal and in the surface, is adequately approximated (Appendix A again) by

$$\sigma = 0.57\left(\frac{fE}{R}\right)^{1/2} \tag{6.31}$$

The constraint: The seal must remain elastic—that is, σ must be less than the yield or failure strength, σ_f, of the material of which it is made. Combining the last two equations with this condition gives

$$b \leq 4.0\,R\left(\frac{\sigma_f}{E}\right) \tag{6.32}$$

The contact width is maximized by maximizing the index

$$M_1 = \frac{\sigma_f}{E}$$

It is also required that the contact stress σ be kept low to avoid damage to the flat surfaces. Its value when the maximum contact force is applied (to give the biggest width) is simply σ_f, the failure strength of the seal. Suppose the flat surfaces are damaged by a stress of greater than 100 MPa. The contact pressure is kept below this by requiring that

$$M_2 = \sigma_f \leq 100\,\text{MPa}$$

The selection The two indices are plotted on the $\sigma_f - E$ chart in Figure 6.16, isolating elastomers, foams, and cork. The candidates are listed in Table 6.16 with comments. The value of M_2 = 100 MPa admits all elastomers as candidates. If M_2 is reduced to 10 MPa, all but the most compliant elastomers are eliminated, and foamed polymers become the best bet.

Postscript The analysis highlights the functions that seals must perform: large contact area, limited contact pressure, environmental stability. Elastomers maximize the contact area; foams and cork minimize the contact pressure; PTFE and silicone rubbers best resist heat and organic solvents. The final choice depends on the conditions under which the seal will be used.

Related case studies

6.7 "Materials for springs"

6.8 "Elastic hinges and couplings"

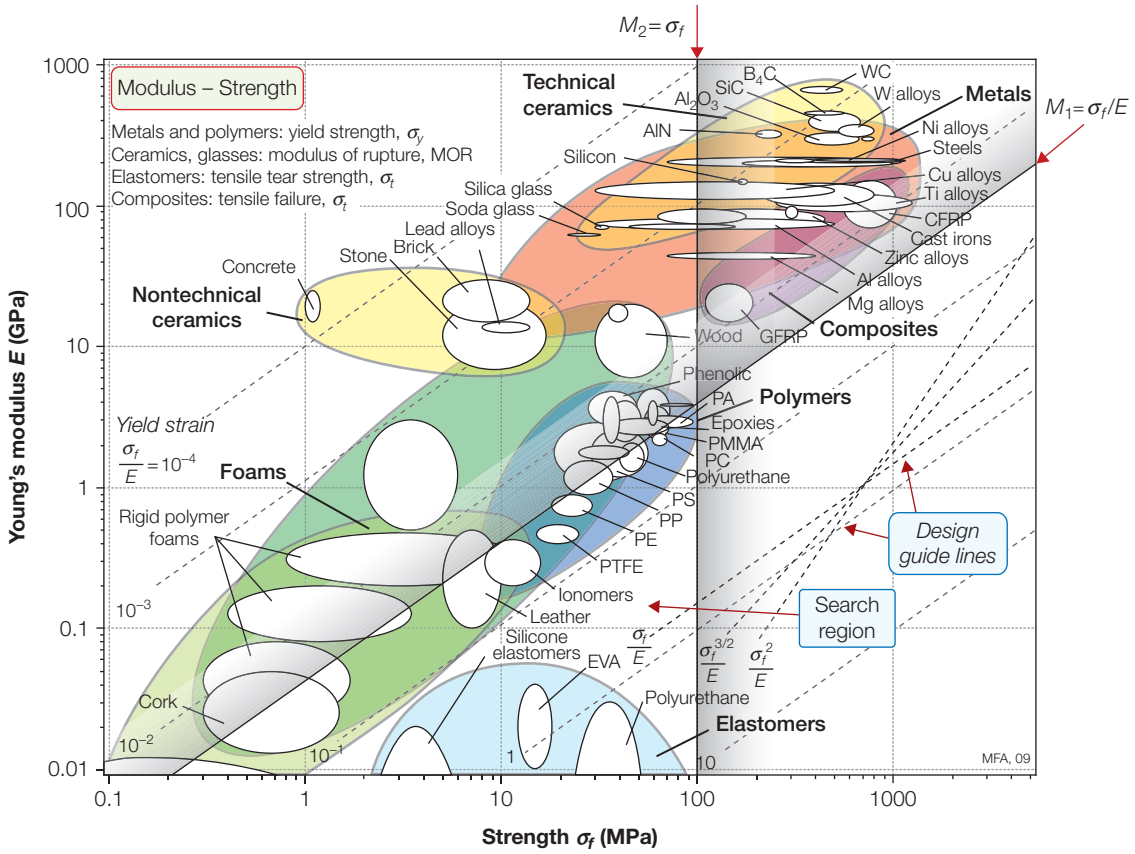


FIGURE 6.16
Materials for elastic seals. Elastomers, compliant polymers, and foams make good seals.

Table 6.16 Materials for Reusable Seals

Material	$M_1 = \frac{\sigma_f}{E}$	Comment
Elastomeric EVA	0.7–1	The natural choice; poor resistance to heat and to some solvents
Polyurethanes	2–5	Widely used for seals
Silicone rubbers	0.2–0.5	Higher temperature capability than carbon-chain elastomers, chemically inert
PTFE	0.05–0.1	Expensive but chemically stable and with high temperature capability
Polyethylenes	0.02–0.05	Inexpensive but liable to take a permanent set
Polypropylenes	0.2–0.04	Inexpensive but liable to take a permanent set
Nylons	0.02–0.03	Near upper limit on contact pressure
Cork	0.03–0.06	Low contact stress, chemically stable
Polymer foams	Up to 0.03	Very low contact pressure; delicate seals

6.10 DEFLECTION-LIMITED DESIGN WITH BRITTLE POLYMERS

The resistance of a material to the propagation of a crack is measured by its plane-strain fracture toughnesses K_{1c}. Among mechanical engineers there is a rule of thumb: avoid materials with $K_{1c} < 15$ MPa.m$^{1/2}$. Almost all metals pass: They have values of K_{1c} in the range of 20 to 100 in these units. White cast iron and some powder-metallurgy products fail; they have values as low as 10 MPa.m$^{1/2}$. Ordinary engineering ceramics have values in the range 1 to 6 MPa.m$^{1/2}$; mechanical engineers view them with deep suspicion. But engineering polymers are even less tough, with K_{1c} in the range 0.5 to 3 MPa.m$^{1/2}$ and yet engineers use them all the time. What is going on here?

When a brittle material is deformed, it deflects elastically until it fractures. The stress at which this happens is

$$\sigma_f = \frac{CK_{1c}}{\sqrt{\pi a_c}} \tag{6.33}$$

where K_c is an appropriate fracture toughness, a_c is the length of the largest crack contained in the material, and C is a constant that depends on geometry but is usually about 1. In a *load-limited* design—a tension member of a bridge, say—the part will fail in a brittle way if the stress exceeds that given by Equation (6.33). Here, obviously, we want materials with high values of K_{1c}.

But not all designs are load-limited; some are *energy-limited*, others are *deflection-limited*. Then the criterion for selection changes. Consider, then, the three scenarios created by the three alternative constraints of Table 6.17.

The translation In load-limited design the component must carry a specified load or pressure without fracturing. It is usual to identify K_c with the plane-strain fracture toughness, K_{1c}, corresponding to the most highly constrained cracking conditions, because this is conservative. Then, as Equation (6.33) shows, the best materials for minimum volume design are those with high values of

$$M_1 = K_{1c} \tag{6.34}$$

Table 6.17 Design Requirements

Function	Resist brittle fracture
Constraints	Design load specified or Design energy specified or Design deflection specified
Objective	Minimize volume (mass, cost)
Free variable	Choice of material

For load-limited design using thin sheet, a plane-stress fracture toughness may be more appropriate; and for multilayer materials, it may be an interface fracture toughness that matters. The point, though, is clear enough: The best materials for load-limited design are those with large values of the appropriate K_c.

But, as we have said, not all design is load-limited. Springs and containment systems for turbines and flywheels are *energy*-limited. Take the spring (refer to Figure 6.11) as an example. The elastic energy per unit volume stored in it is the integral over the volume of

$$U_e = \frac{1}{2}\sigma\varepsilon = \frac{1}{2}\frac{\sigma^2}{E}$$

The stress is limited by the fracture stress of Equation (6.33) so that—if "failure" means "fracture"—the maximum energy the spring can store is

$$U_e^{max} = \frac{C^2}{2\pi a_c}\left(\frac{K_{1c}^2}{E}\right)$$

For a given initial flaw size, energy is maximized by choosing materials with large values of

$$M_2 = \frac{K_{1c}^2}{E} \approx J_c \tag{6.35}$$

where J_c is the toughness (usual units: kJ/m^2).

There is a third scenario: that of *displacement*-limited design (Figure 6.17). Snap-on bottle tops, snap-together fasteners and the like, are displacement-limited: They must allow sufficient elastic displacement to permit the snap action without failure, requiring a large failure strain ε_f. The strain is related to the stress by Hooke's law $\varepsilon = \sigma/E$, and the stress is limited by the fracture equation (Eq. 6.33). Thus the failure strain is

$$\varepsilon_f = \frac{C}{\sqrt{\pi a_c}}\frac{K_{1c}}{E}$$

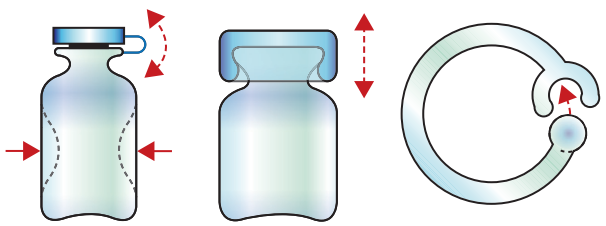


FIGURE 6.17
Load- and deflection-limited design. Polymers, having low moduli, frequently require deflection-limited design methods.

The best materials for displacement-limited design are those with large values of

$$M_3 = \frac{K_{1c}}{E}$$

The selection Figure 6.18 shows the chart of fracture toughness, K_{1c}, plotted against modulus, E. It allows materials to be compared by values of fracture toughness, M_1, by toughness, M_2, and by values of the deflection-limited index M_3. As the engineer's rule of thumb demands, almost all metals have values of K_{1c} that lie above the 15-MPa.m$^{1/2}$ acceptance level for load-limited design, shown as a horizontal selection line in the figure. Polymers and ceramics do not.

The line showing M_2 in Figure 6.18 is placed at the value 1 kJ/m^2. Materials with values of M_2 greater than this have a degree of shock resistance with which engineers feel comfortable (another rule of thumb). Metals,

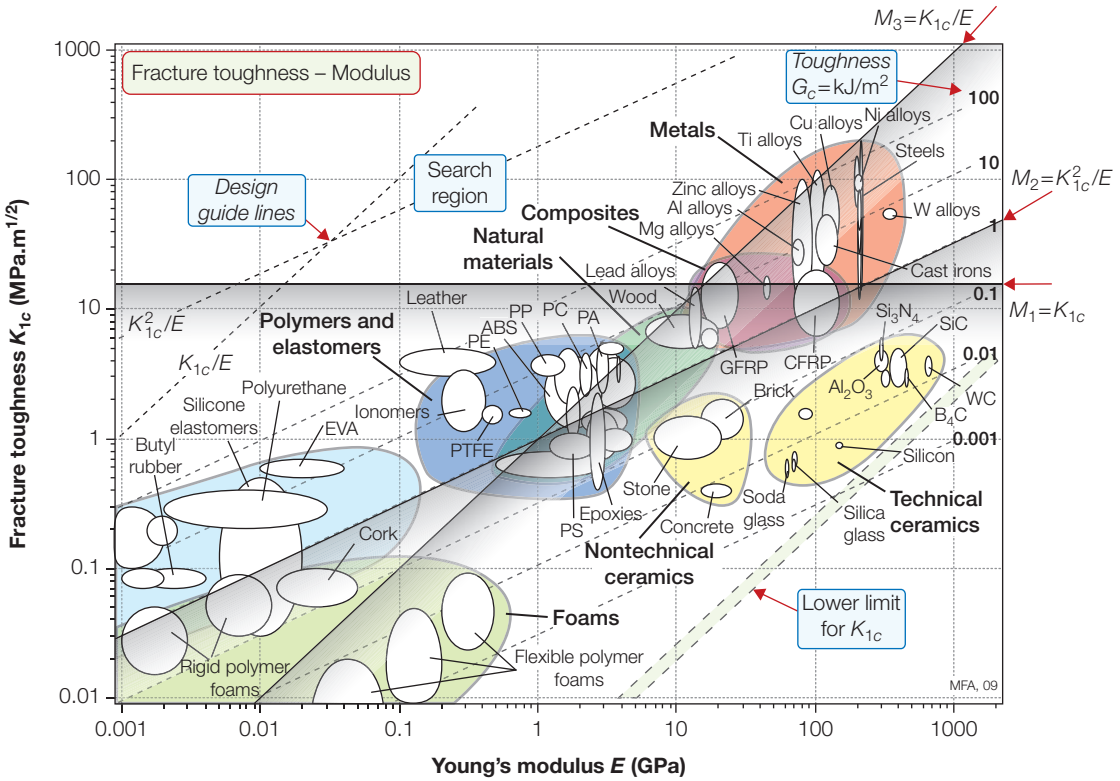


FIGURE 6.18

The selection of materials for load-, deflection-, and energy-limited design. In deflection-limited design, polymers are as good as metals, despite having very low values of fracture toughness.

Table 6.18 Materials for Fracture-Limited Design

Design type and rule of thumb	Material
Load-limited design $K_{1c} > 15$ MPa.m$^{1/2}$	Metals, polymer-matrix composites
Energy-limited design $J_c > 1$ kJ/m^2	Metals, composites, and some polymers
Displacement-limited design $K_{1c}/E > 10^{-3}$m$^{1/2}$	Polymers, elastomers, and the toughest metals

composites, and some polymers qualify; ceramics do not. When we come to deflection-limited design, the picture changes again. The line shows the index $M_3 = K_{1c}/E$ at the value 10^{-3}m$^{1/2}$. It illustrates why polymers find such wide application: When the design is deflection-limited, polymers—particularly polypropylene, ABS, and nylons—are better than the best metals (Table 6.18).

Postscript The figure gives further insights. The mechanical engineers' love of metals (and, more recently, of composites) is inspired not merely by the appeal of their K_{1c} values. They are good by all three criteria (K_{1c}, K_{1c}^2/E, and K_{1c}/E). Polymers have good values of K_{1c}/E and are acceptable by K_{1c}^2/E. Ceramics are poor by all three criteria. Herein lie the deeper roots of the engineers' distrust of ceramics.

Related reading

Background in fracture mechanics and safety criteria can be found in

Brock, D. (1984). *Elementary engineering fracture mechanics*. Martinus Nijoff.
Hellan, K. (1985). *Introduction to fracture mechanics*. McGraw-Hill.
Hertzberg, R.W. (1989). *Deformation and fracture mechanics of engineering materials*. Wiley.

Related case studies

6.7 "Materials for springs"
6.8 "Elastic hinges and couplings"
6.11 "Safe pressure vessels"

6.11 SAFE PRESSURE VESSELS

Pressure vessels, from the simplest aerosol can to the biggest boiler, are designed, for safety, to yield or leak before they break. The details of this design method vary. Small pressure vessels are usually designed to allow general yield at a pressure still too low to cause any crack the vessel may contain to propagate ("yield before break"); the distortion caused by yielding is easy to detect and the pressure can be released safely. With large pressure vessels this may not be possible. Instead, safe design is achieved by ensuring that the

Table 6.19 Design Requirements for Safe Pressure Vessels	
Function	Pressure vessel (contain pressure p safely)
Constraint	Radius R specified
Objective	Maximize safety using yield-before-break criterion or leak-before-break criterion
Free variable	Choice of material

smallest crack that will propagate unstably has a length greater than the thickness of the vessel wall ("leak before break"). The leak is easily detected, and it releases pressure gradually and thus safely (Table 6.19). The two criteria lead to different material indices. What are they?

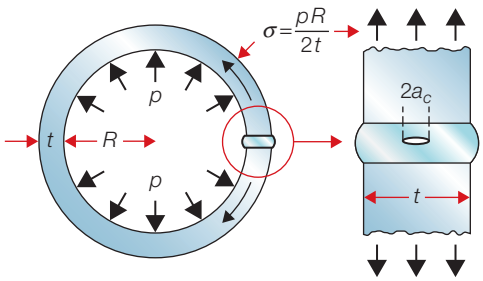


FIGURE 6.19
A pressure vessel containing a flaw. Safe design of small pressure vessels requires that they yield before they break; that of large pressure vessels may require, instead, that they leak before they break.

The translation The stress in the wall of a thin-walled spherical pressure vessel of radius R (Figure 6.19) is

$$\sigma = \frac{pR}{2t} \tag{6.36}$$

In pressure vessel design, the wall thickness, t, is chosen so that, at the working pressure p, this stress is less than the yield strength σ_f of the wall (with a safety factor, of course). A small pressure vessel can be examined ultrasonically, or by X-ray methods, or it can be proof-tested, to establish that it contains no crack or flaw of diameter greater than $2a_c^*$; then the stress required to make the crack propagate[2] is

$$\sigma = \frac{CK_{1c}}{\sqrt{\pi a_c^*}} \tag{6.37}$$

where C is a constant near unity and K_{1c} is the plane-strain fracture toughness. Safety can be achieved by ensuring that the working stress is less than this, giving

$$p \le \frac{2t}{R}\frac{K_{1c}}{\sqrt{\pi a_c^*}} \tag{6.38}$$

The largest pressure (for a given R, t, and a_c^*) is carried by the material with the greatest value of

$$M_1 = K_{1c} \tag{6.39}$$

[2] If the wall is sufficiently thin, and close to general yield, it will fail in a plane-stress mode. Then the relevant fracture toughness is that for plane stress, not the smaller value for plane strain.

But this design is not failsafe. If the inspection is faulty or if, for some other reason a crack of length greater than a_c^* appears, catastrophe follows. Greater security is obtained by requiring that the crack will not propagate even if the stress reaches the general yield stress—for then the vessel will deform stably in a way that can be detected. This condition is expressed by setting σ equal to the yield stress σ_y, giving

$$\pi a_c \leq C^2 \left[\frac{K_{1c}}{\sigma_y}\right]^2$$

The tolerable crack size, and thus the integrity of the vessel, is maximized by choosing a material with the largest value of

$$M_2 = \frac{K_{1c}}{\sigma_y} \tag{6.40}$$

Large pressure vessels cannot always be X-rayed or sonically tested; and proof testing them may be impractical. Further, cracks can grow slowly because of corrosion or cyclic loading, so a single examination at the beginning of service life is not sufficient. Then safety can be ensured by arranging that a crack just large enough to penetrate both the inner and the outer surface of the vessel is still stable, because the leak caused by the crack can be detected. This is achieved by setting $a_c^* = t/2$. Safety is ensured if the stress is always less than or equal to

$$\sigma = \frac{CK_{1c}}{\sqrt{\pi t/2}} \tag{6.41}$$

The wall thickness t of the pressure vessel was, of course, designed to contain the pressure p without yielding. From Equation (6.38), this means that

$$t \geq \frac{pR}{2\sigma_y} \tag{6.42}$$

Substituting this into the previous equation (with $\sigma = \sigma_f$) gives

$$p \leq \frac{4\,C^2}{\pi\,R}\left(\frac{K_{1c}^2}{\sigma_y}\right) \tag{6.43}$$

The maximum pressure is carried most safely by the material with the greatest value of

$$M_3 = \frac{K_{1c}^2}{\sigma_y} \tag{6.44}$$

Both M_2 and M_3 could be made large by making the yield strength of the wall, σ_y, very small: Lead, for instance, has high values of both, but you would not choose it for a pressure vessel. That is because the vessel wall must also be kept thin, both for economy of material and to keep it light. The thinnest wall, from Equation (6.42), is that with the largest yield strength, σ_y. Thus we wish also to maximize

$$M_4 = \sigma_y \tag{6.45}$$

further narrowing the choice of material.

The selection These selection criteria are explored using the chart shown in Figure 6.20: the fracture toughness, K_{1c}, plotted against the elastic limit σ_f (meaning yield strength for metals and polymers). The indices M_1, M_2,

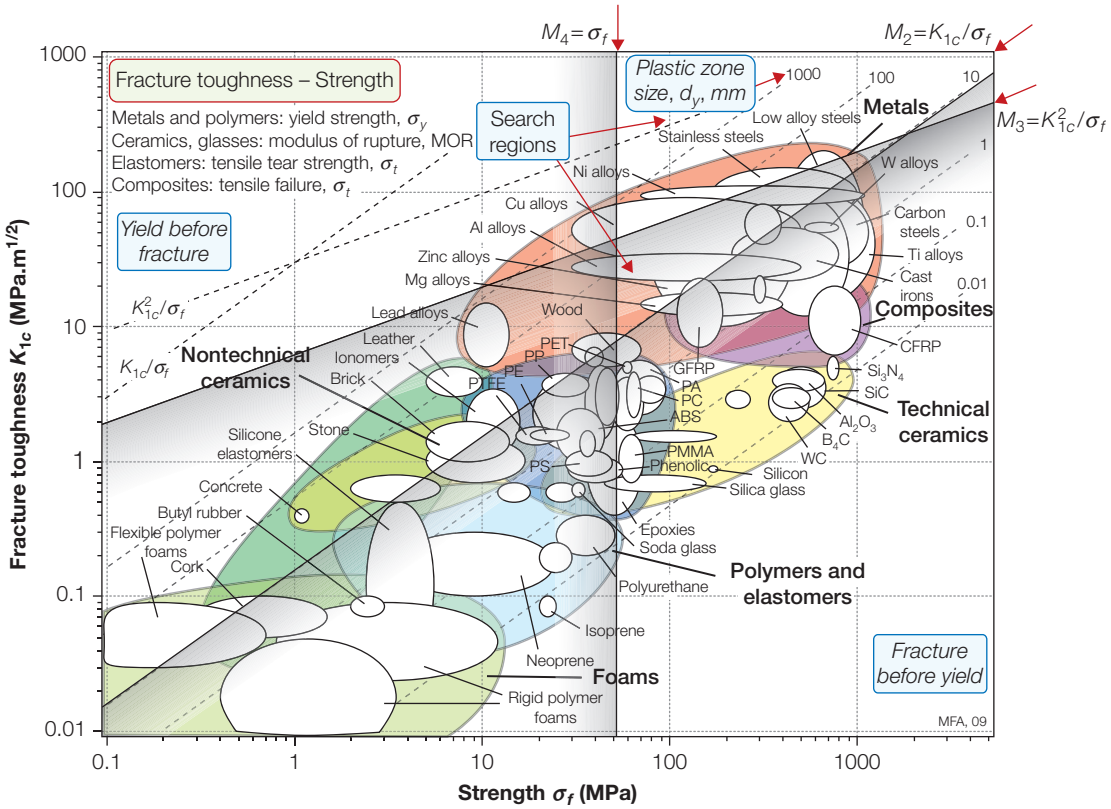


FIGURE 6.20

Materials for pressure vessels. Steel, copper alloys, and aluminum alloys best satisfy the "yield before break" criterion. In addition, a high yield strength allows a high working pressure. The materials in the "search regions" triangle are the best choice. The leak-before-break criterion leads to essentially the same selection.

Table 6.20 Materials for Safe Pressure Vessels

Material	$M_1 = K_{1c}/\sigma_y$ ($m^{1/2}$)	$M_3 = \sigma_y$ (MPa)	Comment
Stainless steels	0.35	300	Nuclear pressure vessels are made of grade 316 stainless steel
Low alloy steels	0.2	800	These are standard in this application
Copper	0.5	200	Hard-drawn copper is used for small boilers and pressure vessels
Aluminum alloys	0.15	200	Pressure tanks of rockets are aluminum
Titanium alloys	0.13	800	Good for light pressure vessels, but expensive

M_3 and M_4 appear as lines of slope 0, 1, and 1/2, and as lines that are vertical. Take "yield before break" as an example. A diagonal line corresponding to a constant value of $M_2 = K_{1c}/\sigma_y$ links materials with equal performance; those above the line are better. The line shown in the figure at a value of M_1, corresponding to a process zone of size 10 mm, excludes everything but the toughest steels, copper, aluminum, and titanium alloys, although some polymers—PP, PE, and PET for instance—nearly make it (pressurized lemonade and beer containers are made of these polymers). Details are given in Table 6.20.

The leak-before-break criterion

$$M_3 = \frac{K_{1c}^2}{\sigma_y}$$

favors low alloy steel and stainless and carbon steels more strongly. Polymers no longer qualify as candidates.

Postscript Large pressure vessels are always made of steel. The ones for models—a model steam engine, for instance—are made of copper. Copper is chosen even though it is more expensive, because of its greater resistance to corrosion. Corrosion rates do not scale with size. The loss of 0.1 mm through corrosion is not serious in a pressure vessel that is 10 mm thick; if it is only 1 mm thick, it becomes a concern.

Boiler failures used to be commonplace—there are even songs about it. Now they are rare, although when safety margins are pared to a minimum (rockets, new aircraft designs) pressure vessels still occasionally fail. This (relative) success is one of the major contributions of fracture mechanics to engineering practice.

Related reading

Background in fracture mechanics and safety criteria can be found in

Brock, D. (1984). *Elementary engineering fracture mechanics*. Martinus Nijoff.
Hellan, K. (1985). *Introduction to fracture mechanics*. McGraw-Hill.
Hertzberg, R.W. (1989). *Deformation and fracture mechanics of engineering materials*. Wiley.

Related case studies

6.6 "Materials for flywheels"
6.10 "Deflection-limited design with brittle polymers"
8.2 "Multiple constraints: Light pressure vessels"

6.12 STIFF, HIGH-DAMPING MATERIALS FOR SHAKER TABLES

Shakers are the members of an obscure and declining New England religious sect, noted for their austere wooden furniture. To those who live elsewhere they are devices for vibration testing (Figure 6.21). This second sort of shaker consists of an electromagnetic actuator driving a table, at frequencies up to 1000 Hz, to which the test object (a space probe, an automobile, an aircraft component, or the like) is clamped. The shaker applies a spectrum of vibration frequencies, f, and amplitudes, A, to the test object to explore its response.

A big table operating at high frequency dissipates a great deal of power. The primary objective is to minimize this, but subject to a number of constraints itemized in Table 6.21. What materials make good shaker tables?

The translation The power p (watts) consumed by a dissipative vibrating system with a sinusoidal input is

$$p = C_1 m A^2 \omega^3 \qquad (6.46)$$

where m is the mass of the table, A is the amplitude of vibration, ω is the frequency (rad/s) and C_1 is a constant. Provided the operating frequency ω is significantly less than the resonant frequency of the table, $C_1 \approx 1$. The amplitude A and the frequency ω are prescribed. To minimize the power lost in shaking the table itself, we must minimize its mass m. We idealize the table as a disk of given radius, R. Its thickness, t, is a free variable. Its mass is shown in Equation (6.47).

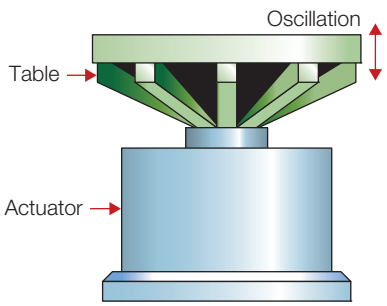


FIGURE 6.21
A shaker table. It is required to be stiff, but have high intrinsic "damping" or loss coefficient.

Table 6.21 Design Requirements for Shaker Tables

Function	Table for vibration tester ("shaker table")
Constraints	Radius, R, specified Must be stiff enough to avoid distortion by clamping forces Natural frequencies above maximum operating frequency (to avoid resonance) High damping to suppress resonance and natural vibrations Tough enough to withstand mishandling and shock
Objective	Minimize power consumption
Free variables	Choice of material Table thickness, t

$$m = \pi R^2 t \rho \tag{6.47}$$

where ρ is the density of the material of which it is made. The thickness influences the bending stiffness of the table—and this is important both to prevent the table flexing too much under clamping loads and because it determines its natural vibration frequencies. The bending stiffness, S, is

$$S = \frac{C_2 E I}{R^3}$$

where C_2 is a constant. The second moment of the section, I, is proportional to $t^3 R$. Thus, for a given stiffness S and radius R,

$$t = C_3 \left(\frac{S R^2}{E} \right)^{1/3}$$

where C_3 is another constant. The thinnest table is the one made from the material with the greatest value of

$$M_1 = E$$

Inserting this expression for t into Equation (6.47), we obtain

$$m = C_3 \pi R^{8/3} S^{1/3} \left(\frac{\rho}{E^{1/3}} \right) \tag{6.48}$$

The mass of the table, for a given stiffness and a minimum vibration frequency, is therefore minimized by selecting materials with high values of

$$M_2 = \frac{E^{1/3}}{\rho}$$

There are three further requirements. The first is high mechanical damping, measured by the loss coefficient, η, to suppress resonance. The second that

the yield strength and fracture toughness, K_{1c}, of the table be sufficient to withstand mishandling and clamping forces. And it must not be too thick.

The selection If we had a chart with $E^{1/3}/\rho$ on one axis and η on the other, we could read off the materials with high values of both. Computer-based methods, illustrated in later case studies, allow charts with any desired combination of properties as axes. But for now we will stick with the charts in Chapter 4, requiring a 2-step selection. Figure 6.4 is an $E - \rho$ chart with a contour of $E^{1/3}/\rho$ already plotted on it. Materials with high values lie above or just below it. They are listed in the first column of Table 6.22. We now turn to the $\eta - E$ chart, reproduced as Figure 6.22. Materials with high values of M_1 lie to the right of the vertical line (here set at 30 GPa); those with $\eta > 0.001$ lie above the horizontal line. The search region contains a number of the candidates in the table: CFRP and alloys of magnesium, aluminum and titanium. All are possible candidates. Table 6.22 compares their properties.

Postscript Stiffness, high natural frequencies, and damping are qualities often sought in engineering applications such as engine and machine-tool mountings, supports for precision instruments, and building foundations. The shaker table found its solution (in real life as well as in this case study) in the choice of a cast magnesium alloy.

Sometimes a solution is possible by combining materials (more on this in Chapter 11). The loss coefficient chart shows that polymers and elastomers have high damping. Sheet steel panels, prone to vibration, can be damped by coating one surface with a polymer, a technique exploited in automobiles, typewriters, and machine tools. Aluminum structures can be stiffened (raising natural frequencies) by bonding carbon fiber to them: an approach sometimes use in aircraft design. And structures loaded in bending or torsion can be made lighter, for the same stiffness (again increasing

Table 6.22 Materials for Shaker Tables

Material	$M_1 = E^{1/3}/\rho$ $GPa^{1/3}/(Mg/m^3)$	Loss Coefficient, η	Fr. Toughness K_{1c} $MPa.m^{1/2}$	Comment
Mg-alloys	Up to 2.3	Up to 0.03	15	The best combination of properties
Al-alloys	Up to 1.7	Up to 0.002	30	Less damping than Mg or Ti
Ti-alloys	Up to 1.1	Up to 0.003	60	Good damping but expensive
CFRP	Up to 3.4	Up to 0.003	15	Less damping than Mg-alloys, but possible
Various ceramics	Up to 3.0	About 0.0002	3	Ruled out by low damping and toughness
Various foams	Up to 10	Up to 0.5	0.1	Lacks strength and toughness to carry service loads

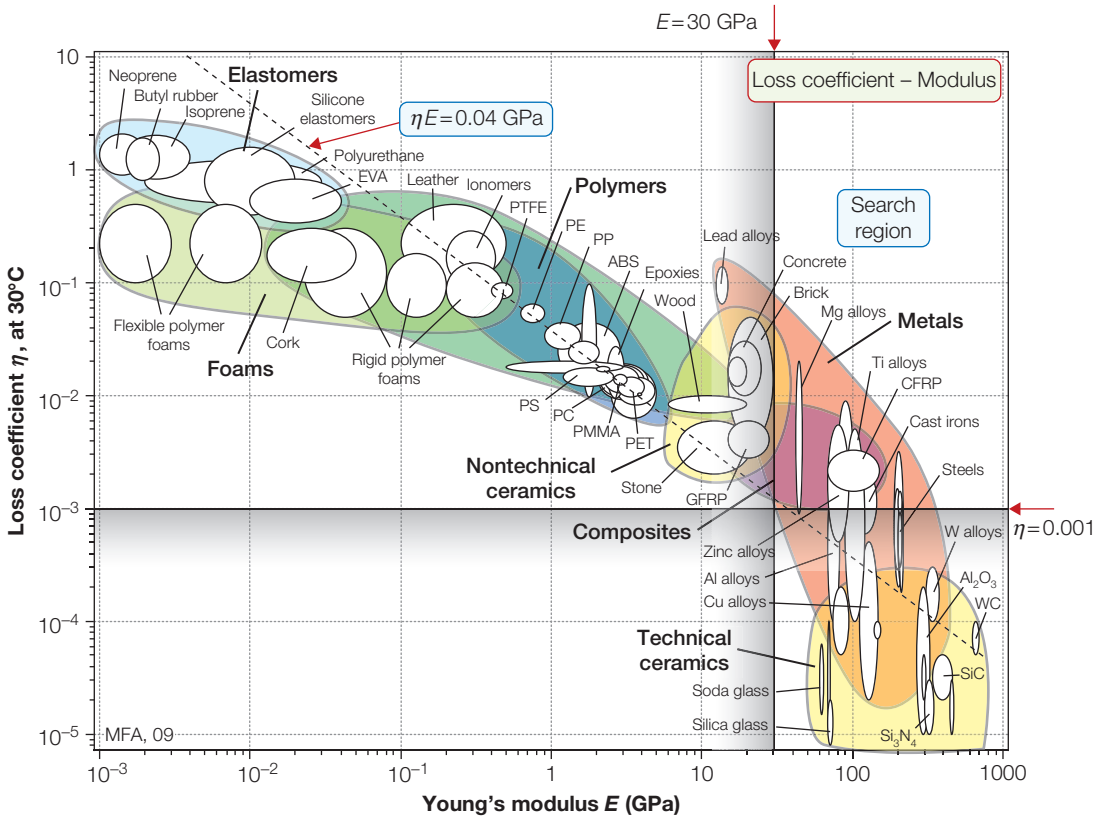


FIGURE 6.22
Selection of materials for the shaker table. Magnesium alloys, cast irons, GFRP, concrete, and the special high-damping Mn-Cu alloys are candidates.

natural frequencies), by shaping them efficiently: by attaching ribs to their underside, for instance. Shaker tables—even the austere wooden tables of the New England Shakers—exploit shape in this way.

Related reading

Tustin, W, & Mercado, R. (1984). *Random vibrations in perspective.* Tustin Institute of Technology Inc., Santa Barbara.

Cebon, D., & Ashby, M.F. (1994). Materials selection for precision instruments. *Meas. Sci. and Technol., 5,* 296–306.

Related case studies

6.4 "Materials for table legs"
6.7 "Materials for springs"
6.16 "Materials to minimize thermal distortion in precision devices"

6.13 INSULATION FOR SHORT-TERM ISOTHERMAL CONTAINERS

Each member of the crew of a military aircraft carries, for emergencies, a radio beacon. If forced to eject, the crew member could find himself in trying circumstances—in water at 4°C, for example (much of the earth's surface is ocean with a mean temperature of roughly this). The beacon guides friendly rescue services, minimizing exposure time.

But microelectronic metabolisms (like those of humans) are upset by low temperatures. In the case of the beacon, its transmission frequency that starts to drift. The design specification for the egg-shaped package containing the electronics (Figure 6.23) requires that, when the temperature of the outer surface is changed by 30°C, the temperature of the inner surface should not change significantly for an hour. To keep the device small, the wall thickness is limited to a w of 20 mm. What is the best material for the package? A dewar system is out—it is too fragile.

A foam of some sort, you might think. But here is a case in which intuition leads you astray. So let us formulate the design requirements (Table 6.23) and do the job properly.

The translation We model the container as a wall of thickness w, thermal conductivity λ. The heat flux q through the wall, once a steady state has been established, is given by Fick's first law:

$$q = -\lambda \frac{dT}{dx} = \lambda \frac{(T_i - T_o)}{w} \qquad (6.49)$$

where T_o is the temperature of the outer surface, T_i is that of the inner one, and dT/dx is the temperature

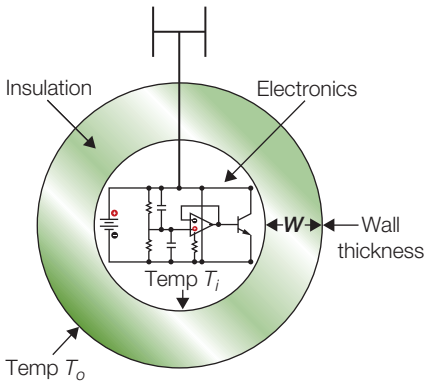


FIGURE 6.23
An isothermal container. It is designed to maximize the time before the inside temperature changes after the outside temperature has suddenly changed.

Table 6.23 Design Requirements for Short-term Insulation

Function	Short-term thermal insulation
Constraint	Wall thickness must not exceed w
Objective	Maximize time t before internal temperature changes when external temperature suddenly drops
Free variable	Choice of material

gradient (Figure 6.23). The only free variable here is the thermal conductivity, λ. The flux is minimized by choosing a wall material with the lowest possible value of λ. The $\lambda - \alpha$ chart (Figure 6.24) shows that this is, indeed, a foam.

But the incorrect question has been answered. The design brief was not to minimize the *heat flux* through the wall, but to maximize the *time* before the temperature of the inner wall changed appreciably. When the surface temperature of a body is suddenly changed, a temperature wave, so to speak, propagates inwards. The distance x it penetrates in time t is approximately $\sqrt{2at}$. Here a is the thermal diffusivity, defined by

$$a = \frac{\lambda}{\rho C_p} \tag{6.50}$$

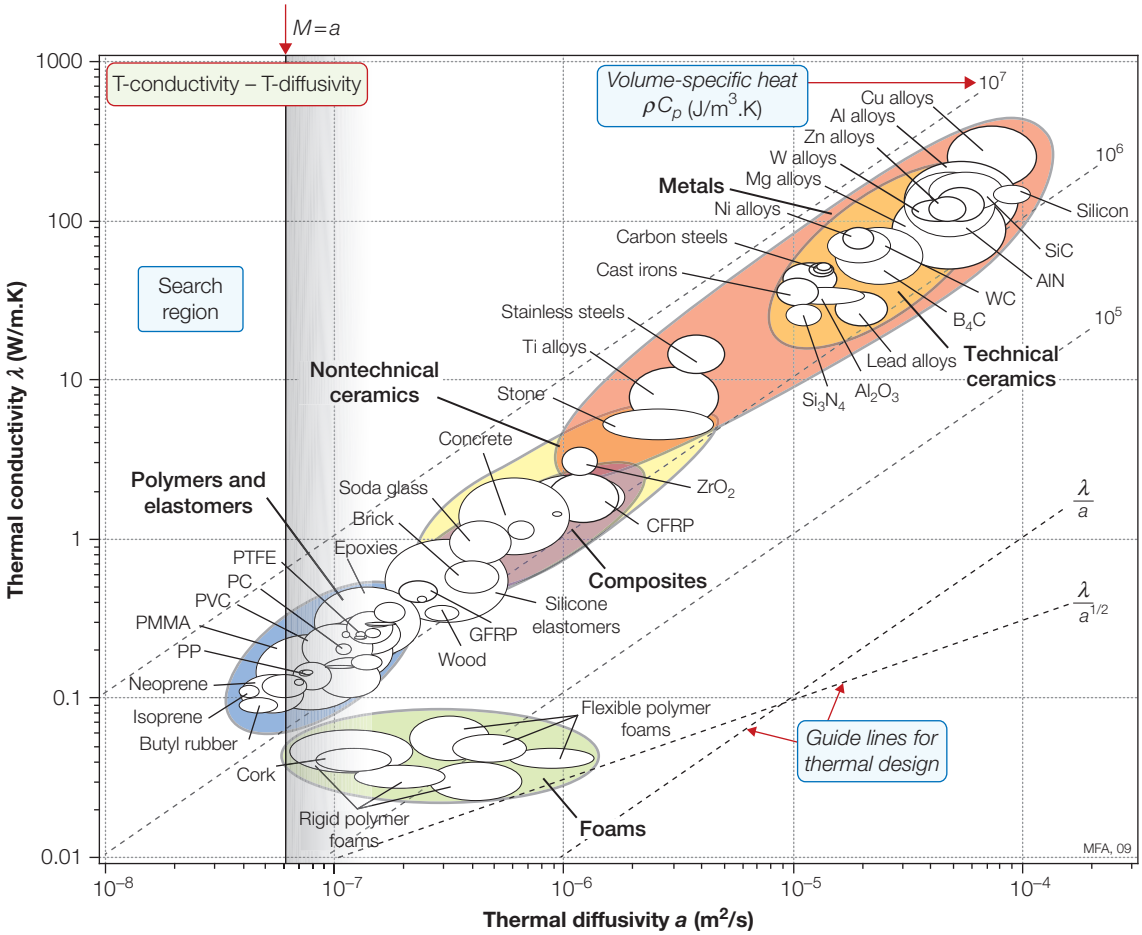


FIGURE 6.24
Materials for short-term isothermal containers. Elastomers are good; foams are not.

where ρ is the density and C_p is the specific heat (see Appendix B). Equating this to the wall thickness w gives

$$t \approx \frac{w^2}{2a} \tag{6.51}$$

The time is maximized by choosing the smallest value of the thermal diffusivity, a, not the conductivity, λ.

The selection Figure 6.24 shows that the thermal diffusivities of foams are not particularly low; it is because they have so little mass and thus heat capacity. The diffusivity of heat in a solid polymer or elastomer is much lower because these materials have specific heats that are exceptionally large. A package made of solid rubber, neoprene, or isoprene would—if of the same thickness—give the beacon a life 10 times greater than one made of (say) a polystyrene foam—though of course it would be heavier. Table 6.24 summarizes the conclusions. The reader can confirm, using Equation (6.51), that 22 mm of neoprene ($a = 5 \times 10^{-8}$ m^2/s, read from Figure 6.24) will allow a time interval of more than 1 hour after an external temperature change before the internal temperature shifts much.

Postscript One can do better than this. The trick is to exploit other ways of absorbing heat. If a liquid—a low-melting wax, for instance—can be found that solidifies at a temperature equal to the minimum desired operating temperature for the transmitter (T_i), it can be used as a "latent-heat sink." Channels in the package are filled with the liquid; the inner temperature can only fall below the desired operating temperature when all the liquid has solidified. The latent heat of solidification must be supplied to do this, giving the package a large (apparent) specific heat and thus an exceptionally low diffusivity for heat at the temperature T_i. The same idea is used, in reverse, in "freezer packs" that solidify when placed in the freezer compartment of a refrigerator and remain cold (by melting, at 4°C) when packed around warm beer cans in a portable cooler.

Table 6.24 Materials for Short-term Thermal Insulation

Material	Comment
Elastomers: Butyl rubber, neoprene and isoprene are examples	Best choice for short-term insulation
Commodity polymers: polyethylenes and polypropylenes	Less expensive than elastomers, but not quite as good for short-term insulation
Polymer foams	Much less good than elastomers for short-term insulation; best choice for long-term insulation at steady state

Related reading

Holman, J.P. (1981). *Heat transfer* (5th ed.). McGraw-Hill.

Related case studies

6.14 "Energy-efficient kiln walls"

6.15 "Materials for passive solar heating"

6.14 ENERGY-EFFICIENT KILN WALLS

The energy cost of one firing cycle of a large pottery kiln (Figure 6.25) is considerable. Part is the cost of the energy that is lost by conduction through the kiln walls; it is reduced by choosing a wall material with a low thermal conductivity, λ, and by making the wall thick. The rest is the cost of the energy used to raise the kiln to its operating temperature; it is reduced by choosing a wall material with a low heat capacity, C_p, and by making the wall thin. Is there a material index that captures these apparently conflicting design goals? And if so, what is a good choice of material for kiln walls? The choice is based on the requirements of Table 6.25.

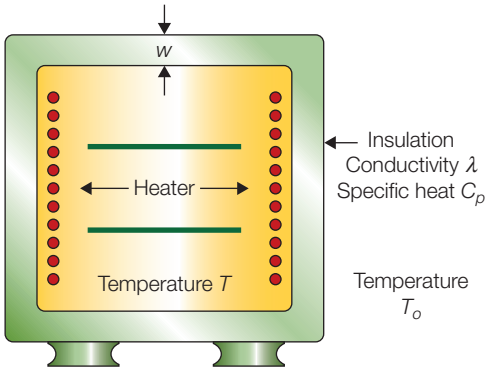


FIGURE 6.25
A kiln. On firing, the kiln wall is first heated to the operating temperature, then held at this temperature. A linear gradient is then expected through the kiln wall.

The translation When a kiln is fired, the internal temperature rises quickly from the ambient, T_o, to the operating temperature, T_i, where it is held for the firing time t. The energy consumed in the firing time has, as we have said, two contributions. The first is the heat conducted out: At steady state the heat loss by conduction, Q_1, per unit area, is given by the first law of heat flow. If held for time t, it is

$$Q_1 = -\lambda \frac{dT}{dx} t = \lambda \frac{(T_i - T_o)}{w} t \qquad (6.52)$$

Here dT/dx is the temperature gradient and w is the insulation wall thickness. The second contribution is the heat absorbed by the kiln wall in raising it to T_i, and this can be considerable. Per unit area, it is

$$Q_2 = C_p \rho w \left(\frac{T_i - T_o}{2}\right) \qquad (6.53)$$

Table 6.25 Design Requirements for Kiln Walls

Function	Thermal insulation for kiln (cyclic heating and cooling)
Constraints	Maximum operating temperature 1000°C Possible limit on kiln-wall thickness for space reasons
Objective	Minimize energy consumed in firing cycle
Free variables	Kiln wall thickness, w Choice of material

where C_p is the specific heat of the wall material and ρ is its density. The total energy consumed per unit area is the sum of these two:

$$Q = Q_1 + Q_2 = \frac{\lambda (T_i + T_o)t}{w} + \frac{C_p \rho w (T_i - T_o)}{2} \tag{6.54}$$

A wall that is too thin loses much energy by conduction, but absorbs little energy in heating the wall itself. One that is too thick does the opposite. There is an optimum thickness, which we find by differentiating Equation (6.54) with respect to wall thickness w and equating the result to zero, giving

$$w = \left(\frac{2\lambda t}{C_p \rho}\right)^{1/2} = (2at)^{1/2}, \tag{6.55}$$

where $a = \lambda/\rho C_p$ is the thermal diffusivity. The quantity $(2at)^{1/2}$ has dimensions of length and is a measure of the distance heat can diffuse in time t. Equation (6.55) says that the most energy-efficient kiln wall is one that only gets really hot on the outside as the firing cycle approaches completion. Substituting Equation (6.55) back into Equation (6.54) to eliminate w gives

$$Q = (T_i - T_o)(2t)^{1/2}(\lambda C_p \rho)^{1/2}$$

Q is minimized by choosing a material with a low value of the quantity $(\lambda C_p \rho)^{1/2}$, that is, by maximizing

$$M = (\lambda C_p \rho)^{-1/2} = \frac{a^{1/2}}{\lambda} \tag{6.56}$$

By eliminating the wall thickness w, we have lost track of it. It could, for some materials, be excessively large. Before accepting a candidate material, we must check, by evaluating Equation (6.55), how thick the wall made from it will be.

The selection Figure 6.26 shows the $\lambda - a$ chart, here with additional refractories and foams, with a selection line corresponding to $M = a^{1/2}/\lambda$ plotted on it. Polymer foams, cork, and solid polymers are good, but only if the internal temperature is less than 150°C. Real pottery kilns operate near 1000°C, requiring materials with a maximum service temperature above this value. The figure suggests brick (Table 6.26), but here the limitation of the hard-copy charts becomes apparent: There is not enough room to show specialized materials (e.g., refractory bricks and concretes). The limitation is overcome by the computer-based methods mentioned in Chapter 5, allowing a search over a much larger number of materials.

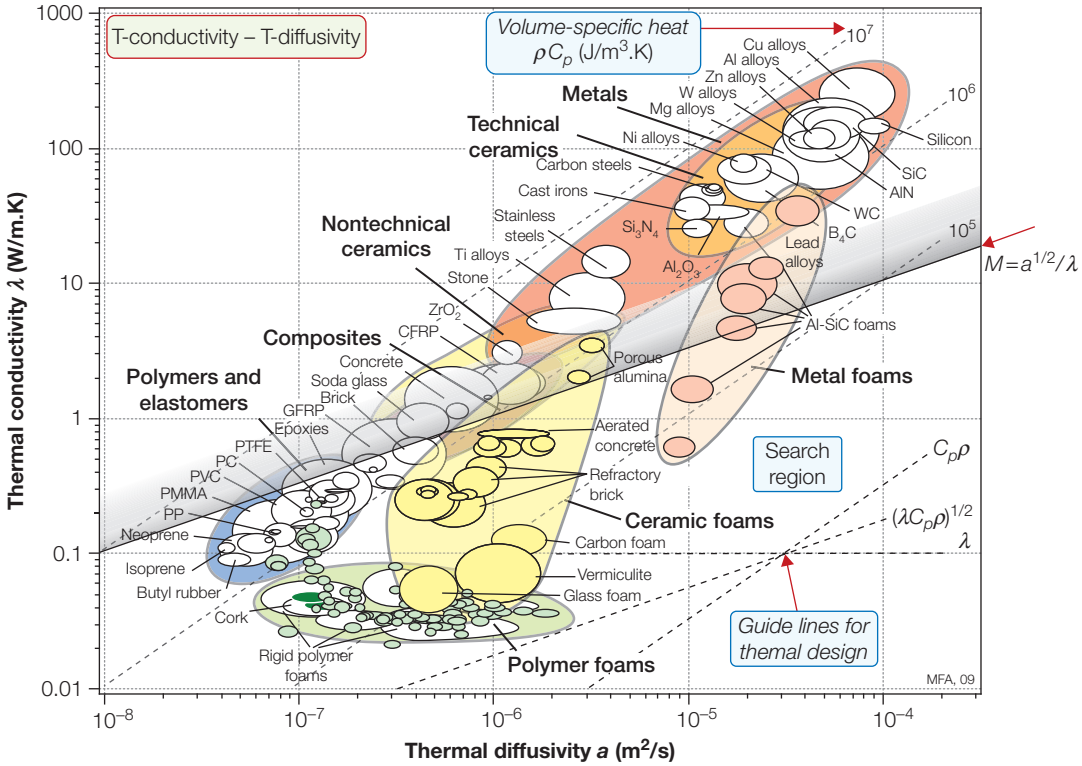


FIGURE 6.26
Materials for kiln walls. Low-density, porous, or foam-like ceramics are the best choice.

Table 6.26 Materials for Energy-efficient Kilns

Material	$M = a^{1/2}/\lambda$ (m^{2}K/W.s$^{1/2}$)	Thickness w (mm)	Comment
Refractory brick	5×10^{-3}	100	The obvious choice: The lower the density, the better the performance. Special refractory bricks have values of M as high as 3×10^{-3}.
Aerated concrete	2×10^{-3}	110	High-temperature concrete can withstand temperatures up to 1000°C.
Glass and carbon foams	Up to 10^{-2}	140	Both offer exceptional thermal insulation, but are limited to temperatures below 800°C.
Woods	2×10^{-3}	60	The boiler of Stevenson's "Rocket" steam engine was insulated with wood.
Solid elastomers and solid polymers	$2 \times 10^{-3} - 3 \times 10^{-3}$ 2×10^{-3}	50	Good values of material index. Useful if the wall must be very thin. Limited to temperatures below 150°C.
Polymer foam, cork	$3 \times 10^{-3} - 6 \times 10^{-2}$	50–140	The highest value of M—hence their use in house insulation. Limited to temperatures below 150°C.

Having chosen a material, the acceptable wall thickness is calculated from Equation (6.55). It is listed, for a firing time of three hours (approximately 10^4 seconds) in Table 6.26.

Postscript It is not generally appreciated that, in an efficiently designed kiln, as much energy goes in heating up the kiln itself as is lost by thermal conduction to the outside environment. It is a mistake to make kiln walls too thick; a little is saved in reduced conduction loss, but more is lost in the greater heat capacity of the kiln itself.

That, too, is the reason that foams are good: They have a low thermal conductivity *and* a low heat capacity. Centrally heated houses in which the heat is turned off at night suffer a cycle like that of the kiln. Here (because T_i is lower) the best choice is a polymeric foam, cork, or fiberglass (which has thermal properties like those of foams). But, as this case study shows, turning the heat off at night doesn't save you as much as you think because you have to supply the heat capacity of the walls in the morning.

Related reading

Holman, J.P. (1981). *Heat transfer* (5th ed.). McGraw-Hill, ISBN 0-07-029618-9.

Related case studies

6.13 "Insulation for short-term isothermal containers"
6.15 "Materials for passive solar heating"

6.15 MATERIALS FOR PASSIVE SOLAR HEATING

There are a number of schemes for capturing solar energy for home heating: solar cells, liquid-filled heat exchangers, and solid heat reservoirs. The simplest of these is the heat-storing wall: a thick wall, the outer surface of which is heated by exposure to direct sunshine during the day, and from which heat is extracted at night by blowing air over its inner surface (Figure 6.27). An essential of such a scheme is that the time constant for heat flow through the wall be about 12 hours; then the wall first warms on the inner surface roughly 12 hours after the sun first warms the outer surface giving out at night what it took in during the day. We will suppose that, for architectural reasons, the wall must not be more than ½ m thick. What materials maximize the thermal

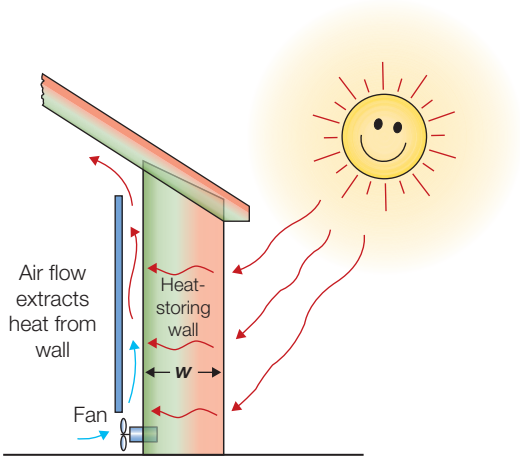


FIGURE 6.27
A heat-storing wall. The sun heats it during the day; heat is extracted from it at night. The heat-diffusion time through the wall must be about 12 hours.

Table 6.27 Design Requirements for Passive Solar Heating

Function	Heat-storing medium
Constraints	Heat diffusion time through wall $t \approx 12$ hours Wall thickness ≤ 0.5 m Adequate working temperature $T_{max} > 100°C$
Objective	Maximize thermal energy stored per unit material cost
Free variables	Wall thickness w Choice of material

energy captured by the wall while retaining a heat-diffusion time of up to 12 hours? Table 6.27 summarizes the requirements.

The translation The heat content, Q, per unit area of wall, when heated through a temperature interval ΔT gives the objective function

$$Q = w \rho C_p \Delta T \tag{6.57}$$

where w is the wall thickness, and $\rho\ C_p$ is the specific heat per unit volume (the density ρ times the specific heat C_p). The 12-hour time constant is a constraint. It is adequately estimated by the approximation used earlier for the heat-diffusion distance in time t (see Appendix B)

$$w = \sqrt{2at} \tag{6.58}$$

where a is the thermal diffusivity. Eliminating the free variable w gives

$$Q = \sqrt{2t}\Delta T a^{1/2} \rho C_p \tag{6.59}$$

or, using the fact that $a = \lambda/\rho C_p$ where λ is the thermal conductivity,

$$Q = \sqrt{2t}\Delta T \left(\frac{\lambda}{a^{1/2}}\right)$$

The heat capacity of the wall is maximized by choosing material with a high value of

$$M = \frac{\lambda}{a^{1/2}} \tag{6.60}$$

This is the reciprocal of the index of the previous case study. The restriction on thickness w requires (from Equation (6.58)) that

$$a \leq \frac{w^2}{2t}$$

with $w \leq 0.5$ m and $t = 12$ h (4×10^4 s), we obtain an attribute limit

$$a \leq 3 \times 10^{-6} \text{m}^2/\text{s} \qquad (6.61)$$

The selection Figure 6.28 shows thermal conductivity λ plotted against thermal diffusivity a with M and the limit on a plotted on it. It identifies the group of materials, listed in Table 6.28: They maximize M_1 while meeting the constraint on wall thickness. Solids are good; porous materials and foams (often used in walls) are not.

Postscript All this is fine, but what of cost? If this scheme is to be used for housing, cost is an important consideration. The approximate costs per unit volume, read from Tables A.3 and A.11 in Appendix A, are listed in the table—it points to the selection of concrete, with stone and brick as alternatives.

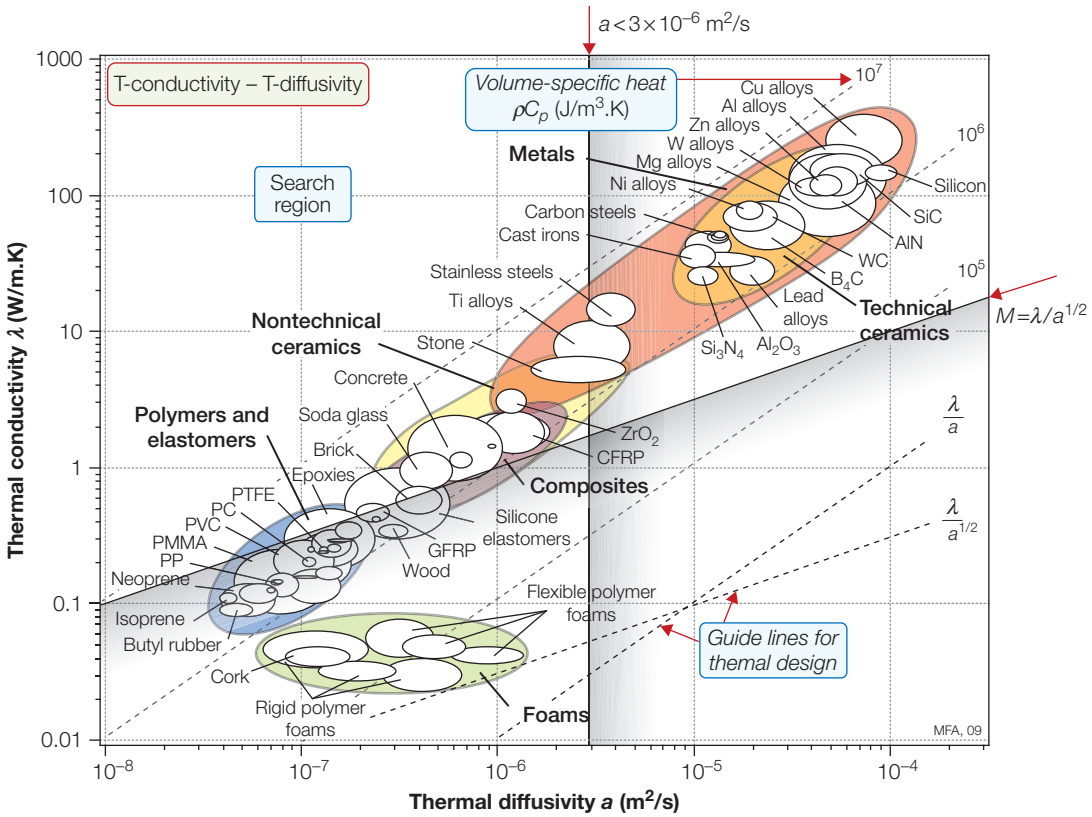


FIGURE 6.28
Materials for heat-storing walls. Cement, concrete, and stone are practical choices; brick is less good.

Table 6.28 Materials for Passive Solar Heat Storage

Material	$M_1 = \lambda/a^{1/2}$ ($W.s^{1/2}/m^2.K$)	Approximate Cost $/m^3	Comment
Concrete	2.2×10^3	200	The best choice—good performance at minimum cost
Stone	3.5×10^3	1400	Better performance than concrete because specific heat is greater, but more expensive
Brick	10^3	1400	Not as good as concrete
Glass	1.6×10^3	10,000	Useful—part of the wall could be glass
Titanium	4.6×10^3	200,000	An unexpected, but valid, selection; expensive

Related case studies

6.13 "Insulation for short-term isothermal containers"

6.14 "Energy-efficient kiln walls"

6.16 MATERIALS TO MINIMIZE THERMAL DISTORTION IN PRECISION DEVICES

The precision of a measuring device, like a submicrometer displacement gauge, is limited by its stiffness and by the dimensional change caused by temperature gradients. Compensation for elastic deflection can be arranged; and corrections to cope with thermal expansion are possible, too—provided the device is at a uniform temperature. *Thermal gradients* are the real problem: They cause a change of shape—that is, a distortion of the device—for which compensation is not possible. Sensitivity to vibration is also a problem: Natural excitation introduces noise and thus imprecision into the measurement. So it is permissible to allow expansion in precision instrument design, provided distortion does not occur (Chetwynd, 1987). Elastic deflection is allowed, provided natural vibration frequencies are high.

What, then, are good materials for precision devices? Table 6.29 lists the requirements.

The translation Figure 6.29 shows, schematically, such a device: It consists of a probe, a force loop, an actuator, and a sensor. We want a material for the force loop. It will, in general, support heat sources: the fingers of the operator of the device in the figure, or, more usually, electrical or electronic components that generate heat. The relevant material index is found by considering the simple case of one-dimensional heat flow through a rod insulated except at its ends, one of which is at ambient temperature and the other is connected to the heat source. In the steady state, Fourier's law is

$$q = -\lambda \frac{dT}{dx} \tag{6.62}$$

Table 6.29 Design Requirements for Precision Devices	
Function	Force loop (frame) for precision device
Constraints	Must tolerate heat flux Must tolerate vibration
Objective	Maximize positional accuracy (minimize distortion)
Free variable	Choice of material

where q is heat flux per unit area, λ is the thermal conductivity, and $\frac{dT}{dx}$ is the resulting temperature gradient. The strain is related to temperature by

$$\varepsilon = \alpha(T - T_o) \qquad (6.63)$$

where α is the thermal expansion coefficient and T_o is ambient temperature. The distortion is proportional to the gradient of the strain:

$$\frac{d\varepsilon}{dx} = \frac{\alpha\, dT}{dx} = \left(\frac{\alpha}{\lambda}\right) q$$

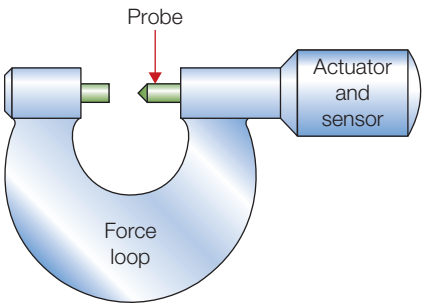


FIGURE 6.29
A schematic of a precision measuring device. Super-accurate dimension-sensing devices include the atomic-force microscope and the scanning-tunneling microscope.

Thus for a given geometry and heat flow, the distortion $d\varepsilon/dx$ is minimized by selecting materials with large values of the index

$$M_1 = \frac{\lambda}{\alpha} \qquad (6.64)$$

The other problem is vibration. The sensitivity to external excitation is minimized by making the natural frequencies of the device as high as possible. The flexural vibrations have the lowest frequencies; they are proportional to

$$M_2 = \frac{E^{1/2}}{\rho}$$

A high value of this index will minimize the problem. Finally, of course, the material must not cost too much.

The selection Figure 6.30 reproduces the chart of expansion coefficient, α, and thermal conductivity, λ. Contours show constant values of the quantity λ/α. A search region is isolated by the line $\lambda/\alpha = 10^7$ W/m, giving the shortlist of Table 6.30. Values of $M_2 = E^{1/2}/\rho$ read from the $E - \rho$ chart in

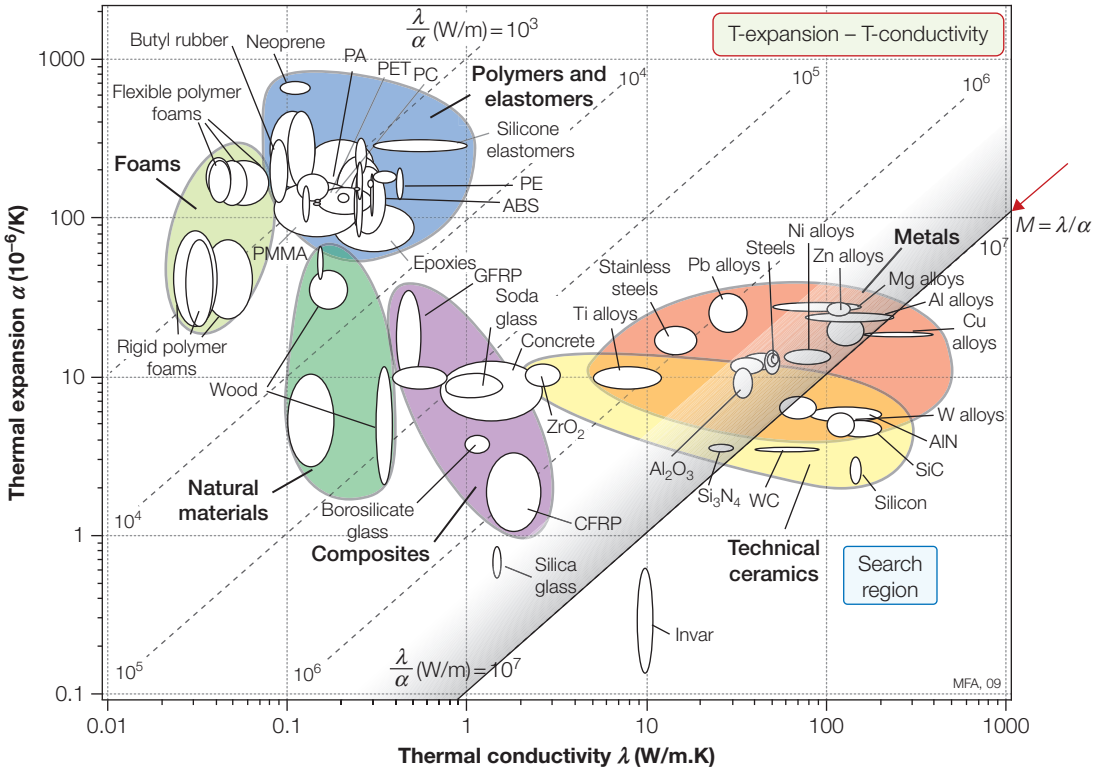


FIGURE 6.30

Materials for precision measuring devices. Metals are less good than ceramics because they have lower vibration frequencies. Silicon may be the best choice.

Table 6.30 Materials to Minimize Thermal Distortion

Material	$M_1 = \lambda/\alpha$ (W/m)	$M_2 = E^{1/2}/\rho$ $(GPa^{1/2})/(Mg/m^3)$	Comment
Silicon	6×10^7	5.2	Excellent M_1 and M_2
Silicon carbide	3×10^7	6.4	Excellent M_1 and M_2 but more difficult to shape than silicon
Copper	2×10^7	1.3	High density gives poor value of M_2
Tungsten	3×10^7	1.1	Better than copper, silver, or gold, but not as good as silicon or SiC
Aluminum alloys	10^7	3.3	The least expensive and the most easily shaped choice

Figure 4.3 are included in the table. Among metals, copper, tungsten, and the special nickel alloy Invar have the best values of M_1 but are disadvantaged by having high densities and thus poor values of M_2. The best choice is silicon, available in large sections, with high purity. Silicon carbide is an alternative.

Postscript Nanoscale measuring and imaging systems suffer from the problem analyzed here. The atomic-force microscope and the scanning-tunneling microscope both rely on a probe, supported on a force loop, typically with a piezo-electric actuator and electronics to sense the proximity of the probe to the test surface. Closer to home, the mechanism of a video recorder and that of a hard-disk drive qualify as precision instruments; both have a sensor (the read head) attached, with associated electronics, to a force loop. The materials identified in this case study are the best choice for force loop.

Related reading

Chetwynd, D.G. (1987). Selection of structural materials for precision devices. *Precision Engineering*, *9*(1), 3.

Cebon, D., & Ashby, M.F. (1994). Materials selection for precision instruments. *Meas. Sci. and Technol.*, *5*, 296.

Related case studies

6.3 "Mirrors for large telescopes"

6.13 "Insulation for short-term isothermal containers"

6.17 MATERIALS FOR HEAT EXCHANGERS

This and the next two case studies illustrate how the CES software, described in Section 5.5, is used to explore material selection in greater depth.

Heat exchangers take heat from one fluid and pass it to a second (Figure 6.31). The fire-tube array of a steam engine is a heat exchanger, taking heat from the hot combustion gases of the firebox and transmitting it to the water in the boiler. The network of finned tubes in an air conditioner is a heat exchanger, taking heat from the air of the room and dumping it into the working fluid of the conditioner. A key element in all heat exchangers is the tube wall or membrane that separates the two fluids. It is required to

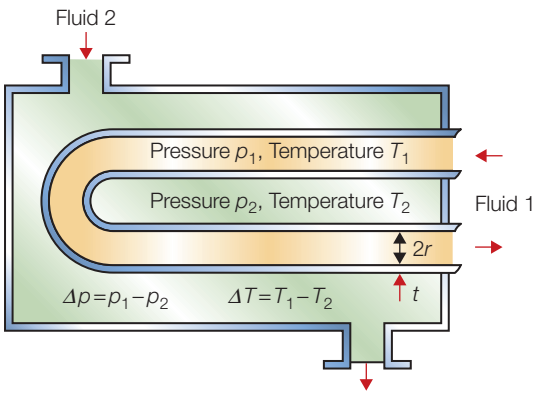


FIGURE 6.31
A heat exchanger. There is a pressure difference Δp and a temperature difference ΔT across the tube wall that also must resist attack by chloride ions.

Table 6.31 Design Requirements for a Heat Exchanger

Function	Heat exchanger
Constraints	Support pressure difference Δp Withstand chloride ions Operating temperature up to 150°C Modest cost
Objectives	Maximize heat flow per unit area (minimum volume exchanger), or maximize heat flow per unit mass (minimum mass exchanger)
Free variables	Tube-wall thickness, t Choice of material

transmit heat, and there is frequently a pressure difference across it, which can be large.

What are the best materials for making heat exchangers? Or, to be specific, what are the best materials for a conduction-limited exchanger with substantial pressure difference between the two fluids, one of them containing chloride ions (seawater)? Table 6.31 contains a summary of these requirements.

The translation First, a little background on heat flow. Heat transfer from one fluid, through a membrane to a second fluid, involves *convective* transfer from fluid 1 into the tube wall, *conduction* through the wall, and *convection* again to transfer it into fluid 2. The heat flux into the tube wall by convection (W/m^2) is described by the heat transfer equation:

$$q = h_1 \Delta T_1 \tag{6.65}$$

in which h_1 is the heat transfer coefficient and ΔT_1 is the temperature drop across the surface from fluid 1 into the wall. Conduction is described by the conduction (or Fourier) equation, which, for one-dimensional heat flow takes the form

$$q = \lambda \frac{\Delta T}{t} \tag{6.66}$$

where λ is the thermal conductivity of the wall (thickness t) and ΔT is the temperature difference across it. It is helpful to think of the *thermal resistance* at surface 1 as $1/h_1$; that of surface 2 as $1/h_2$; and that of the wall itself as t/λ. Then continuity of heat flux requires that the total resistance $1/U$ is

$$\frac{1}{U} = \frac{1}{h_1} + \frac{t}{\lambda} + \frac{1}{h_2} \tag{6.67}$$

where U is called the "total heat transfer coefficient." The heat flux from fluid 1 to fluid 2 is then given by

$$q = U(T_1 - T_2) \tag{6.68}$$

where $(T_1 - T_2)$ is the difference in temperature between the two working fluids.

When one of the fluids is a gas—as in an air conditioner—convective heat transfer at the tube surfaces contributes most of the resistance; then fins are used to increase the surface area across which heat can be transferred. But when both working fluids are liquid, convective heat transfer is rapid and conduction through the wall dominates the thermal resistance; $1/h_1$ and $1/h_2$ are negligible compared with t/λ. In this case, simple tube or plate elements are used, making the wall as thin as possible to minimize t/λ. We will consider the second case: conduction-limited heat transfer. Then, the heat flow is adequately described by Equation (6.66).

Consider, then, a heat exchanger with n tubes of length L, each of radius r and wall thickness t. Our aim is to select a material to maximize the total heat flow:

$$Q = qA = \frac{A\lambda}{t}\Delta T \tag{6.69}$$

where $A = 2\pi r L n$ is the total surface are of tubing.

This is the objective function. The constraint is that the wall thickness must be sufficient to support the pressure Δp between the inside and outside, as in Figure 6.31. This requires that the stress in the wall remain below the elastic limit, σ_y, of the material of which the tube is made (multiplied by a safety factor—which we can leave out):

$$\sigma = \frac{\Delta p r}{t} < \sigma_y \tag{6.70}$$

This constrains the minimum value of t. Eliminating t between Equations (6.69) and (6.70) gives

$$Q = \frac{A\Delta T}{r\Delta p}(\lambda\sigma_y) \tag{6.71}$$

The heat flow per unit area of the tube wall, Q/A, is maximized by maximizing

$$M_1 = \lambda\sigma_y \tag{6.72}$$